



Affirmed Networks Gy Interface Specification

Release 9.4.0.0

Updated: October 2019

Inside ...

- Interface Overview
- Message Flows
- Configuring
- Monitoring
- Failure Handling
- Message Definitions
- AVP Descriptions

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Figures

Preface

This specification contains information on the Gy interface used for reporting and charging.

Note: *The Affirmed Networks Mobile Content Cloud™ is the mobile services software solution that operates on virtual platforms such as blade systems.*

Audience

This guide is intended for network operators (sometimes referred to as Users or Operators in this guide) working with the Gy interface. The audience should have basic knowledge of the Offline Charging System (OFCS), Online Charging System (OCS), and GTP' protocol. The audience should also be familiar with 3G or 4G networks.

Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

How to Use This Guide

Read the following chapters and appendices in the order in which they appear:

Chapter/Appendix	Describes
Chapter 1, Gy Interface	The message flows, message definitions, and failure handling used to control and monitor the Gy interface.
Appendix A, Diameter AVP Descriptions	The attribute value pair definitions for the Gy interface.

References and Specifications

The system supports the following references and specifications for the Gy interface:

- 3GPP TS 29.212 V15.1.0 (2017-12): *Technical Specification Group Core Network and Terminals; Policy and Charging Control (PCC); Reference points*
- 3GPP TS 29.214 V15.1.0 (2017-09): *Technical Specification Group Core Network and Terminals; Policy and Charging Control over Rx reference point*
- 3GPP TS 32.299 V10.7.0 (2012-09): *Telecommunication management; Charging management; Diameter charging application*
- 3GPP TS 32.299 V15.2.0 (2018-06): *Telecommunication management; Charging management; Diameter charging application*
- 3GPP TS 32.251 V9.9.0 (2012-06): *Telecommunication management; Charging management; Packet Switched (PS) domain charging*
- RFC 3588: *Diameter Based Protocol*
- RFC 3539: *Authentication, Authorization and Accounting (AAA) Transport Profile*
- RFC 4006: *Diameter Credit-Control Application*
- RFC 6733: *Diameter Base Protocol*
- Vodafone Variants

Features in this Release

This release includes the following new features. Note the following:

- For a complete list of the features supported in this release, see the *Software Release Notes*.
- For a list of additional Affirmed Networks documentation, see the *Affirmed Networks Documentation Library*.
- For a comprehensive list of terms and acronyms used in Affirmed Networks documentation, see *Affirmed Networks Terms and Acronyms*.

Feature	Description
Added support for the TMO-Virtual-Gi-ID VSA on RADIUS, Gx, and Gy Interfaces See TMO-Virtual-Gi-ID (page A-128) .	This feature adds support for the TMO-Virtual-Gi-ID vendor-specific attribute (VSA) on RADIUS, Gx, and Gy interfaces. This feature also adds two new configuration parameters under the APN object to specify whether to process the TMO-Virtual-Gi-ID VSA when received in the Access-Accept message from the RADIUS server. <pre>an3000-mcm-slot7cpu1(config)# zone default gateway apn <apn-name> vsa-virtual-gi-id-enabled [true false] vsa-virtual-gi-id <default-value></pre> ■ vsa-virtual-gi-id-enabled – true – If the MCC gateway receives the TMO-Virtual-Gi-ID VSA in the Access-Accept message, it sends the VSA in Accounting-Request messages (Start/Interim/Stop) on the RADIUS interface and in CCR-I messages on the Gx and Gy interfaces. – false – (default) If the MCC gateway receives the TMO-Virtual-Gi-ID VSA in the Access-Accept message, it does not send the VSA in Accounting-Request messages on the RADIUS interface or in CCR-I messages on the Gx and Gy interfaces. ■ vsa-virtual-gi-id – Specifies a default value for the TMO-Virtual-Gi-ID VSA. If the vsa-virtual-gi-id-enabled parameter is set to true but the MCC gateway does not receive the TMO-Virtual-Gi-ID VSA in the RADIUS Access-Accept message, the MCC gateway uses this default value to fill the TMO-Virtual-Gi-ID VSA sent in RADIUS Accounting-Request messages (Start/Interim/Stop) and in CCR-I messages on the Gx and Gy interfaces. This parameter is only visible in the CLI when the vsa-virtual-gi-id-enabled parameter is set to true.

Feature	Description
Added support for the TMO-Virtual-Gi-ID VSA on RADIUS, Gx, and Gy Interfaces <i>Continued</i> See TMO-Virtual-Gi-ID (page A-128) .	You can configure whether to include or exclude the TMO-Virtual-Gi-ID VSA in RADIUS Access-Accept and Accounting-Request (Start/Stop/Interim) messages and Gx/Gy CCR-I messages. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw auth radius access-accept <template-name> tmo_virtual_gi_id presentFlag [conditional no yes] an3000-mcm-slot7cpu1(config)# service-construct data-record-template pdngw accounting radius [start interim stop] <template-name> tmo_virtual_gi_id presentFlag [conditional no yes] an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw [pcrf ocs] diameter start <template-name> tmovirtualgiid presentFlag [yes no conditional]</pre>

Feature	Description
Added support for dynamic rating groups (quota-id not pre-configured on the MCC) and the provisioning of rating sub-groups and a new parameter to configure a dynamic rating group template See the <i>Affirmed Networks Gx Interface Specification</i>.	Added the ability to dynamically allocate a rating group when the quota-id received in the Charging-Rule-Definition AVP is not pre-configured on the MCC. When the PCRF dynamically allocates a rating group, the MCC uses the Online and Offline AVPs received in the Charging-Rule-Definition AVP to enable the corresponding charging method for the rating group/service rule. This feature also adds the ability to configure a list of separate rating sub-groups under the same rating group (quota-id), provided the <code>charging-method</code> for the rating group is set to <code>subgroup</code>. A rating sub-group can have any of the following charging methods: ■ Only online charging ■ Only offline charging ■ Both offline and online charging ■ No charging Use the following commands to configure a rating group with rating sub-groups. 1 Configure a rating sub-group. <pre>an3000-mcm-slot7cpul(config)# service-construct rating-sub-group <sub-group-name></pre> The following parameters are configurable for a rating sub-group. offline-enabled – Specifies whether to enable/disable offline charging for the rating sub-group. The default is true. online-enabled – Specifies whether to enable/disable online charging for the rating sub-group. The default is true. offline-volume-threshold – Specifies the volume threshold, in kilobytes, after which a new service-rule level service data container is generated for rules associated with this rating sub-group. offline-time-threshold – Specifies the time threshold, in seconds, after which a new service-rule level service data container is generated for rules associated with this rating sub-group. service-rule – Specifies the service rule(s) to associate with this rating sub-group. A service rule can only be associated with one rating sub-group within the same rating group. A service rule can be associated with two rating sub-groups as long as the two rating sub-groups are not associated with the same rating group. The service rule must already exist. 2 Associate the rating sub-groups with a rating group. <pre>an3000-mcm-slot7cpul(config)# services charging rating-group <rating-group> charging-method subgroup rating-sub-group <list-of-sub-groups></pre> If the <code>charging-method</code> for the rating group is set to <code>subgroup</code>, you cannot configure a service rule directly under the rating group. Service rules must be associated with a rating sub-group. <i>Note: The Override Control feature and synchronizing Gz/Rf and Gy Charging Data does not apply to rating sub-groups.</i>

Feature	Description
Added support for dynamic rating groups (quota-id not pre-configured on the MCC) and the provisioning of rating sub-groups and a new parameter to configure a dynamic rating group template <i>Continued</i> See the <i>Affirmed Networks Gx Interface Specification</i>.	This feature also adds a new configuration parameter to specify a rating group template to use when a dynamic rating group is created because of a charging rule install from the PCRF. <pre>an3000-mcm-slot7cpu1(config)# workflow data-profile <data-profile-name> dynamic-rg-template <rating-group-name></pre> dynamic-rg-template – Specifies the name of a rating group to be associated with the data profile. The rating group must already be configured and will be used as a template when a dynamic rating group is created because of a charging rule install from the PCRF. The associated rating group must be configured with a <code>charging-method</code> of <code>subgroup</code>.

Feature	Description
Enhanced the provisioning of the default charging method See the <i>Affirmed Networks Gx Interface Specification</i> .	This feature enhances the provisioning of the default charging method for a session. When the Online/Offline AVPs are included in the CCR-Initial (CCR-I) message at the command level, these AVPs indicate the default charging method for the session based on the configuration at the PCEF. This feature also adds support for the Online and Offline AVPs at the RAR command-level on the Gx interface. Currently, the <code>user-subscription-charging-type</code> parameter configured under the control-profile object is used to advertise the default charging method (online/offline) supported on the MCC. With this feature, the default charging method can also be configured under the APN object and/or the crbn-profile object and used to advertise the session-wide default charging method in the Online/Offline AVPs in the CCR-I message at the command level. However, the default charging method provided by the PCRF in the CCA-Initial (CCA-I) at the command level takes precedence over any pre-configured default charging method on the MCC gateway (PCEF). After processing the CCA-I, the MCC will derive the session-wide default charging method using the following order of priority. 1 Online/Offline AVP provided by the PCRF in the CCA-I at the command level. 2 Online/Offline configuration under the first activated CRBN profile in the CCA-I. 3 Online/Offline configuration under the APN. 4 Online/Offline configuration under the Control Profile. The MCC also now supports a PCRF-initiated mid-session change to the session-wide default charging method by including the Online/Offline AVPs in the RAR message at the command level. ■ If the PCRF changes online charging from enabled to disabled mid-session, the MCC reports all current outstanding usage to the OCS and disables online charging for the entire session. When online-charging is changed from disabled to enabled, the MCC starts tracking usage from that moment and reports the usage to the OCS. ■ If the PCRF changes offline charging from enabled to disabled mid-session, the MCC reports all current outstanding usage in service-rule level service data containers and disables offline charging for the entire session. When offline-charging is changed from disabled to enabled, the MCC starts tracking usage from that moment and reports the usage in offline CDRs. Set parameters for this feature using the following commands. The default is true. <pre>an3000-mcm-slot7cpu1(config)# zone default gateway apn <apn-name> charging-methods offline-enabled [true false] online-enabled [true false] an3000-mcm-slot7cpu1(config)# services crbn-profile <crbn-profile-name> charging-methods offline-enabled [true false] online-enabled [true false]</pre>

Feature	Description
Added a new configuration parameter to control the charging method for a service rule based on both the session-wide default charging method and the service-rule default charging method See the <i>Affirmed Networks Gx Interface Specification</i> and Configuring the Gy Online Charging Interface (page 1-11).	This feature adds a new configuration parameter under the gateway-common profile to control the charging method for a service rule based on both the session-wide default charging method and the service-rule default charging method. <pre>an3000-mcm-slot7cpu1(config)# zone default gateway profile gateway-common <name> use-session-wide-default-charging-state-as-gating [enable disabled]</pre> ■ enabled – A rule will be charged using the online/offline charging interface if the Online/Offline AVP is enabled at both the session level and the service-rule level. Otherwise, the rule is not charged. ■ disabled – (default) A rule will be charged using the online/offline charging interface if the Online/Offline AVP is enabled at the service-rule level. This feature also adds a new configuration parameter to specify whether multiple Service-Identifiers can be included within the Multiple-Services-Credit-Control AVP on the Gy interface. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> avp-customization mscc-serviceid num-instances [multiple single]</pre>

Feature	Description
Added ability to select a new control profile and online/offline interface using the CRBN from the PCRF See the <i>Affirmed Networks Gx Interface Specification</i>.	This feature adds the ability to select a new control profile and online/offline interface using the Charging-Rule-Base-Name (CRBN) AVP received from the PCRF. When the MCC gateway receives a CCA-I message from the PCRF containing the CRBN AVP, the MCC gateway can use the CRBN as a subscriber analyzer key to select a new subscriber analyzer. In doing so, the subscriber sessions use the control profile defined for the new subscriber analyzer. When a new control profile is selected using this method, the new control profile can only be used to make the following online/offline interface changes: ■ Online – Switch from a control profile with no online interface to a control profile with an online interface. ■ Offline – Switch to a control profile with a different offline interface. Only the following parameters can differ between the old offline interface and the new offline interface. – data-volume-limit – interim-update-interval – cdr-close-time Set parameters for this feature using the following existing commands. 1 Ensure the <code>cbrn-mode</code> parameter for the <code>pcrf-charging</code> interface is set to <code>data-profile</code>. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface pcrf-charging <interface-name> cbrn-mode data-profile</pre> 2 Enable the MCC gateway to use the CRBN AVP received from the PCRF as a key to select a new subscriber analyzer. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface pcrf-charging <interface-name> allow-cbrn-dataprof-based-subscriber-analyzer-reselection enabled</pre> 3 Configure a subscriber analyzer with a list of CRBNs (locally configured data-profiles) as one of the key attributes and associate a control profile with this subscriber analyzer. The control profile must already exist. <pre>an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer <subscriber-analyzer-name> analyzer-type workflow-profile action control-profile <control-profile-name> key <key-name> cbrn-dataprof-name-list values <list-of-data-profiles> match-type [match-any match-none]</pre>

Feature	Description
Added a new configuration option to allocate a default time quota for use with the Default Gy Quota feature See Default Gy Quota (page 1-72).	Previously, the Default Gy Quota feature could only be used with rating groups that had the <code>measurement-method</code> set to <code>volume</code>. With this feature, the Default Gy Quota feature can also be used with rating groups that have the <code>measurement-method</code> set to <code>time</code> or <code>timeAndVolume</code>. Use the following command to configure a default time quota for use with the Default Gy Quota feature: <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> ocs-down-default-time-quota <value></pre> ocs-down-default-time-quota – Specifies the default time threshold, in seconds, allocated during an OCS failure when the rating group <code>measurement-method</code> set to <code>time</code> or <code>timeAndVolume</code>. The default is 90 seconds.
Added support for new TMO-Transparent Data attribute for the RADIUS Access-Accept message and in CCR messages on the Gy interface, and added support for a new CDR field on the Ga interface See the <i>Affirmed Networks Gateway Administration Guide</i> and TMO-Transparent-Data (page A-128).	This feature lets the PGW select MCC/MNC (PLMN ID) values in a configured priority order by using the GTP message from the SGSN/SGW, a sig-info-table, or the RADIUS Access-Accept message from the RADIUS authentication server. This feature also lets you configure the selection of TMO-Transparent-Data values on the PGW when the RADIUS Access-Accept message does not return a transparent data value. The PGW selects a value from a sig-info-table with PLMN ID mappings or from a sig-info-table with SGSN/SGW subnet mappings. This is enabled at the APN level. Added support for the following: ■ TMO Transparent Data AVP in CCR messages. This AVP is configurable in Gy CCR data record templates and by default is set to no, indicating the field is not sent. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw ocs diameter [event interim start stop] <template-name> tmotransparentdata presentFlag [yes no]</pre> ■ TMO-Transparent-Data attribute in RADIUS Access-Accept message from the RADIUS authentication server. This attribute is configurable in RADIUS authorization access-accept data record templates and by default is set to no, indicating the field is not processed. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw auth radius access-accept <template-name> tmo_transparent_data presentFlag [yes no]</pre> ■ TMO Transparent Data subfield in the RecordExtensions field of GTPP CDRs (sent via the Ga interface). This subfield, transparentVSA, is configurable in GTPP data record templates and by default is set to no, indicating the field is not present in CDRs. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template [pgw sgw] ofcs gtp [interim stop] <template-name> extensioninfo presentFlag [yes no]</pre> Three other subfields have also been added to support the transparentVSA subfield: <code>extensionType</code>, <code>length</code>, and <code>serviceList</code>. To view the TMO transparent data string for a subscriber on a PGW or SGW: <pre>an3000-mcm-slot7cpu1# show subscriber pdn-session <id></pre>

Feature	Description
Added new configuration parameters to control whether the Destination-Host and Destination-Realm AVPs are updated based on information received in the CCA-I/CCA-U/RAR messages See Configuring the Gy Online Charging Interface (page 1-11).	With this feature, you can control whether the Destination-Host/Destination-Realm AVPs are updated in CCR-U/CCR-T messages on the Gx/Gy interfaces based on information received in the CCA-I/CCA-U/RAR messages. The default is <code>enabled</code> for the CCA-I/CCA-U messages and <code>disabled</code> for the RAR message. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface [pcrf-charging online-charging] <interface-name> allow-dest-host-realm-change [cca-i cca-u rar] [enabled disabled]</pre> ■ enabled – If the received Origin Host/Origin Realm in the CCA-I/CCA-U/RAR is different from the current destination host and realm, the MCC uses the Origin-Host/Origin-Realm AVPs received in the CCA-I/CCA-U/RAR message to fill the Destination-Host/Destination-Realm AVPs in subsequent CCR-U/CCR-T messages. ■ disabled – The MCC gateway continues to use the existing values to fill the Destination-Host/Destination-Realm AVPs in subsequent CCR-U/CCR-T messages and does not update the Destination-Host/Destination-Realm AVPs with the values received in the CCA-I/CCA-U/RAR messages. <i>Note: The MCC gateway always sends the RAA message with the Destination Host and Realm values of the Origin Host and Realm received in the RAR message.</i> <i>Note: For the CCA-I/CCA-U, the MCC will only update the Destination-Host and Destination-Realm in CCR-U/CCR-T messages if the result code received in the CCA-I/CCA-U is DIAMETER_SUCCESS (2001).</i>
Added a new online charging configuration option to restrict the PGW from establishing the online-charging session and requesting quota with the OCS until after it receives the first service data flow packet for rating groups configured with <code>credit-authorization-event</code> set to <code>bearerSetup</code> See Setting up the Gy Session After Receiving the First SDF Packet (page 1-45).	Typically, the online-charging session with the OCS is established as part of the PDN session setup. With this feature, you can restrict the PGW from establishing an online-charging session and requesting quota until after it receives the first service data flow (SDF) packet for rating groups configured with the <code>credit-authorization-event</code> parameter set to <code>bearerSetup</code>. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> first-packet-credit-auth-rating-groups [firstsdf all]</pre> ■ firstsdf – (default) The credit authorization for rating groups with the <code>credit-authorization-event</code> set to <code>firstSdfPkt</code> will be deferred until the PGW receives the first SDF packet. ■ all – For all active online rating groups, the credit authorization will be deferred until the PGW receives the first SDF packet. When the PGW receives the first SDF packet on any active online rating group, the quota request (MSCC AVP) is included in the CCR-U for all rating groups with <code>credit-authorization-event</code> set to <code>bearerSetup</code> and for the rating group (<code>bearerSetup/firstSdfPkt</code>) that received the first packet. For rating-groups with the <code>credit-authorization-event</code> set to <code>firstSdfPkt</code>, the credit authorization is requested when the PGW receives the first SDF packet for the rating group. <i>Note: This parameter is only visible when <code>interface-compliance</code> is set to <code>3GPPVersion2</code>.</i>

Feature	Description
Added support for a new configuration option to configure the validity-time on the Gy interface See Configuring the Gy Online Charging Interface (page 1-11).	This feature adds a new online-charging configuration option to configure a validity time that is used when the OCS does not send the Validity-Time AVP within the MSCC AVP. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> attributes validity-time <value-in-seconds></pre>
Added support for the CGI-SAI change and ECGI change event-triggers and increased the maximum number of SDCs that the gateway maintains per CDR from 20 to 50 See Configuring the Gy Online Charging Interface (page 1-11).	This feature adds two new event-triggers to the <code>event-trigger-template</code>, which is used to specify the events that trigger an interim update on offline/online charging interfaces and to trigger the closure of an interim SGW/PGW/GGSN CDR. <i>Applies to the Gy, Gz/Rf and Ga (GTPP), local (ASN.1) interfaces.</i> For all event-triggers in the <code>event-trigger-template</code>, the maximum service data container (SDC) limit per CDR closure event is increased from 20 to 50. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface-trigger-template event-trigger-template <template-name> [cgi-sai-change-close-trigger ecgi-change-close-trigger]</pre> ■ cgi-sai-change-close-trigger – Specifies whether a change in the Cell Global Identification- Service Area Identifier (CGI-SAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of CGI-SAI change events to trigger a CDR close. ■ ecgi-change-close-trigger – Specifies whether a change in the E-UTRAN Cell Global Identifier (ECGI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of ECGI change events to trigger a CDR close. This feature also increased the maximum number of SDCs that the gateway maintains per CDR at the bearer-level from 20 to 50 if interim-updates are enabled on the interface. Upon reaching this limit, the gateway transfers the SDCs in an interim CDR to the offline server. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface offline-charging <interface-name> [pgw sgw ggsn wag epdg] num-service-container [1..50]</pre>

Feature	Description
Added a new configuration option under the online-charging interface to specify either incremental or accumulative volume-based usage reporting to the OCS See Configuring the Gy Online Charging Interface (page 1-11).	This feature adds a new configuration option under the online-charging interface to specify whether the MCC reports the CC-Total-Octets, CC-Input-Octets, and CC-Output-Octets (within the Used-Service-Units AVP) either accumulatively or incrementally. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> usage-reporting-mode [accumulative-mode incremental-mode]</pre> ■ accumulative-mode – The accumulated CC-Total-Octets, CC-Input-Octets, and CC-Output-Octets is reported to the OCS in the Used-Service-Units AVP when the subscriber requests quota or when the PGW sends a CCR-U to report the final usage for the rating group. The CC-Total-Octets is reset to 0 only when the Gy session is terminated or the subscriber is cleared. ■ incremental-mode – (default) The incremental CC-Total-Octets, CC-Input-Octets, and CC-Output-Octets is reported to the OCS in the Used-Service-Units AVP each time the subscriber requests quota. The CC-Total-Octets is reset to 0 every time the PGW reports usage to the OCS. <i>Note: Changes to this parameter take effect immediately. If this parameter is changed mid-session, it applies to all new subscriber sessions.</i>
Added ability to select an online-charging interface using the CRBN profile and to configure the time-threshold and volume-threshold for the online-charging interface See Using the CRBN Profile to Select an Online-Charging Interface (page 1-63) and Configuring the Gy Online Charging Interface (page 1-11).	This feature allows you to select an online-charging interface using the CRBN profile if that CRBN profile has an online-charging interface configured. <pre>an3000-mcm-slot7cpu1(config)# services crbn-profile <profile-name> online-interface <interface-name></pre> Note the following guidelines when using this feature: ■ The online-charging interface configured for the CRBN profile overrides the online-charging interface configured for the workflow control profile. ■ If the MCC gateway receives multiple CRBN profiles from the PCRF or RADIUS server, the MCC uses the first CRBN profile in the list to select the online-charging interface. This feature also adds two new online-charging configuration options to configure the time-threshold and volume-threshold. The MCC uses these configured values when the OCS does not send the Time-Quota-Threshold and Volume-Quota-Threshold AVPs within the MSCC AVP and these values are not configured under the rating group. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> attributes time-threshold <value> Volume-threshold <value></pre> Note the following guidelines when using this feature: ■ The time-threshold and volume-threshold configured for the rating group take precedence over the time-threshold and volume-threshold configured for the online-charging interface. ■ If the GSU is received with result code 4012, the MCC does not apply the time-threshold and volume-threshold configured for the rating group or online-charging interface.

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menus	Times New Roman bold font	Use configure_ha to configure... Enter y . To continue, press Enter .
Selecting a menu item	Menu => Option	Select Security => Group Management .
Variable parameters for which you supply values	<i><courier bold italic blue font></i>	configure_dr convert slave --masterhost <i><host IP></i>
User input	Courier bold blue font	cd /opt/Affirmed/NMS/bin
Screen messages, system output, sample source code	Courier regular blue font	Checking for License...
Required keywords and arguments in square brackets	[]	cluster [id] node [id] reboot
Optional keywords and arguments in curly brackets	{ }	{cdr-file-name}
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.		Is this correct? [y n]
Keywords representing a single element in angle brackets	< >	<i><host name></i>
Asterisk after a CLI object indicates a required object.	*	name*
Variables, book and chapter titles, file path names, new terms, and emphasized text	<i>Times New Roman italic font</i>	See the <i>Acuitas User's Guide</i> .

Tip: Helpful suggestions.

Note: *Additional information that may apply to the subject text.*



CAUTION: Proceed carefully to avoid possible equipment damage or data loss.

How to Obtain Product Documentation

Documentation is available for currently supported product releases. Documentation is available online in Adobe PDF format. You can view PDFs online using the latest version of Adobe Reader®. To download the latest version of the Adobe Reader software from the Adobe web site, go to <http://get.adobe.com/reader/>

Product documentation is available from the Affirmed Networks SFTP server. Contact your sales representative or the TAC for instructions on accessing the SFTP server.

See the *Affirmed Networks Documentation Library* for a complete list of Affirmed Networks product documentation.

Technical Assistance Center

Affirmed Networks Technical Assistance Center (TAC) provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an active maintenance contract.

To contact the TAC:



E-mail: prodsupport@affirmednetworks.com



24 x 7 Telephone support:

Local: 1-978-268-0871

Toll free: 1-888-283-2838

Gy Interface

This section contains the following information used to control and monitor the Gy interface:

- [Interface Overview \(page 1-2\)](#)
- [Message Flows \(page 1-5\)](#)
- [Configuring the Gy Online Charging Interface \(page 1-11\)](#)
- [Optional Gy Interface Configuration Tasks \(page 1-37\)](#)
- [Monitoring \(page 1-64\)](#)
- [Failure Handling \(page 1-68\)](#)
- [Message Definitions \(page 1-107\)](#)
- [Result Codes \(page 1-129\)](#)

See [Appendix A, Diameter AVP Descriptions](#) for a description of the Attribute Value Pairs (AVPs) supported on the Gy interface.

Interface Overview

The 3GPP standards define the Gy interface as the online-charging interface between the Gateway GPRS Support Node/Packet Data Network Gateway (GGSN/PGW) and the Online Charging System (OCS). The Gy interface uses the Diameter Credit Control Application (DCCA) as defined in RFC 4006. The DCCA provides a framework for real-time charging by supporting communication between subscriber management platforms and back-end billing systems. The application specifies methods for quota management (reserve, reauthorize, abandon, etc.).

Prepaid billing plans use online charging systems to authorize and manage a subscriber's quota and report data usage against a quota. The use of a quota ensures that a subscriber cannot access data services unless those services are paid in advance (otherwise known as prepaid billing). The Gy interface can also be used in parallel with offline (Charging Data Record-based) billing to ensure postpaid subscribers spend only a predefined amount of money each month.

Affirmed Networks Mobile Content Cloud (MCC) devices support session-based multi-service (rating-group level) credit control and initiate the credit management session for each IP-CAN session as part of the context activation procedure.

MCC devices support the following online charging functions:

- Multi-service rating-group level credit control
- Quota requesting and usage reporting on time, volume, and unit based thresholds
- Quota Holding Time (QHT) and Quota Consumption Time (QCT)
- Validity time
- Tariff time change reporting
- Envelope reporting
- Event trigger on rating condition change
- Network-initiated credit reauthorization from the OCS
- Network-initiated session termination from the OCS
- Local configured, RADIUS, and PCRF provided OCS parameters
- Advice of charge

Gy Interface Support for CUPS

The MCC Control and User Plane Separation (CUPS) architecture supports the Gy interface. The Gy interface is configured between the PGW-C and the OCS. There is no change to the Gy interface configuration.

The PGW-C sends charging information over PFCP to the PGW-U via the Sx interface using vendor-specific information elements (IEs). The PGW-U reports traffic usage information to the PGW-C via the Sx interface. The PGW-C then reports the usage information to the OCS via the Gy interface. See the *Affirmed Networks Gateway Administration Guide* for more information on the MCC CUPS architecture.

Note: The user-plane cluster configuration must match the configuration for the control-plane cluster, including data profiles, charging-instances, billing-plans, rating-groups, service-rules, and packet-filters.

Diameter Routing Agent

A Diameter Routing Agent (DRA) provides routing capabilities between a PGW and an OCS when multiple OCS nodes are deployed in a Diameter realm. When the PGW sends the Credit-Control-Request-Initial (CCR-I) message to a DRA, it includes the DRA host name and realm within the Destination-Host and Destination-Realm AVPs. However, the PGW uses the values from the Origin-Host/Origin-Realm AVPs received in the Credit-Control-Answer-Initial (CCA-I) response to fill the Destination-Host/Destination-Realm AVPs in subsequent CCR-Update (CCR-U), CCR-Terminate (CCR-T), Abort-Session-Answer (ASA), and Re-Auth-Answer (RAA) messages. This feature ensures the same OCS manages all messages related to a particular session.

Rating Group Memory Allocation

When a billing plan is applied to a session, the MCC:

- Allocates memory for all rating groups that have at least one service rule configured with the `service-activation` parameter set to `always-on`.
- Does not allocate memory for rating groups that only have service rules configured with the `service-activation` parameter set to `external-activation`.

When applying the billing plan for a Charging-Rule-Base-Name (CRBN) profile, the MCC only allocates memory for rating groups that have at least one service rule configured with the `service-activation` parameter set to `always-on` or that have a service rule that is part of the CRBN service rule list. See the *Affirmed Networks Gx Interface Specification* for more information on CRBN profiles.

If a rating group memory allocation was skipped while applying a billing plan (because all the rules in the rating group were set to `external-activation`), memory for this rating group gets allocated when the PCRF activates the first service rule in this rating group.

Note: The MCC does not deallocate memory for a rating group within a billing plan even if the last service rule is removed from that rating group.

Online Charging on the MCC TWAG/TWAP

The MCC Trusted WLAN Access Gateway/Proxy (TWAG/TWAP) supports the Gy interface between the MCC TWAG/TWAP and the OCS. Gy interface support on the MCC TWAG/TWAP allows for online charging when trusted non-3GPP subscribers access the Internet using Non-Seamless WLAN Offload (NSWO). For more information on the MCC TWAG/TWAP, see the *Affirmed Networks TWAG/TWAP Administration Guide*.

The MCC TWAG/TWAP supports Gx-based dynamic Gy session set up based on predefined and dynamic rules. The MCC TWAG/TWAP does not support Gy session set up independent of the Gx interface. For more information, see [Gx-Based Dynamic Gy Session Set Up \(page 1-7\)](#).

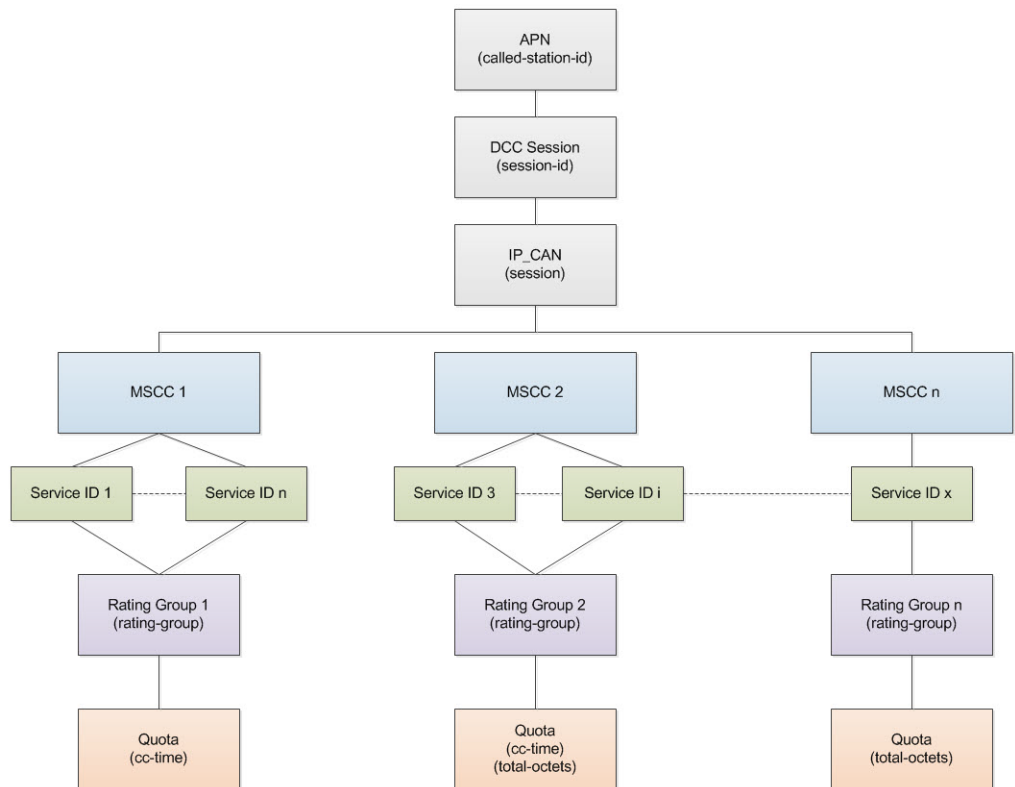
Message Flows

This section describes the relationship between service identifiers, rating groups, credit pools, and credit control sessions and shows the following message flows:

- [Typical OCS Session \(page 1-6\)](#)
- [Gx-Based Dynamic Gy Session Set Up \(page 1-7\)](#)

Figure 1-1 shows the OCS session flow.

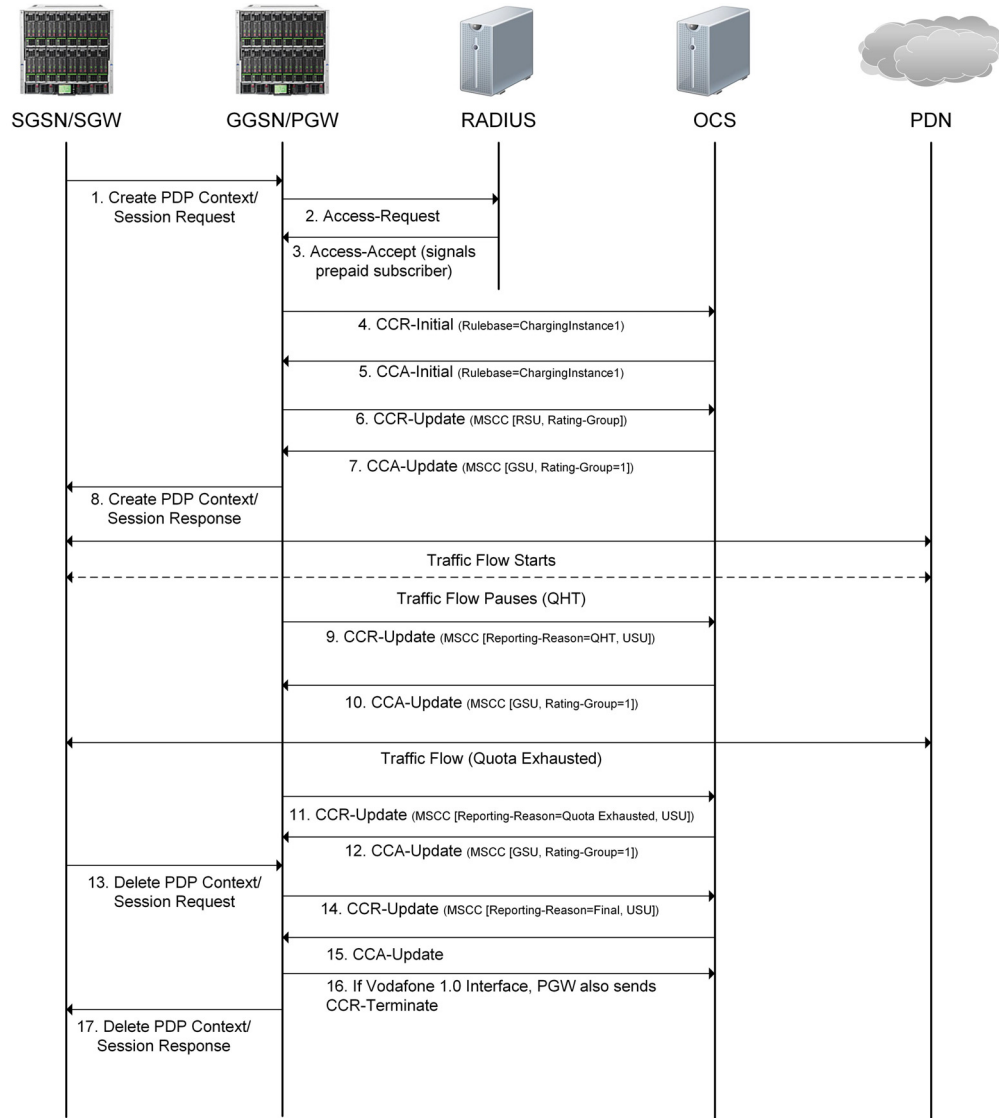
Figure 1-1 OCS Session Flow



Typical OCS Session

Figure 1-2 shows the message flow for a typical OCS session.

Figure 1-2 Typical OCS Session Message Flow



Gx-Based Dynamic Gy Session Set Up

Operators can configure the PGW/TWAG to dynamically activate, terminate and restore Gy online-charging sessions based on predefined PCC rules received from the PCRF via the Gx interface. Predefined PCC rules are pre-configured on the MCC.

Operators can also configure the PGW/TWAG to dynamically install, remove and update Gy online-charging sessions based on dynamic PCC rules received from the PCRF via the Gx interface. Dynamic PCC rules are provisioned on the PCRF and sent over the Gx interface to the PGW/TWAG.

When the PGW/TWAG receives a Create Session Request, the PGW/TWAG initiates a Gx session by sending a CCR-I to the PCRF, requesting PCC rules to establish the PDN session. If the PCC rules indicate that online charging is required (Online AVP set to enabled for a charging rule), the PGW/TWAG notifies the OCS via the Gy interface. The PGW/TWAG requests a grant for the rating group associated with the online rules via the Gy interface.

To enable the PGW/TWAG to dynamically provision a Gy session based on PCC rules from the PCRF, note the following guidelines:

- All the service rules associated with the online rating group for a PDN session must have the `service-activation` parameter set to `external-activation` to dynamically activate/install and deactivate/remove a Gy session from the PCRF. If at least one service rule associated with the online rating group for a PDN session has the `service-activation` parameter set to `always-on`, the Gy session for that PDN session remains active for the duration of the PDN session.
- If none of the service rules associated with the online rating group for a PDN session are activated/installed, the PGW/TWAG will not dynamically establish the Gy session.
- The PGW/TWAG terminates a dynamic Gy session when all externally activated/installed service rules are deactivated/removed or the PCRF sets the Online AVP to disabled. In this case, the PDN session can still progress with the active Gx session.
- The PGW/TWAG restores a dynamic Gy session when Gx redirection is removed and the Online AVP is set to enabled for any service rule. In this case, the PGW/TWAG dynamically sets up a Gy session with the OCS. Gx redirection is removed when the PCRF sends the Charging-Rule-Install AVP with the Online AVP set to enabled.
- Gx-based dynamic Gy session set up is only supported when the `interface-compliance` parameter for the online-charging interface is set to `3GPPVersion2`.

Figure 1-3 shows the message flow for dynamically activating a Gy online-charging session based on predefined PCC rules received from the PCRF via the Gx interface on the PGW.

Figure 1-3 Gx-based Dynamic Gy Session Set Up on the PGW

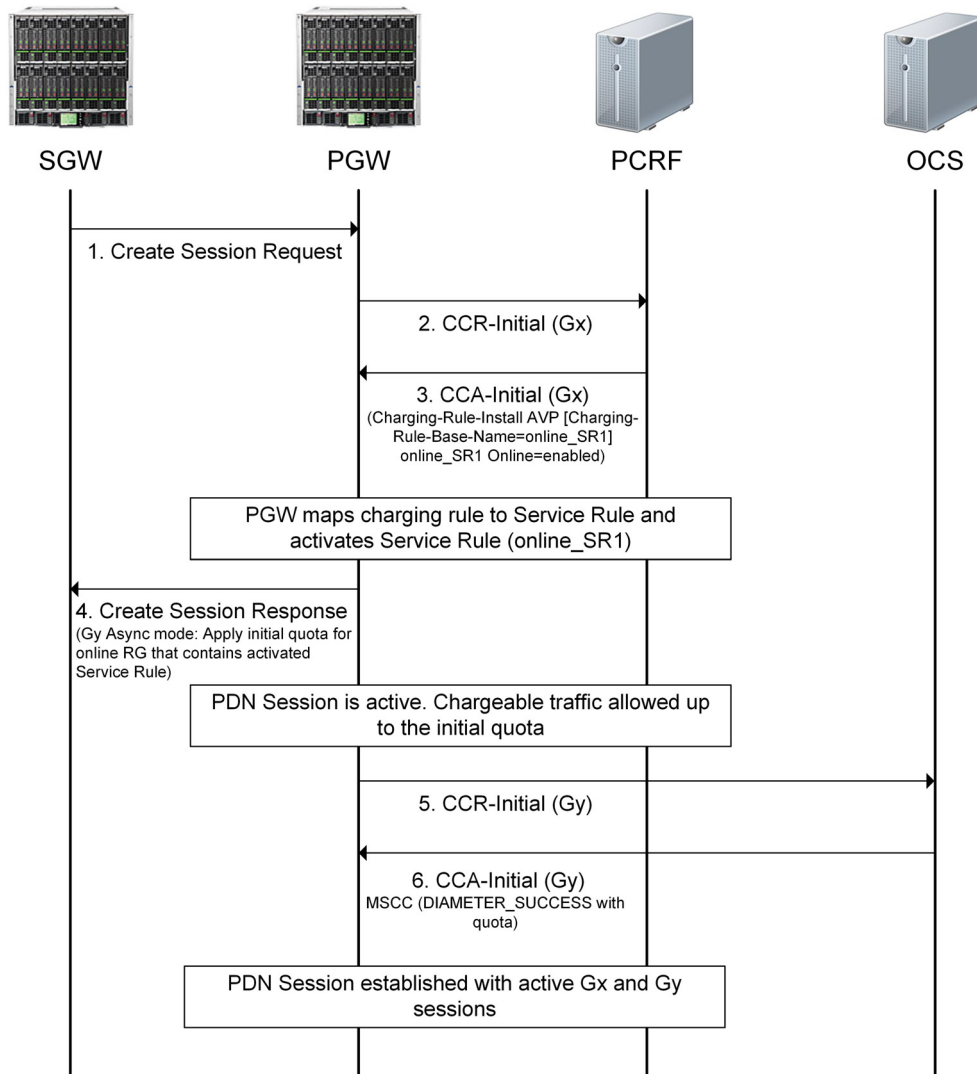
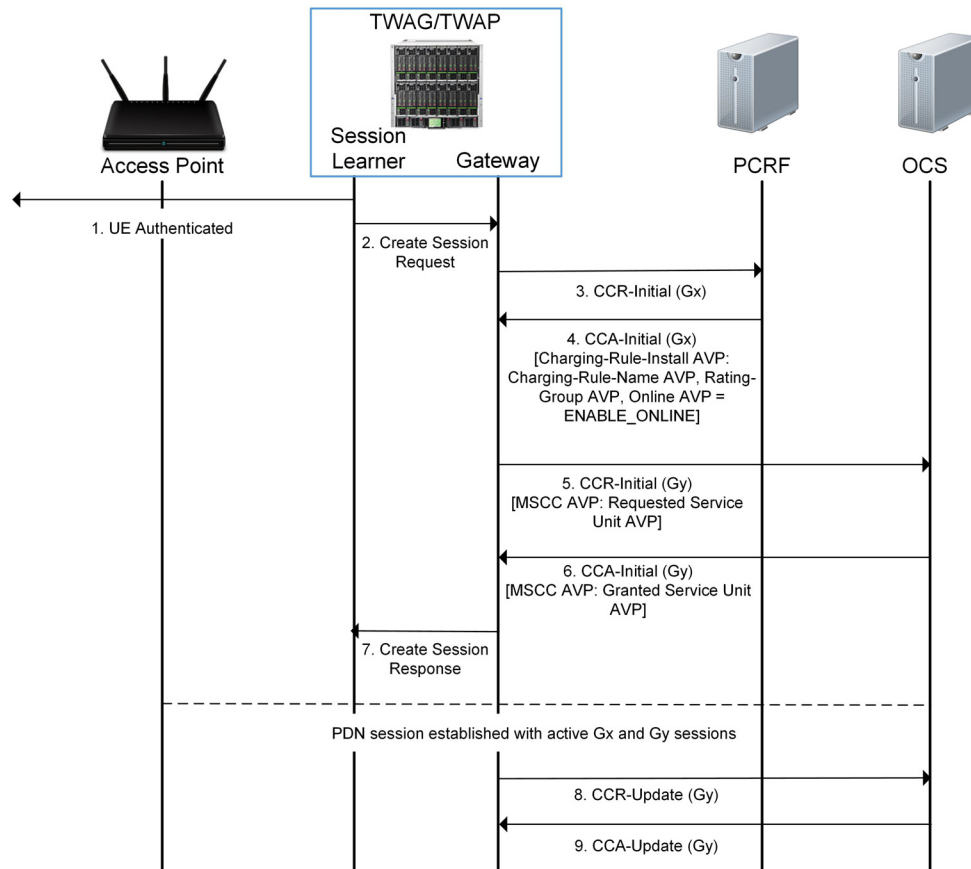


Figure 1-4 shows the message flow for dynamically activating a Gy online-charging session based on predefined PCC rules received from the PCRF via the Gx interface on the TWAG/TWAP.

Figure 1-4 Gx-based Dynamic Gy Session Set Up on the TWAG/TWAP



- 1 The TWAG/TWAP and AAA server exchange messages to authenticate the UE.
- 2 Once UE is authenticated, the session learner sends a Create Session Request message to the gateway for a new PDN session.
- 3 The TWAG/TWAP initiates a Gx session by sending the CCR-I message to the PCRF, requesting PCC rules to establish the PDN session.
- 4 The PCRF responds with a CCA-I message, including the Charging-Rule-Install AVP with charging rules for offline and/or online traffic (Online AVP set to enabled). The TWAG/TWAP maps these charging rules to service rules that are associated with online/offline rating groups.
- 5 If there is an online rating group with at least one activated service rule, the TWAG/TWAP initiates a Gy session by sending the CCR-I message to the OCS, with the Requested-Service-Unit AVP within the MSCC AVP.
- 6 The OCS responds with a CCA-I message containing the Granted-Service-Unit AVP within the MSCC AVP and with result code DIAMETER_SUCCESS.

- 7 The gateway sends a Create Session Response message to the session learner. The gateway applies the received quota and the PDN session is established with both Gx and Gy sessions.
- 8 If the quota is exhausted, the subscriber can request additional quota from the OCS by sending a CCR-U message containing the Requested-Service-Unit AVP within the MSCC AVP.
- 9 The OCS responds with a CCA-U message containing the Granted-Service-Unit AVP within the MSCC AVP. If the subscriber has reached the credit limit, the OCS responds with result code CREDIT_LIMIT_REACHED, which places the online rating group in redirect mode. If all online rating groups are in redirect mode, the Gy session will be terminated.

Configuring the Gy Online Charging Interface

This section describes the steps for configuring an MCC gateway to communicate with an OCS via the Gy interface.

Note: When configuring the Gy interface on the GTP Proxy or TWAG/TWAP, use the PGW object.

Use the following sequence to configure the system to send charging records to the OCS via the Gy interface.

- 1 Configure the local diameter node.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name>
```

where **<node-name>** specifies the name of the Diameter service profile on the gateway.

Note: The maximum number of supported Diameter server groups is 128 and the maximum number of supported diameter peers is 32 peers (32 Geo-active and 32 Geo-standby).

The following table describes the configurable parameters.

Parameter	Description
<code>cea-timeout</code>	Specifies the wait time (in seconds) for receiving a Capability-Exchange-Answer (CEA).
<code>cer-origin-state-id-mode</code>	Specifies whether to increase the value of the Origin-State-Id AVP, sent in CER messages, for every new Diameter connection. ■ static – (default) The MCC updates the value of the Origin-State-Id AVP sent in CER messages with the static value, configured using the <code>cer-origin-state-id-static-value</code> parameter. ■ timestamp – For every new Diameter connection, the MCC updates the value of the Origin-State-Id AVP sent in CER messages with the current timestamp. The Diameter peer that receives the message can infer that sessions associated with a lower Origin-State-ID value are no longer active.
<code>cer-origin-state-id-static-value</code>	Specifies the static Origin-State-Id value sent in CER messages. The default value is 25. Note: This parameter only applies when <code>cer-origin-state-id-mode</code> is set to <i>static</i>.

Parameter	Description
connection-attempt-count	Specifies the maximum number of connection retry attempts from the PGW to the OCS.
connection-attempt-timeout	Specifies the Tc timer, in seconds, as described in RFC 6733. Specifies how frequently to send transport connection attempts to a peer with whom no active transport connection exists.
dscp-value	Specifies the Differentiated Services Code Point (DSCP) value with which to mark all IP packets on this Diameter server group. For more information, see Configuring DSCP Values for Control Plane Traffic (page 1-53) .
dwr-retry-count	Specifies the maximum number of Device-Watchdog-Request (DWR) attempts to the diameter peer before declaring the peer down.
dwr-retry-timer	Specifies the number of seconds the diameter node waits for a Device-Watchdog-Answer (DWA) after retransmitting a DWR once the first dwr-timer expires. <i>Note: The dwr-retry-timer is typically set at a shorter interval than the dwr-timer.</i>
dwr-timer	Specifies the number of seconds the diameter node waits for a DWA after sending a DWR to the diameter peer. <i>Note: When the dwr-timer expires, the diameter node retransmits the DWR to the diameter peer and the dwr-retry-timer is started.</i>
geo-redundant-group	Specifies the Geographic Redundancy Group name. The Geographic Redundancy Group must already exist. For information on configuring a Geographic Redundancy Group, see Configuring Geo-Redundancy Health Checks on the Gy Interface (page 1-49) .
loopback-ip	Specifies the signaling port IP address.
network-context	Specifies the network context, selected from the VPN Group list. Restricts the inbound diameter messages by accepting only inbound messages on the IP interfaces in the interface group. The network context must already exist.
node-role	Specifies the local node role: ■ client ■ server
origin-host	Specifies the local host name in Fully Qualified Domain Name (FQDN). The value is used in the Origin-Host AVP in the request message. <i>Note: You can configure multiple Diameter server groups with the same origin host.</i> <i>Note: The origin-host configured at the peer level overrides the origin-host configured at the Diameter server-group level.</i>

Parameter	Description
<code>origin-realm</code>	Specifies the local realm name. The value is used in the Origin-Realm AVP in the request message. <i>Note: You can configure multiple Diameter server groups with the same origin realm.</i> <i>Note: The origin-realm configured at the peer level overrides the origin-realm configured at the Diameter server-group level.</i>
<code>port</code>	Specifies the TCP listening port number used when <code>node-role</code> is set to <code>server</code>. The default is 3868. <i>Note: This is the TCP listening port, not the Gy source port used when <code>node-role</code> is set to <code>client</code>. The Gy source port is dynamically assigned and is not configurable.</i>
<code>product-name</code>	Specifies the diameter product name.
<code>send-destination-host-avp</code>	Specifies whether the PGW sends the statically configured Destination-Host and Destination-Realm on the Gz/Rf interface in the initial Accounting-Request (ACR) message. <i>Note: Applies to the Gz/Rf interface only.</i>
<code>transport</code>	Specifies the transport type. ■ <code>tcp</code> (default) ■ <code>sctp</code> – To use Stream Control Transmission Protocol (SCTP) as a transport protocol on the Gy interface, you must configure an SCTP profile and associate the profile with the local Diameter node. See Configuring an SCTP Profile (page 1-57).
<code>vendor-id</code>	Specifies the diameter vendor ID.
<code>sctp-profile</code>	Specifies the name of the SCTP profile. The SCTP profile must already exist. See Configuring an SCTP Profile (page 1-57). <i>Note: The <code>sctp-profile</code> parameter is only visible when the local Diameter node transport type is set to SCTP.</i>
<code>secondary-loopback-ip</code>	Specifies a secondary loopback IP address for the local Diameter node to support SCTP multi-homing. See Configuring an SCTP Profile (page 1-57) for more information. <i>Note: The <code>secondary-loopback-ip</code> parameter is only visible when the local Diameter node transport type is set to SCTP.</i>

2 Configure the list of peers with which the local diameter node communicates.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name>
```

The following table describes the configurable parameters.

Parameter	Description
node-admin	Specifies the name of the Diameter service profile on the gateway.
peer	Specifies the peer host name. Each peer is configured with a unique peer host name. The peer host name is used by the MCC for load balancing between peers and for statistics.
admin-state	Specifies the administrative state of the diameter peer. When enabled, the system attempts to establish peer connection. ■ enabled (default) ■ disabled
backup-peer-name	Identifies the secondary peer host name used during a failover from the primary to the secondary peer. <i>Note: Only primary peers can be configured with a backup peer. If failover to a secondary peer is supported, this parameter must be configured with the host name of a secondary peer.</i>
dest-ip-address	Specifies the destination IP address of the peer (can be an IPv4 or IPv6 address).
dest-port	Specifies the signaling port number.
dest-realm	Specifies the destination realm name. Realm is used in the Destination-Realm AVP in Diameter requests.
origin-host	Specifies the local host name in Fully Qualified Domain Name (FQDN). The value is used in the Origin-Host AVP in the request message. See Configuring Multiple Pairs of Primary and Secondary Peers (page 1-94) for more information. <i>Note: The origin-host configured at the peer level overrides the origin-host configured at the Diameter server-group level.</i>
origin-realm	Specifies the local realm name. The value is used in the Origin-Realm AVP in the request message. See Configuring Multiple Pairs of Primary and Secondary Peers (page 1-94) for more information. <i>Note: The origin-realm configured at the peer level overrides the origin-realm configured at the Diameter server-group level.</i>
override-dest-hostname	Specifies the host name used to override the configured peer host name in the Destination-Host AVP within CCR-I messages. <i>Note: For more information, see Overriding the Destination-Host and Destination-Realm Static Configuration (page 1-39).</i>
peer-identifier	Specifies the external identifier for this peer.

Parameter	Description
role	Specifies the type of peer. When multiple primary, active peers are configured, the system uses a round-robin approach to select peers. ■ Primary – (default) Active peers used for diameter sessions. ■ Secondary – Used for diameter sessions, only when the primary peers are down. <i>Note: If failover to a secondary peer is supported, you must enter the peer host name of the secondary server in the <code>backup-peer-name</code> parameter.</i>
secondary-loopback-ip	Specifies a secondary loopback IP address for the Diameter peer to support SCTP multi-homing. See Configuring an SCTP Profile (page 1-57) for more information. <i>Note: The <code>secondary-loopback-ip</code> parameter is only visible when the local Diameter node transport type is set to SCTP.</i>
weight	Specifies the weight allocated to this Diameter peer. The Diameter peer with the highest weight (10) receives the most distribution of new session requests. The default is 1 (lowest weight). When set to 0, the MCC places the peer in “maintenance mode” and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-U/CCR-T) for existing sessions. <i>Applies to Gy/Gx/Gz/STa/Swm/S6b interfaces.</i> See Weighted Round-Robin Distribution to Multiple Diameter Peers (page 1-62) for more information.

- 3 Configure the data record templates to filter optional Diameter AVPs in the CCR and CCA messages.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|ggsn] ocs diameter [start|stop|interim|event|cca] <template-name>
<attribute-name> presentFlag [yes|no|conditional]
```

where

Parameter	Description
[pgw ggsn]	Specifies the gateway type for the online charging interface.
[start stop interim event cca]	Specifies the type of data record template.
<template-name>	Specifies the name of the data record template.
<attribute-name>	Specifies the name of the AVP field you are modifying.

Parameter	Description
presentFlag	CCR Start/Stop/Interim/Event templates: ■ yes – Indicates the field is sent. ■ no – Indicates the field is not sent. ■ conditional – Indicates the field is sent if it is available. CCA template (Initial/Update/Final/Event templates): ■ yes – (default) The field is processed when received in the CCA response. ■ no – The field is ignored when received in the CCA response.

To view a list of the configurable AVPs in the CCR and CCA data record templates, use the following CLI command:

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|ggsn] ocs diameter [start|stop|interim|event|cca] <template-name>
?
```

- 4 Configure the triggers used to send CCR-U messages.

```
an3000-mcm-slot7cpu1(config)# service-construct
interface-trigger-template event-trigger-template <template-name>
```

The following table describes the configurable parameters.

Parameter	Description
cgi-sai-change-close-trigger	Specifies whether a change in the Cell Global Identification-Service Area Identifier (CGI-SAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of CGI-SAI change events to trigger a CDR close.
change-in-up-to-ue-close-trigger	Applies to offline interfaces only. Specifies whether a change in the User Plane (UP) to User Equipment (UE) (for example, a switch between S11-U and S1-U) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of UP to UE Change events to trigger a CDR close. <i>Note: Applies to the SGW only.</i>
ecgi-change-close-trigger	Specifies whether a change in the E-UTRAN Cell Global Identifier (ECGI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of ECGI change events to trigger a CDR close.

Parameter	Description
qos-change-close-trigger	Specifies whether a QoS Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of QoS Change events to trigger a CDR close.
rat-type-change-close-trigger	Specifies whether a RAT Type Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of RAT Type Change events to trigger a CDR close.
routing-area-change-close-trigger	Specifies whether a change in the Routing Area Identity (RAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of RAI change events to trigger a CDR close.
serving-node-change-close-trigger	Specifies whether a Serving Node Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of Serving Node Change events to trigger a CDR close. <i>Note: Applies to the GGSN/PGW when the context is transferred to a new SGW/SGSN or to the SGW when the Mobility Management Entity (MME) is changed.</i>
serving-node-plmn-change-close-trigger	Specifies whether a Serving Node PLMN Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of Serving Node PLMN Change events to trigger a CDR close. <i>Note: Applies to PGW and GGSN only.</i>
time-limit-close-trigger	Applies to offline interfaces only. If interim updates are enabled on the interface, specifies the number of events to trigger a CDR close. Enter a value between 1-50.
tracking-area-change-close-trigger	Specifies whether a change in the Tracking Area Identity (TAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of TAI change events to trigger a CDR close.
ue-time-zone-change-close-trigger	Specifies whether a UE Time Zone Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of UE Time Zone Change events to trigger a CDR close.
user-csg-change-close-trigger	Specifies whether a CSG change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of CSG change events to trigger a CDR close.

Parameter	Description
user-location-change-close-trigger	Specifies whether a User Location Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of User Location Change events to trigger a CDR close.
volume-limit-close-trigger	Applies to offline interfaces only. If Data Volume Limit is enabled on the interface, specifies the number of events to trigger a CDR close. Enter a value between 1-50.

5 Configure the Gy online charging interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name>
```

where **<interface-name>** specifies the online charging interface name.

The following table describes the configurable parameters.

Parameter	Description
admin-state	Specifies the administrative state of the online charging interface. ■ enabled (default) ■ disabled
allow-dest-host-realm-change [cca-i cca-u rar]	Specifies whether the Destination-Host/Destination-Realm AVPs are updated in CCR-U/CCR-T messages based on information received in the CCA-I/CCA-U/RAR messages. The default is enabled for the CCA-I/CCA-U messages and disabled for the RAR message. ■ enabled – If the received Origin Host/Origin Realm in the CCA-I/CCA-U/RAR is different from the current destination host and realm, the MCC uses the Origin-Host/Origin-Realm AVPs received in the CCA-I/CCA-U/RAR message to fill the Destination-Host/Destination-Realm AVPs in subsequent CCR-U/CCR-T messages. ■ disabled – The MCC gateway continues to use the existing values to fill the Destination-Host/Destination-Realm AVPs in subsequent CCR-U/CCR-T messages and does not update the Destination-Host/Destination-Realm AVPs with the values received in the CCA-I/CCA-U/RAR messages. <i>Note:</i> The MCC gateway always sends the RAA message with the Destination Host and Realm values of the Origin Host and Realm received in the RAR message. <i>Note:</i> For the CCA-I/CCA-U, the MCC will only update the Destination-Host and Destination-Realm in CCR-U/CCR-T messages if the result code received in the CCA-I/CCA-U is DIAMETER_SUCCESS (2001).

Parameter	Description
apn-reporting	<i>Applies to the PGW only.</i> Specifies the Access Point Names (APNs) to be reported over the Gy interface. ■ original – Initial APN. ■ mapped – Final / current APN. ■ prior-mapped – APN mapped prior to the mapped (final/current) APN. This setting allows APN Reporting to continue to be based on the previously-mapped APN (even following a remap to a final APN). The apn-reporting value configured at the interface level overrides the apn-reporting value configured for the Control-Profile and is used to fill the Called-Station-ID AVP and User-Name AVP sent in messages to external devices. If the apn-reporting value is not configured at the interface level, the Called-Station-ID AVP and User-Name AVP will be filled based on the apn-reporting value configured for the Control Profile. For more information, see Configuring APN Reporting at the Interface Level (page 1-61). Note: The interface-level apn-reporting parameter does not support the <i>custom</i> option.
attributes	Specifies default values for certain parameters that were not provided by the OCS. ■ validity-time – Specifies the validity time, in seconds, used when the OCS does not send the Validity-Time AVP within the MSCC AVP. ■ time-threshold – Specifies the time-threshold used when the OCS does not send the Time-Quota-Threshold AVP within the MSCC AVP and this value is not configured under the rating group. ■ volume-threshold – Specifies the volume-threshold used when the OCS does not send the Volume-Quota-Threshold AVP within the MSCC AVP and this value is not configured under the rating group. Note: The <i>time-threshold</i> and <i>volume-threshold</i> configured for the rating group take precedence over the <i>time-threshold</i> and <i>volume-threshold</i> configured for the online-charging interface. Note: If the GSU is received with result code 4012, the MCC does not apply the <i>time-threshold</i> and <i>volume-threshold</i> configured for the rating group or online-charging interface.
avp-customization mscc-serviceid num-instances	Specifies whether multiple Service-Identifiers can be included within the Multiple-Services-Credit-Control AVP on the Gy interface. ■ multiple ■ single

Parameter	Description
avp-customization user-equipment-info- imeisv format-tbcd	Specifies whether to encode the IMEISV, sent within the User-Equipment-Info AVP in the CCR message on the Gy interface, in Telephony Binary Coded Decimal (TBCD) format. ■ enabled – The IMEISV sent within the User-Equipment-Info AVP in the CCR message on the Gy interface is encoded in TBCD format. ■ disabled – (default) The IMEISV sent within the User-Equipment-Info AVP in the CCR message on the Gy interface is encoded in Octet String format. <i>Note: Changes to this parameter take effect immediately.</i>
avp-customization user-equipment-info- imeisv allow-zero-value	Specifies whether to enable the MCC to send the IMEISV with a value of 0 within the User-Equipment-Info AVP. <i>Applies to the SGW and PGW only.</i> ■ enabled – If the IMEI/IMEISV received in the Create Session Request has a value of 0, the MCC sends the IMEISV with a value of 0 within the User-Equipment-Info AVP on the Gy interface. ■ disabled – (default) If the IMEI/IMEISV received in the Create Session Request has a value of 0, the MCC does not send the IMEISV within the User-Equipment-Info AVP on the Gy interface.
avp-customization 3gpp-imeisv allow-zero-value	Specifies whether to enable the MCC to send the IMEISV with a value of 0 within the 3GPP-IMEISV AVP. <i>Applies to the SGW and PGW only.</i> ■ enabled – If the IMEI/IMEISV received in the Create Session Request has a value of 0, the MCC sends the IMEISV with a value of 0 within the 3GPP-IMEISV AVP on the Gy interface. ■ disabled – (default) If the IMEI/IMEISV received in the Create Session Request has a value of 0, the MCC does not send the IMEISV within the 3GPP-IMEISV AVP on the Gy interface.
cca-error-send- terminate	Controls whether the GGSN/PGW sends the OCS a CCR-T message upon receipt of a CCA-U message with a permanent failure (result code 5xxx). ■ true (default) ■ false
charge-rating-group- initial-quota	Controls whether to charge for the traffic passed using the initial quota grant per rating group. ■ true ■ false (default) <i>Note: All traffic, including the initial quota traffic, is included in offline CDRs regardless of whether charging is enabled for the initial quota.</i> <i>Note: For more information, see Setting up the PDN Session and Gy Session Asynchronously (page 1-43).</i>

Parameter	Description
<code>command-level-result-code-list</code>	Specifies the name of the command-level result code list. Note: The command-level result code list must already exist. For more information, see Configuring the Diameter Result Code List (page 1-78).
<code>defer-gy-session-until-first-sdf-packet</code>	Specifies whether to restrict the PGW from establishing an online-charging session until after it receives the first service data flow (SDF) packet. In this case, the PGW waits for the first SDF packet to arrive before sending a credit authorization request (CCR-I) to the OCS. Configure separate actions for home and roaming subscribers. Applies when all online rating groups have the <code>credit-authorization-event</code> parameter set to <code>firstSdfPkt</code>. ■ home-subscriber – For home subscribers, the PGW/GGSN establishes the online-charging session and requests quota with the OCS (MSCC AVP in the CCR-I) upon receiving the first SDF packet. ■ roaming-subscriber – For roaming subscribers, the PGW/GGSN establishes the online-charging session and requests quota with the OCS (MSCC AVP in the CCR-I) upon receiving the first SDF packet. ■ both – For both home and roaming subscriber sessions, the PGW/GGSN establishes the online-charging session and requests quota with the OCS (MSCC AVP in the CCR-I) upon receiving the first SDF packet. Note: By default, the <code>defer-gy-session-until-first-sdf-packet</code> parameter is not configured. Note: This parameter is only visible when <code>interface-compliance</code> is set to <code>3GPPVersion2</code>. Note: For more information, see Setting up the Gy Session After Receiving the First SDF Packet (page 1-45).
<code>delete-session-on-fui-terminate</code>	Specifies whether to terminate the session when the FUI AVP is set to TERMINATE, irrespective of whether a grant is available for other online rating groups. By default, this parameter is set to false, indicating the session will not be terminated. ■ true ■ false (default) Note: The configured action based on the result code always takes precedence over the FUI AVP received at the MSCC level from the OCS. See Configuring the Diameter Result Code List (page 1-78) for more information on configuring a specific action based on the result code received in the CCA.

Parameter	Description
first-packet-credit-auth-rating-groups	Specifies whether to restrict the PGW from establishing an online-charging session and requesting quota until after it receives the first SDF packet for rating groups configured with the <code>credit-authorization-event</code> parameter set to <code>bearerSetup</code>. ■ firstsdf – (default) The credit authorization for rating groups with the <code>credit-authorization-event</code> set to <code>firstSdfPkt</code> will be deferred until the PGW receives the first SDF packet. ■ all – For all active online rating groups, the credit authorization will be deferred until the PGW receives the first SDF packet. When the PGW receives the first SDF packet on any active online rating group, the quota request (MSCC AVP) is included in the CCR-U for all rating groups with <code>credit-authorization-event</code> set to <code>bearerSetup</code> and for the rating group (<code>bearerSetup/firstSdfPkt</code>) that received the first packet. For rating-groups with the <code>credit-authorization-event</code> set to <code>firstSdfPkt</code>, the credit authorization is requested when the PGW receives the first SDF packet for the rating group. <i>Note: This parameter is only visible when <code>interface-compliance</code> is set to <code>3GPPVersion2</code>.</i> <i>Note: For more information, see Setting up the Gy Session After Receiving the First SDF Packet (page 1-45).</i>
gy-action-per-result-code	Specifies whether to enable or disable the command-level or MSCC-level result code policy lists. ■ enabled – The result code returned in the CCA message is compared to the configured result code list, and if there is a match, the PGW/GGSN applies the configured action. ■ disabled – (default) The result code list functionality is not used. <i>Note: For more information, see Configuring the Diameter Result Code List (page 1-78).</i>
gy-async	Controls whether to enable or disable the Gy Async feature. ■ enabled – The PGW completes the PDN session setup by sending the Create Session Response and setting up the online charging session with the OCS asynchronously (not as part of the PDN session setup). ■ disabled – (default) The PGW completes the PDN session setup after establishing the online charging session with the OCS. <i>Note: This parameter is only configurable when <code>interface-compliance</code> is set to <code>3GPPVersion2</code>. By default, <code>gy-async</code> is disabled for all other interfaces.</i> <i>Note: For more information on Gy Async, see Setting up the PDN Session and Gy Session Asynchronously (page 1-43).</i>

Parameter	Description
<code>gy-close-time-for-online-disable-on-lte-wifi-handover</code>	Specifies the Gy close timer, in seconds. The Gy close timer supports dynamic Gy session termination when the PCRF disables all online service rules during interface-access handovers (LTE <=> Wi-Fi handovers). When the PCRF disables all online service rules during inter-access handovers, the PGW sends a CCR-U to the OCS to report the final usage for online rating groups and starts the Gy close timer. If the Gy close timer expires before the PCRF installs new online service rules, the PGW terminates the Gy session. However, if the PCRF installs new online service rules before the Gy close timer expires, the PGW resumes the Gy session by sending a CCR-U to the OCS requesting a new grant for the required rating groups.
<code>interface-compliance</code>	Specifies the interface compliance.
<code>mscc-level-result-code-list</code>	Specifies the name of the MSCC-level result code list. Note: The MSCC-level result code list must already exist. For more information, see Configuring the Diameter Result Code List (page 1-78).
<code>multi-crbn-allowed-in-ccr</code>	Controls whether to include multiple CRBN AVPs (within the PS-Information AVP) in CCR-I/CCR-U messages towards the OCS. ■ true - The PGW/GGSN includes multiple CRBN AVPs (within the PS-Information AVP) in CCR-I/CCR-U messages towards the OCS. The value of the CRBN AVPs correspond to both the locally configured charging-instance from the data-profile (<code>workflow -> data-profile -> charging</code>) and all locally configured charging-instances from the PCRF-installed crbn-profiles (<code>services -> crbn-profile -> charging-instance</code>). Note: The MCC gateway will only include multiple CRBN AVPs (within the PS-Information AVP) on the Gy interface if the charging-instances in the data-profile and crbn-profiles are different from each other. ■ false - (default) The PGW/GGSN includes a single CRBN AVP (within the PS-Information AVP) in CCR-I/CCR-U messages towards the OCS. The value of the CRBN AVP corresponds to the locally configured charging-instance from the data-profile on the MCC gateway (<code>workflow -> data-profile -> charging</code>). Note: This parameter is only supported when the online-charging interface-compliance is set to <code>3gpp</code> or <code>3GPPVersion2</code> and the <code>crbn-mode</code> under the <code>pcrf-charging</code> interface is set to <code>crbn-profile</code>.

Parameter	Description
<code>ocs-down-default-gy-call-setup</code>	Specifies whether to enable or disable the Default Gy Quota feature. ■ enabled – When no OCS is available or the OCS returns a temporary failure (result code 3xxx), the GGSN/PGW sets up an online charging session and applies the default quota to the session. ■ disabled – (default) The GGSN/PGW does not set up an online charging session when no OCS is available or the OCS returns a temporary failure (result code 3xxx). <i>Note: For more information on the Default Gy Quota, see Default Gy Quota (page 1-72).</i>
<code>ocs-down-default-gy-quota</code>	Specifies the maximum number of credits, in bytes, that the PGW allocates to a new or existing online charging session in the event of an OCS failure. The default is 10000000.
<code>ocs-down-default-time-quota</code>	Specifies the default time threshold, in seconds, allocated during an OCS failure when the rating group <code>measurement-method</code> set to <code>time</code> or <code>timeAndVolume</code>. The default is 90 seconds.
<code>ocs-down-gy-rejoin</code>	Specifies whether to enable or disable Gy Rejoin. ■ enabled – If the default quota used for an online charging session is consumed or the <code>ocs-down-rejoin-timer</code> expires, the PGW tries to rejoin the OCS. ■ disabled – (default) The PGW does not try to rejoin the OCS even if the OCS comes back up in the case where the default quota is used to set up a session. If the OCS is down when a CCR-U is attempted, the session is terminated. <i>Note: For more information on Gy Rejoin, see Gy Rejoin and Gy Replay (page 1-77).</i>
<code>ocs-down-gy-replay</code>	Specifies whether to enable or disable Gy Replay. ■ enabled – If all OCS servers go down after the Gy session is already established and the session using the default quota is terminated before the OCS comes back up, the GGSN/PGW caches the used service units (USU) based on the OCS granted service units (GSU) and replays the cached session information when the OCS becomes available. ■ disabled – (default) The session usage consumed based on the OCS grant is not cached and replayed to the OCS. <i>Note: For information on Gy Replay, see Gy Rejoin and Gy Replay (page 1-77).</i> <i>Note: Gy Replay and the Gy CCR-T file storage functionality cannot both be enabled. See Storing Gy CCR-T Records in a Local File (page 1-84) for more information on storing Gy CCR-T records.</i>

Parameter	Description
<code>ocs-down-max-num-default-quota</code>	Specifies the maximum number of times a session can be allocated a default quota while the OCS is down. If the user exceeds the maximum number of default quota allocations, the session is terminated. The default value of 0 allows for an unlimited number of default quota allocations.
<code>ocs-down-rejoin-time</code>	Specifies the number of seconds to wait before attempting to connect to the non-responsive OCS.
<code>ocs-down-session-terminate-timer</code>	Specifies the number of seconds to wait before terminating a session when the default Gy quota is exhausted. By default, this parameter is set to 0 (disabled). See Terminating a Session Upon Depletion of Default Gy Quota (page 1-74) for more information. <i>Note: To use this feature, Gy Rejoin (<code>ocs-down-gy-rejoin</code>) must be disabled and the Default Gy Quota (<code>ocs-down-default-gy-call-setup</code>) must be enabled for the online-charging interface.</i>
<code>offline-cdr-for-gy-event</code>	Specifies whether to enable or disable offline charging service data container creation based on Gy reporting events. ■ enabled ■ disabled (default) <i>Note: For more information on synchronizing Gz/Rf and Gy charging data, see the Affirmed Networks Charging Data Records Specification.</i>
<code>override-destination-realm</code>	Specifies the realm name used to override the configured peer realm in the Destination-Realm AVP within CCR-I messages. The <code>override-destination-realm</code> only applies to the Destination-Realm AVP in CCR-I messages. <i>Note: For more information, see Overriding the Destination-Host and Destination-Realm Static Configuration (page 1-39).</i>
<code>pcrf-charging-info-avp-usage</code>	Specifies how the destination host/realm extracted from the Charging-Information AVP (received from the PCRF) is used during Gy session setup. ■ overrideDestHostRealm – The extracted host/realm is used for the Destination-Host AVP and the Destination-Realm AVP in the CCR-I sent to the OCS. The MCC supports two formats for extracting realm information: Consumer Realm Format and Reseller Realm Format. See Formats for Extracting Realm Information (page 1-41). ■ peerSelection – (default) The extracted destination host is used for selecting the OCS peer.

Parameter	Description
pcrf-num-labels-destination-realm	Specifies the number of labels to extract from the Charging-Information AVP for the destination-realm. The Charging-Information AVP contains a single FQDN value (with one or more labels separated with dots) for each primary and secondary OCS but does not indicate which part of the FQDN belongs to the parent realm. The default is 0, indicating that the destination-realm is not extracted from the Charging-Information AVP. <i>Note: This parameter is only applicable when pcrf-charging-info-avp-usage is set to <code>overrideDestHostRealm</code>.</i> <i>Note: This parameter does not apply when the Consumer Realm Format is used to extract realm information from the Gx Charging-Information AVP during Gy session setup. See Formats for Extracting Realm Information (page 1-41).</i>
rating-group-initial-quota	Specifies the maximum number of credits, in bytes, that the PGW can allow per rating group before establishing the Gy session and receiving the granted quota from the OCS. If the initial quota is consumed before the PGW receives the granted quota from the OCS, traffic for this session is blocked. The default is 0 (disabled), indicating that traffic is blocked until the granted quota is received from the OCS. <i>Note: For more information, see Setting up the PDN Session and Gy Session Asynchronously (page 1-43).</i>
reauth-response-success-result-code	Specifies which Diameter result code the PGW sends in RAA messages to the OCS to indicate the Re-Auth-Request (RAR) was successfully processed. ■ success_2001 – (default) The PGW sends the DIAMETER_SUCCESS (2001) result code in the RAA message to the OCS. ■ limited_success_2002 – The PGW sends the DIAMETER_LIMITED_SUCCESS (2002) result code in the RAA message to the OCS.
report-all-usage-on-rejoin	Specifies whether to enable or disable the reporting of all interim usage (including all default quota allocations) in a CCR-U message when the PGW rejoins the OCS in the event of an OCS failure. ■ enabled ■ disabled (default)

Parameter	Description
restart-on-session-not-found	Specifies whether the PGW initiates a new Gy session if the OCS returns a 5002 result code when the PGW attempts to rejoin the OCS. If enabled, the PGW sends the CCR-I to initiate a new Gy session followed by the CCR-U to report the usage. ■ enabled ■ disabled (default) <i>Note: The configuration for this parameter takes precedence over the <code>cca-error-send-terminate</code> configuration.</i>
retransmit-count	Specifies the maximum number of CCR retransmit attempts to the OCS.
retransmit-timeout	Specifies the time interval, in milliseconds, before the system retransmits the CCR to the OCS.
send-called-station-id-at-command-level	Specifies whether to include the Called-Station-Id AVP at the command level in 3GPP CCR messages. ■ true – Include the Called-Station-Id AVP at the command level in 3GPP CCR messages. ■ false – (default) Include the Called-Station-Id AVP as a subfield under the Service-Information -> PS-Information AVPs in 3GPP CCR messages. <i>Note: This feature is only applicable when <code>interface-compliance</code> is set to <code>3GPP</code>, <code>3GPPVersion2</code>, <code>3GPPCustom1</code> or <code>RFC4006</code>.</i>
send-final-in-update	Controls whether the GGSN/PGW sends a CCR-U message to the OCS with a Reporting-Reason AVP (at the MSCC level) value of FINAL in response to a CCA message with result code DIAMETER_CREDIT_LIMIT_REACHED (4012). ■ true (default) ■ false
send-RSU-in-update	Controls whether the GGSN/PGW always includes and populates the Requested-Service-Unit (RSU) AVP (at the MSCC level) in the CCR-U message to the OCS. ■ true ■ false (default)
send-zero-usu-on-no-grant	Specifies whether to report Used-Service-Units (USU) as zero for the rating group in the CCR-U/CCR-T to the OCS when the PGW receives a CCA with no grant (no GSU AVP) within the MSCC AVP. ■ true ■ false (default)
service-context-id	Specifies the Service-Context-Id content.

Parameter	Description
subscriptionid-imsi-format-non-3gpp-access	Specifies the IMSI format sent in the Subscription-Id AVP for non-3GPP access. <i>Applicable only to Gx and Gy interfaces.</i> ■ imsiformat (default) ■ imsinaiprefix0format
suppress-final-on-mscc-error	Controls whether the PGW/GGSN suppresses the MSCC AVP in the CCR-T message with a FINAL Reporting-Reason AVP (at the MSCC level) when the PGW/GGSN receives a CCA-U with result code DIAMETER_CREDIT_LIMIT_REACHED (4012) or DIAMETER_END_USER_SERVICE_DENIED (4010) and no GSU AVP for a rating group. ■ enabled ■ disabled (default)
suppress-invalid-uli	Specifies whether to suppress the User Location Information (ULI) on the Gy interface if the ULI received from the MME/SGSN is invalid. ■ true ■ false (default) <i>Note: ULI changes from valid to invalid and vice versa are considered valid ULI changes and are reported for offline/PCRF/OCS event-triggers.</i>
terminate-gy-if-no-active-online-rule	Specifies whether a CCR-T or a CCR-U is sent when no active online rules exist. ■ true (default) – MCC sends a CCR-T. ■ false – MCC sends a CCR-U instead of a CCR-T.
user-name-type	Specifies the User-Name AVP format used in CCR messages on the Gy interface if not present in the Protocol Configuration Options (PCO). This setting overrides the user-name-type present at the APN level. However, if the Gy interface user-name-type is set to none, the APN level user-name-type is used. ■ apn ■ default-username ■ imsi ■ imsi-nai-no-prefix ■ imsi-nai-prefix-0 ■ imsi-plus-apn ■ imsi-plus-apnnionly ■ msisdn ■ msisdn-plus-apn ■ msisdn-plus-apnnionly ■ none

- 6 Associate the diameter server group and CCR/CCA data record templates with the Gy online charging interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name> [pgw|ggsn] diameter server-group
<service-profile> start-template <template-name> stop-template
<template-name> interim-template <template-name> event-template
<template-name> cca-initial <template-name> cca-update <template-name>
cca-final <template-name> cca-event <template-name>
```

where

Parameter	Description
online-charging	Specifies the online charging interface profile name.
server-group	Specifies the Service Profile for the selected streaming protocol.
start-template	Specifies the CCR Start record template.
stop-template	Specifies the CCR Stop record template.
interim-template	Specifies the CCR Interim record template.
event-template	Specifies the CCR Event record template.
cca-initial	Specifies the CCA Initial template.
cca-update	Specifies the CCA Update template.
cca-final	Specifies the CCA Final template.
cca-event	Specifies the CCA Event template.

- 7 Associate the event trigger template with the Gy online charging interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name> [pgw|ggsn] event-trigger-template
<template-name>
```

where

Parameter	Description
online-charging	Specifies the online charging interface profile name.
event-trigger-template	Specifies the name of the event trigger template to use.

- 8 Configure the rating group used for quota reporting.

```
an3000-mcm-slot7cpu1(config)# services charging rating-group
<rating-group>
```

where **<rating-group>** specifies the rating group name. The rating group defines the basis for charging service data flows.

The following table describes the configurable parameters.

Parameter	Description
admin-state	Specifies the administration state of this rating group. ■ enabled ■ disabled
advice-of-charge	Specifies the Advice of Charge name. Defines the action to take after online quota is exhausted. The Advice of Charge must already exist. See the <i>Affirmed Networks Gateway Administration Guide</i> for more information.
charging-method	Specifies the charging method. ■ local ■ offline ■ online ■ nocharge – Subscriber traffic that uses this rating group is either not charged or CDRs are not generated for this rating group. This feature is useful when an Operator wants to exclude certain flows such as DNS requests or portal requests used to charge subscribers. ■ metering ■ nometering
credit-authorization-event	Specifies the credit authorization beginning stage: ■ bearerSetup – The PGW establishes the online-charging session and requests quota with the OCS upon receiving the first control signaling packet. In this case, the quota request (MSCC AVP) is included in the CCR-I sent to the OCS. ■ firstSdfPkt – The PGW establishes the online-charging session during the PDN session setup but does not include the quota request (MSCC AVP) in the CCR-I sent to the OCS. In this case, the quota request (MSCC AVP) is included in the CCR-U once the PDP session is set up and the PGW receives the first service data flow packet.
credit-pool-id	Specifies the credit pool used to grant credit blocks. Enter 0 if no credit pool is used.
enable-cdr	Specifies whether to enable CDRs for the online rating group: ■ enabled ■ disabled <i>Note: Applies when charging-method is set to Online.</i>
home-subscriber-charging	Specifies whether to use the rating group for home subscriber charging: ■ enabled ■ disabled

Parameter	Description
measurement-method	Specifies the measurement method. ■ time ■ volume ■ timeAndVolume ■ event – Only applies when charging-method is set to offline or local.
mms-mt-ocs-size	Message size (in bytes) to be used for MMS-MT charging. <i>Note: Applies when service-type is set to mms-mt.</i>
multiplier	Specifies the multiplier applied to time, volume and event charging.
ocs-response-grace-period	Specifies the number of seconds to pass traffic while waiting for an OCS response. <i>Note: Applies when charging-method is set to Online.</i>
priority	Specifies the priority. This value resolves conflicts in the case of multiple service instances. Priority is hierarchical in that the priority of the higher-level object takes precedence over the lower-level object. Thus, the rating group priority takes precedence over the associated service rule priority.
quota-black-list-timer	Specifies the Quota Black List timer, in minutes. Specifies how frequently to send grant requests after termination or suspension for this rating group. For more information, see Reducing the Load on the Gy Interface (page 1-38) .
quota-consumption-time	Specifies the inactivity timeout for time-based accounting, in seconds. <i>Note: Applies when measurement-method is set to time or timeAndVolume.</i>
quota-hold-time	Specifies the hold time after inactivity, in seconds.
quota-id	Specifies the unique rating group ID.
reporting-level	Specifies the policy control reporting level. ■ ratingGroup ■ serviceRule
requested-unit-value	Specifies the requested unit value for this rating group.
roamer-subscriber-charging	Specifies whether to use the rating group for roaming subscriber charging: ■ enabled ■ disabled

Parameter	Description
service-activation	Specifies when to use or skip the service flow for packet classification: ■ always-on – Always use this service flow for classification. ■ external-activation – Use the activation state of the service flow to determine whether to use or skip the service flow for packet classification. This setting allows the gateway to dynamically activate/deactivate rating groups.
service-rule	Specifies the service rule(s) to link with the rating group. The service rule must already exist. See the <i>Affirmed Networks Gateway Administration Guide</i> for information on adding a service rule.
service-type	Specifies the service type: ■ data ■ mms-mo (inbound, mobile originated) ■ mms-mt (outbound, mobile termination)
tcp-retransmission	Inclusion or exclusion of TCP retransmissions in volume counts. ■ charge ■ noCharge <i>Note: Applies when measurement-method is set to volume or timeAndVolume.</i>
time-threshold	Specifies the time threshold, in seconds. Specify 0 if thresholding is not desired. <i>Note: Applies when measurement-method is set to time or timeAndVolume.</i>
usage-reporting-mode	Specifies whether the MCC reports the CC-Total-Octets, and CC-Input-Octets, CC-Output-Octets (within the Used-Service-Units AVP) either accumulatively or incrementally. ■ accumulative-mode – The accumulated CC-Total-Octets, CC-Input-Octets, and CC-Output-Octets is reported to the OCS in the Used-Service-Units AVP when the subscriber requests quota or when the PGW sends a CCR-U to report the final usage for the rating group. The CC-Total-Octets is reset to 0 only when the Gy session is terminated or the subscriber is cleared. ■ incremental-mode – (default) The incremental CC-Total-Octets, CC-Input-Octets, and CC-Output-Octets is reported to the OCS in the Used-Service-Units AVP each time the subscriber requests quota. The CC-Total-Octets is reset to 0 every time the PGW reports usage to the OCS. <i>Note: Changes to this parameter take effect immediately. If this parameter is changed mid-session, it applies to all new subscriber sessions.</i>

Parameter	Description
volume-measurement-count	Specifies the volume measurement count. ■ inputOctets ■ outputOctets ■ totalOctets <i>Note: Applies when <code>measurement-method</code> is set to <code>volume</code> or <code>timeAndVolume</code>.</i>
volume-measurement-layer	Specifies the volume measurement layer. ■ layer3 – Counts the entire IP datagram. ■ layer4 – Excludes the IP header. ■ layer7 – Excludes the IP header and the UDP/TCP/SCTP header. <i>Note: Applies when <code>measurement-method</code> is set to <code>volume</code> or <code>timeAndVolume</code>.</i>
volume-threshold	Specifies the volume threshold in bytes. Specify 0 if thresholding is not desired. <i>Note: Applies when <code>measurement-method</code> is set to <code>volume</code> or <code>timeAndVolume</code>.</i>

- 9 Configure the billing plan with the rating group. The billing plan defines the charging method to be applied to the rating group(s).

```
an3000-mcm-slot7cpu1(config)# services charging billing-plan
<billing-plan-name> rating-group <rating-group-name> fraud-charging
[true|false]
```

where

Parameter	Description
billing-plan	Specifies the billing plan name.
rating-group	Specifies the rating group name.
fraud-charging	Specifies whether to designate this rating group for charging of fraudulent activity within this billing plan. ■ true ■ false (default)

- 10 Configure the charging instance. The charging instance binds one or more billing plans.

```
an3000-mcm-slot7cpu1(config)# services charging charging-instance
<charging-instance-name> priority-evaluation-mode [flat|hierarchical]
state [enabled|disabled]
```

where

Parameter	Description
charging-instance	Specifies the name of the charging instance.
priority-evaluation-mode	Specifies the Priority Evaluation Mode when classifying traffic: ■ flat – Use service rule priority (ignore rating group priority). To make a match, examine all of the service rules on all of the rating groups for best priority. ■ hierarchical – (default) Use rating group priority and service rule priority hierarchically. To make a match, examine the rating group in priority order and within each of them, examining the service rules in priority order.
state	Specifies the administration state of this charging instance. ■ enabled ■ disabled

- 11 Associate the charging instance with the billing plan.

```
an3000-mcm-slot7cpu1(config)# services charging charging-instance
<charging-instance-name> billing-plan-map <billing-plan-name>
charging-choice charging-characteristic <value>
```

where

Parameter	Description
charging-instance	Specifies the charging instance name. The charging instance binds one or more billing plans.
billing-plan-map	Specifies the billing plan name. The billing plan defines the charging method to be applied to the list of rating groups.
charging-characteristic	Specifies the charging characteristic. This number could be one provisioned in each user's subscription data, per-APN, or for the entire profile at HSS. It is significant inside the user's home network. It can be used by the Operator to specify data as CDF address, OCS address, interim reporting interval/volume for online charging, tariff times, and so on. It is attached to the billing plan as a method to associate the charging profile to a particular subscriber.

- 12 Enable charging on the data profile by adding the new charging instance name. The data profile must already exist.

```
an3000-mcm-slot7cpu1(config)# workflow data-profile
<data-profile-name> charging <charging-instance-name>
```

- 13 Associate the Gy interface with a workflow control profile. The control profile must already exist.

```
an3000-mcm-slot7cpu1(config)# workflow control-profile
<control-profile-name> default-online-interface <interface-name>
is-ocs-enabled [true|false]
```

where

Parameter	Description
control-profile	Specifies the workflow control profile name.

Parameter	Description
<code>default-online-interface</code>	Specifies the online charging interface name. If the Control Profile with which an online rating group is associated does not have a <code>default-online-interface</code> configured, the MCC still allows the PDN session to be established if there is no active online service rule during session setup. In this case, traffic that does not match the online rating group's service rules, such as offline rating group traffic, will be allowed and CDRs will be generated. However, if a rating group has at least one static online service rule (<code>always-on</code> rule) or if a predefined/dynamic online service rule is installed by the PCRF during/after PDN session setup, the MCC terminates the PDN session if there is no <code>default-online-interface</code> configured.
<code>is-ocs-enabled</code>	Specifies whether online charging is enabled per subscriber by default: ■ true – The MCC establishes an online charging session with the OCS by default when the PDN session is established. ■ false – The MCC does not, by default, establish an online charging session with the OCS when the PDN session is established. <i>Note: For Event-based charging when the HTTP Proxy triggers an online charging session with the OCS, set this parameter to <code>false</code>.</i>

Optional Gy Interface Configuration Tasks

This section describes optional Gy interface configuration tasks.

[Reducing the Load on the Gy Interface \(page 1-38\)](#)

[Overriding the Destination-Host and Destination-Realm Static Configuration \(page 1-39\)](#)

[Using Gx Charging-Information AVP for Gy Session Setup \(page 1-40\)](#)

[Setting up the PDN Session and Gy Session Asynchronously \(page 1-43\)](#)

[Setting up the Gy Session After Receiving the First SDF Packet \(page 1-45\)](#)

[Configuring Geo-Redundancy Health Checks on the Gy Interface \(page 1-49\)](#)

[Configuring DSCP Values for Control Plane Traffic \(page 1-53\)](#)

[Mapping a Filter-ID to a Rating Group \(page 1-54\)](#)

[Modifying Default Diameter AVP Flags \(page 1-55\)](#)

[Configuring an SCTP Profile \(page 1-57\)](#)

[Redirecting the First HTTP Request of a Session Based on the FUI AVP \(page 1-59\)](#)

[Customizing the APN Reported on External Interfaces \(page 1-60\)](#)

[Configuring APN Reporting at the Interface Level \(page 1-61\)](#)

[Weighted Round-Robin Distribution to Multiple Diameter Peers \(page 1-62\)](#)

[Using the CRBN Profile to Select an Online-Charging Interface \(page 1-63\)](#)

Reducing the Load on the Gy Interface

MCC devices support the following methods to help reduce the load on the Gy interface when a subscriber runs out of credits.

- *Using the Quota Blacklist Timer* – The `quota-black-list-timer` parameter specifies how frequently to send grant requests after the termination or suspension of a rating group.

If a subscriber reaches the credit limit during a session, the MCC device activates the quota blacklist timer, configurable at the rating-group level. This timer prevents the MCC device from sending additional quota requests to the OCS for a given rating group until the timer expires. When the timer expires, the system again sends a grant request to the OCS. This timer ensures the MCC only sends periodic quota requests to the OCS on behalf of this subscriber.

Use the following command to configure the quota blacklist timer:

```
an3000-mcm-slot7cpu1(config)# services charging rating-group
<rating-group-name> quota-black-list-timer <time-in-minutes>
```

- *Using the Smart Recharge Feature* – The feature is implemented in accordance with the reauthorization procedures defined in 3GPP TS 32.299.

If a subscriber reaches the credit limit during a session, the MCC device does not send periodic grant requests to the OCS. Instead, once the subscriber's credit limit is increased, the OCS initiates an unsolicited credit reauthorization (for the specific subscriber) by sending an RAR message. Upon receiving this message, the system responds with an RAA message and activates the rating groups requested as part of the RAR and starts sending quota requests. Note that this method requires that both the OCS and PGW/GGSN support the reauthorization functionality.

Note: Operators can use one or both of these methods to help reduce the load on the Gy interface. If you are using both methods, Affirmed Networks recommends setting the quota blacklist timer to a value of at least 24 hours.

Note: If the OCS does not include any rating group names in the RAR message, the MCC PGW reauthorizes all rating groups, including blocked rating groups, upon receiving the RAR message. In this case, the MCC PGW includes both active and blocked rating groups in the CCR-U transaction used for new quota requests.

Overriding the Destination-Host and Destination-Realm Static Configuration

The Diameter node on the PGW communicates with statically configured Diameter peers via the Gy interface. When configuring a Diameter peer, you specify the peer host name and realm, which are used in the Destination-Host and Destination-Realm AVPs in Diameter request messages by default.

You can override these default settings by configuring the `override-hostname` and `override-destination-realm` parameters. When these override parameters are configured, the PGW sends the CCR-I to the selected Diameter peer but includes the configured override destination host/realm in the Destination-Host and Destination-Realm AVPs. This feature enables Operators to provision a Diameter Routing Agent (DRA) to select a specific OCS when multiple OCS nodes are deployed within a given Diameter realm.

Note: The static destination host/realm override configuration only applies to the Destination-Host and Destination-Realm AVPs in CCR-I messages. In CCR-U/CCR-T/ASA/RAA messages, the gateway uses the values from the Origin-Host/Origin-Realm AVPs received in the CCA-I message to fill the Destination-Host/Destination-Realm AVPs. If the `override-dest-hostname` is configured and if the CCA-I response does not have a different Origin-Host, the gateway continues using the `override-dest-hostname`.

Use the following CLI command to override the static configuration for the Destination-Host AVP sent in CCR-I messages on the Gy interface:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name> override-dest-hostname <destination-host>
```

Use the following CLI command to override the static configuration for the Destination-Realm AVP sent in CCR-I messages on the Gy interface:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name> override-destination-realm
<destination-realm>
```

Using Gx Charging-Information AVP for Gy Session Setup

You can configure the PGW to extract information from the Primary/Secondary-Event-Charging-Function-Name AVPs, contained within the Charging-Information AVP received from the PCRF, and either use this information in the Destination-Host and Destination-Realm AVPs sent in the CCR-I to the OCS during Gy session setup or use the extracted destination host to select the OCS peer.

Use the following CLI commands to configure the PGW to extract information from the Charging-Information AVP received from the PCRF.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name> pcrf-charging-info-avp-usage
[overrideDestHostRealm|peerSelection] pcrf-num-labels-destination-
realm <value>
```

The following table describes the configurable parameters.

Parameter	Description
online-charging	Specifies the online charging interface name.
pcrf-charging-info-avp-usage	Specifies how the destination host/realm extracted from the Charging-Information AVP (received from the PCRF) is used during Gy session setup. ■ overrideDestHostRealm – The extracted host/realm is used for the Destination-Host AVP and the Destination-Realm AVP in the CCR-I sent to the OCS. The MCC supports two formats for extracting realm information: Consumer Realm Format and Reseller Realm Format. See Formats for Extracting Realm Information (page 1-41). ■ peerSelection – (default) The extracted destination host is used for selecting the OCS peer.
pcrf-num-labels-destination-realm	The Charging-Information AVP contains a single FQDN value (with one or more labels separated with dots) for each primary and secondary OCS but does not indicate which part of the FQDN belongs to the parent realm. Specify the number of labels to extract from the Charging-Information AVP for the destination-realm. The default is 0, indicating that the destination realm is not extracted from the destination host. <i>Note:</i> This parameter is only applicable when <i>pcrf-charging-info-avp-usage</i> is set to <i>overrideDestHostRealm</i>. <i>Note:</i> This parameter does not apply when the Consumer Realm Format is used to extract realm information from the Gx Charging-Information AVP during Gy session setup. See Formats for Extracting Realm Information (page 1-41).

Formats for Extracting Realm Information

When the `pcrf-charging-info-avp-usage` parameter is set to `overrideDestHostRealm`, the MCC uses the extracted information to fill the Destination-Host AVP and the Destination-Realm AVP in the CCR-I message sent to the OCS. The MCC supports two formats for extracting realm information: Consumer Realm Format and Reseller Realm Format.

Consumer Realm Format

If there is a `@` symbol present in the FQDN received in the Charging-Information AVP from the PCRF, the MCC uses the Consumer Realm Format to extract the destination-realm from the Charging-Information AVP. For example:

```
pcrf-charging-info-avp-usage = overrideDestHostRealm
```

```
Charging-Information AVP = aaa://ocs1.location1@ocs.amdocs.com
```

- Gy Destination-Host = `ocs1.location1`
The sub-string between the final slash (/) and the `@` symbol is the host. The MCC gateway overrides the destination-host in the CCR-I to the OCS with the parsed value.
- Gy Destination-Realm = `ocs.amdocs.com`
The sub-string after the `@` symbol is the realm. The MCC gateway overrides the destination-realm in the CCR-I to the OCS with the parsed value.
- The `pcrf-num-labels-destination-realm` parameter does not apply when the Consumer Realm Format is used.

Reseller Realm Format

If the `@` symbol is absent in the received FQDN, the MCC uses the Reseller Realm Format to extract the destination-realm from the Charging-Information AVP. For example:

```
pcrf-charging-info-avp-usage = overrideDestHostRealm
```

```
pcrf-num-labels-destination-realm = 3
```

```
Charging-Information AVP = aaa://ocs1.location1.ocs.amdocs.com:1813;  
transport=tcp;protocol=diameter
```

- The MCC determines the `@` symbol is absent, so the Reseller Realm Format is used.
- FQDN = `ocs1.location1.ocs.amdocs.com`
The sub-string between the final slash (/) and the first colon (:) is the FQDN.
- Gy Destination-Host = `ocs1.location1.ocs.amdocs.com`

The MCC gateway overrides the destination-host in the CCR-I to the OCS with the parsed value.

- Gy Destination-Realm = `ocs.amdocs.com`
 - For the Reseller Realm Format, the `pcrf-num-labels-destination-realm` parameter is used to determine the realm from the FQDN. The parameter specifies the number of labels to extract from the Charging-Information AVP for the destination-realm. The labels are separated with dots.
 - The MCC gateway overrides the destination-realm in the CCR-I to the OCS with the parsed value.

Setting up the PDN Session and Gy Session Asynchronously

With the Gy Async feature, you can configure the PGW to proceed with the PDN session setup and permit traffic to pass up to a predefined initial quota while the online-charging session is setup in parallel. There is no disruption in traffic if the granted quota is received from the OCS in the CCA-I before the initial quota is consumed. However, if the initial quota is consumed before the PGW receives the granted quota from the OCS, traffic for this session is blocked.

The initial quota is applicable per rating group and is uniform across all rating groups in the session. When configuring the rating group, you can select whether the OCS credit authorization begins upon the receipt of the first control signaling packet (`bearerSetup`) or the first service data flow (SDF) packet (`firstSdfPkt`).

When first SDF packet (`firstSdfPkt`) is selected, the PGW sets up the session with the OCS first and waits for the first SDF packet to arrive before sending a credit authorization request to the OCS. In this case, there could be a delay before the credit authorization from the OCS is received which could result in user traffic being blocked. Since the Gy Async feature was designed to avoid user disruptions, Affirmed Networks recommends using the `bearerSetup` credit authorization option when Gy Async is enabled.

The following configuration is used when setting up a new online charging session or restoring an online charging session that was previously suspended based on information from the PCRF.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name> gy-async [enabled|disabled]
rating-group-initial-quota <value> charge-rating-group-initial-quota
[true|false]
```

The following table describes the configurable parameters.

Parameter	Description
<code>online-charging</code>	Specifies the online charging interface name.
<code>gy-async</code>	Controls whether to enable or disable the Gy Async feature. ■ enabled – The PGW completes the PDN session setup by sending the Create Session Response and sets up the online charging session with the OCS asynchronously (not as part of the PDN session setup). ■ disabled – (default) The PGW completes the PDN session setup after establishing the online charging session with the OCS. <i>Note: The <code>gy-async</code> parameter is only configurable when <code>interface-compliance</code> is set to <code>3GPPVersion2</code>. By default, <code>gy-async</code> is disabled for all other interfaces.</i>

Parameter	Description
<code>rating-group-initial-quota</code>	Specifies the maximum number of credits, in bytes, that the PGW can allow per rating group before establishing the Gy session and receiving the granted quota from the OCS. If the initial quota is consumed before the PGW receives the granted quota from the OCS, traffic for this session is blocked. The default is 0 (disabled), indicating that traffic is blocked until the granted quota is received from the OCS.
<code>charge-rating-group-initial-quota</code>	Controls whether to charge for the traffic passed using the initial quota per rating group. ■ <code>true</code> ■ <code>false</code> (default)

Note: All traffic, including the initial quota traffic, is included in offline CDRs regardless of whether charging is enabled for the initial quota.

Setting up the Gy Session After Receiving the First SDF Packet

Typically, the online-charging session with the OCS is established as part of the PDN session setup. However, you can restrict the PGW from establishing an online-charging session until after it receives the first service data flow (SDF) packet. In this case, the PGW waits for the first SDF packet to arrive before sending a credit authorization request (CCR-I) to the OCS.

There are two different configuration parameters for the online charging interface that allow you to defer the credit authorization to the OCS. Modifications to these parameters only apply to new sessions created after the configuration change. These parameters are only visible when `interface-compliance` is set to `3GPPVersion2`.

- `defer-gy-session-until-first-sdf-packet` – Applies when all online rating groups have the `credit-authorization-event` parameter set to `firstSdfPkt`.
- `first-packet-credit-auth-rating-groups` – Applies to rating groups that have the `credit-authorization-event` parameter set to `bearerSetup`.

See [Table 1-1 on page 1-47](#) for information on the PGW/OCS interaction when using the `defer-gy-session-until-first-sdf-packet` and `first-packet-credit-auth-rating-groups` parameters together.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name>
defer-gy-session-until-first-sdf-packet
[home-subscriber|roaming-subscriber|both]
first-packet-credit-auth-rating-groups [firstsdf|all]
```

The following table describes the configurable parameters.

Parameter	Description
<code>online-charging</code>	Specifies the online charging interface name.

Parameter	Description
defer-gy-session-until-first-sdf-packet	Specifies whether to restrict the PGW from establishing an online-charging session until after it receives the first SDF packet. Configure separate actions for home and roaming subscribers. Applies when all online rating groups have the <code>credit-authorization-event</code> parameter set to <code>firstSdfPkt</code>. ■ home-subscriber – For home subscribers, the PGW establishes the online-charging session and requests quota with the OCS (MSCC AVP in the CCR-I) upon receiving the first SDF packet. ■ roaming-subscriber – For roaming subscribers, the PGW establishes the online-charging session and requests quota with the OCS (MSCC AVP in the CCR-I) upon receiving the first SDF packet. ■ both – For both home and roaming subscriber sessions, the PGW establishes the online-charging session and requests quota with the OCS (MSCC AVP in the CCR-I) upon receiving the first SDF packet. For the set of subscribers for which the online-charging session setup is not deferred and is setup along with the PDN sessions, the PGW establishes the online-charging session with the OCS as part of the PDN session setup but does not include the quota request (MSCC AVP) in the CCR-I sent to the OCS. In this case, the quota request (MSCC AVP) is included in the CCR-U message once the PDP session is set up and the PGW receives the first service data flow packet. <i>Note: By default, the <code>defer-gy-session-until-first-sdf-packet</code> parameter is not configured.</i>
first-packet-credit-auth-rating-groups	Specifies whether to restrict the PGW from establishing an online-charging session and requesting quota until after it receives the first SDF packet for rating groups configured with the <code>credit-authorization-event</code> parameter set to <code>bearerSetup</code>. ■ firstsdf – (default) The credit authorization for rating groups with the <code>credit-authorization-event</code> set to <code>firstSdfPkt</code> will be deferred until the PGW receives the first SDF packet. ■ all – For all active online rating groups, the credit authorization will be deferred until the PGW receives the first SDF packet. When the PGW receives the first SDF packet on any active online rating group, the quota request (MSCC AVP) is included in the CCR-U for all rating groups with <code>credit-authorization-event</code> set to <code>bearerSetup</code> and for the rating group (<code>bearerSetup/firstSdfPkt</code>) that received the first packet. For rating-groups with the <code>credit-authorization-event</code> set to <code>firstSdfPkt</code>, the credit authorization is requested when the PGW receives the first SDF packet for the rating group.

The following table summarizes the PGW/OCS interaction when configuring the PGW to defer the establishment of the online charging session and quota request using the `defer-gy-session-until-first-sdf-packet` and `first-packet-credit-auth-rating-groups` parameters together.

Table 1-1 Configuring the `defer-gy-session-until-first-sdf-packet` and `first-packet-credit-auth-rating-groups` Parameters Together

<code>defer-gy-session-until-first-sdf-packet</code>	<code>first-packet-credit-auth-rating-groups</code>	PGW/OCS Interaction
Not Configured (default)	<code>firstSdf</code> (default)	The PGW sends the CCR-I during the PDN session setup. The CCR-I might include the MSCC AVP if there are rating groups with the <code>credit-authorization-event</code> set to <code>bearerSetup</code>. The credit authorization for rating groups with the <code>credit-authorization-event</code> set to <code>firstSdfPkt</code> will be deferred until the PGW receives the first SDF packet and the MSCC AVP for these rating groups will be requested in the CCR-U.
	<code>all</code>	The PGW sends the CCR-I (without the MSCC AVP) during the PDN session setup. For all rating groups, the credit authorization will be deferred until the PGW receives the first SDF packet. ■ For rating groups with the <code>credit-authorization-event</code> set to <code>firstSdfPkt</code>, the PGW includes the quota request (MSCC AVP) in the CCR-U when the PGW receives the first SDF packet for the rating group. ■ When the PGW receives the first SDF packet on any rating group, the PGW sends the CCR-U (with MSCC AVP) for all rating groups with the <code>credit-authorization-event</code> set to <code>bearerSetup</code> and for the rating group (<code>bearerSetup/firstSdfPkt</code>) that received the first packet.

Table 1-1 Configuring the `defer-gy-session-until-first-sdf-packet` and `first-packet-credit-auth-rating-groups` Parameters Together

<code>defer-gy-session-until-first-sdf-packet</code>	<code>first-packet-credit-auth-rating-groups</code>	PGW/OCS Interaction
<code>home-subscriber </code> <code>roaming-subscriber </code> <code>both</code>	<code>firstSdf</code> (default)	The PGW will not send the CCR-I during the PDN session setup if all rating groups have the <code>credit-authorization-event</code> set to <code>firstSdfPkt</code>. If there are rating groups with the <code>credit-authorization-event</code> set to <code>bearerSetup</code>, the PGW will send the CCR-I (with MSCC AVP) during the PDN session-setup for all these rating groups. The credit authorization for rating groups with the <code>credit-authorization-event</code> set to <code>firstSdfPkt</code> will be deferred until the PGW receives the first SDF packet and the MSCC AVP for these rating groups will be requested in the CCR-U.
	<code>all</code>	The PGW does not send the CCR-I during the PDN session setup. ■ When the PGW receives the first SDF packet on any rating group, the PGW sends the CCR-I (with MSCC AVP) for all rating groups with the <code>credit-authorization-event</code> set to <code>bearerSetup</code> and for the rating group (<code>bearerSetup/firstSdfPkt</code>) that received the first packet. ■ For rating groups with the <code>credit-authorization-event</code> set to <code>firstSdfPkt</code>, the PGW includes the quota request (MSCC AVP) in the CCR-U when the PGW receives the first SDF packet for the rating group.

Configuring Geo-Redundancy Health Checks on the Gy Interface

Geographic Redundancy network health checks are supported on RADIUS, Gx, Gy and S6b interfaces and use both shared loopback IPs and private loopback IPs. Operators must enable health-check monitoring on a least one interface.

Note: It is recommended that you use the Geographic Redundancy Manager application in the Acuitas Service Management System to configure the Geographic Redundancy group, service endpoint, and server parameters ([step 1](#) through [step 3](#)). For more information on configuring Geographic Redundancy, see the Acuitas User's Guide.

To configure Geographic Redundancy network health checks on the Gy interface:

- 1 Configure the geographic redundancy group.

```
an3000-mcm-slot8cpu1(config)# geographic-redundancy group
<geo-redundancy-group-name> admin-state [enabled|disabled] group-id
<id> manual-state [active|standby] mode
[hot-auto|hot-manual|warm-manual] role [primary|secondary]
```

where

Parameter	Description
geographic-redundancy group	Specifies a Geographic Redundancy group name that is unique across the network.
admin-state	Specifies the administrative state: ■ enabled – Initializes and starts the redundancy mechanism (such as synchronization in hot mode). ■ disabled – Invalidates any synchronized data and sets the mechanism into a non-ready state.
group-id	Specifies the unique group ID that matches with the peer.
manual-state	Specifies the manually configured HA state (the Operator current desired state). ■ active ■ standby
mode	Specifies the instance configured HA mode. ■ warm-manual – No session synchronization occurs between nodes, failovers are Operator controlled. ■ hot-auto – Synchronization of per-session data occurs between nodes, failovers are automatically triggered based on network health monitoring results. ■ hot-manual – Synchronization of per-session data occurs.

Parameter	Description
role	Specifies the instance configured HA backup role. ■ primary – Specifies which node is “active” under normal network conditions. ■ secondary – Specifies which node is “standby” under normal network conditions.

2 Configure the service endpoint for the Geographic Redundancy server.

```
an3000-mcm-slot8cpu1(config)# network-context <network-context-name>
service-endpoint <service-endpoint-name> local-loopback-address
<ip-address> local-port-range-base <port> local-port-range-numports
<value>
```

where

Parameter	Description
network-context	Specifies a network context containing this Geographic Redundancy service endpoint.
service-endpoint	Specifies the name of the Geographic Redundancy service endpoint.
local-loopback-address	Specifies the local node address used for communication between primary and secondary nodes.
local-port-range-base	Specifies the base UDP port number used for communication between primary and secondary nodes. Must be the initial value of a contiguous block of UDP ports.
local-port-range-numports	Specifies the size of the contiguous block of ports.

3 Configure the Geographic Redundancy server parameters.

```
an3000-mcm-slot8cpu1(config)# service-construct geo-redundancy-server
<geo-redundancy-group-name> network-context <network-context-name>
service-endpoint <service-endpoint-name> remote-ip-address
<ip-address> remote-base-port <port> connectivity-monitoring-params
[monitor-AAA|monitor-OCS|monitor-PCRF|monitor-S6B] admin-state
[enabled|disabled]
```

where

Parameter	Description
geo-redundancy-server	Specifies the Geographic Redundancy group name to which this Geographic Redundancy server belongs.
network-context	Specifies the network context containing this Geographic Redundancy service endpoint.
service-endpoint	Specifies the service endpoint for this Geographic Redundancy server.
remote-ip-address	Specifies the IP address for the remote peer in the group.
remote-base-port	Specifies the port number for the remote peer in the group
connectivity-monitoring-params	Specifies network connectivity monitoring for automatic failover. Configure the parameters as follows: ■ monitor-AAA – Set to enabled to monitor the RADIUS Accounting Authentication Authorization (AAA) interface to govern fault detection and automatic failover. ■ monitor-PCRF – Set to enabled to monitor the Gx interface to govern fault detection and automatic failover. ■ monitor-OCS – Set to enabled to monitor the Gy interface to govern fault detection and automatic failover. ■ monitor-S6B – Set to enabled to monitor the S6B interface to govern fault detection and automatic failover.

- 4 Associate the Geographic Redundancy group with the local diameter node associated with the interface and configure health-check parameters on the interface.

```
an3000-mcm-slot8cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <diameter-node-name>
geo-redundant-group <geo-redundancy-group-name>
net-health-report-enabled [true|false] link-type [private|shared]
```

where

Parameter	Description
node-admin	Specifies the local diameter node associated with the interface. The local diameter node must already exist. See Configuring the Gy Online Charging Interface (page 1-11) .
geo-redundant-group	Specifies the Geographic Redundancy group name.
net-health-report-enabled	Specifies whether to monitor network health towards the interface: ■ true ■ false
link-type	Specifies whether the link used to monitor neighboring network elements is a private (local) or shared IP address among the primary and secondary NE's: ■ private ■ shared

- 5 Specify a default Gy interface in the workflow control profile.

```
an3000-mcm-slot8cpu1(config)# workflow control-profile <profile-name>
default-charging-interface <interface-name>
```

Note: The interface must already exist. See [Configuring the Gy Online Charging Interface \(page 1-11\)](#).

- 6 Use the following command to view the parameters of the diameter node:

```
an3000-mcm-slot8cpu1# show service-construct aaa-interface-group
diameter node-admin <diameter-node-name>
```


Configuring DSCP Values for Control Plane Traffic

Operators can assign different levels of service to control plane traffic on Diameter interfaces using Differentiated Services Code Point (DSCP).

Diameter interfaces (S6b, Gx, Gy, Gz/Rf) – DSCP value configured at the Diameter server group level.

```
an3000-mcm-slot7cpu1(config)# service-construct interface  
aaa-interface-group diameter node-admin <name> dscp-value <value>
```

Operators can also configure a DSCP value at the network-context level applicable to end marker packets, error indication messages, DNS queries and GTP-U Echo messages.

```
an3000-mcm-slot7cpu1(config)# network-context <name>  
control-dscp-value <value>
```

Note: The DSCP value configured at the Diameter server-group level overrides the network-context DSCP value.

Mapping a Filter-ID to a Rating Group

When the FUI AVP (embedded within the MSCC AVP), received in the CCA from the OCS, contains the Final-Unit-Action (FUA) AVP set to `RESTRICT_ACCESS` and contains one or more Filter-ID AVPs, the PGW restricts user access according to the IP packet filters identified by the Filter-ID AVP. The Filter-ID AVP sent in the CCA references a locally configured rating group with an IP address filter list. Multiple Filter-IDs are supported by configuring a rating group for each Filter-ID. When configuring a restrict-access rating group on the MCC, the rating group must be configured as follows:

- Set the `charging-method` parameter to `offline`.
- Set the `service-activation` parameter to `external-activation`.
- Set the `priority` parameter at a higher priority (lower number) than the corresponding online rating group that requires restricted access.
- All the configured service rules/packet filters associated with the restrict-access rating group must have the `service-activation` parameter set to `always-on`.

Activating the restrict-access rating group – When the PGW receives the FUA AVP set to `RESTRICT_ACCESS`, the PGW uses the Filter-ID to dynamically activate the matching restrict-access rating group(s) and places the corresponding online rating group in a blocked state. The PGW maintains a restrict-access rating group list for each online rating group as needed for activation and deactivation. The PGW allows traffic matching the restrict-access rating group to pass and drops the traffic matching the online rating group. The PGW reports the usage on the restrict-access rating group for offline charging.

Note: *If no matching rating group is found for the Filter-ID, the PGW ignores the Filter-Id AVP and takes no action for the restrict-access setting in the FUA AVP.*

Deactivating the restrict-access rating group – The PGW deactivates the corresponding restrict-access rating group(s) when one of the following events occur:

- A grant is received for the corresponding online rating group. A restrict-access rating group (offline rating group with external-activation) is active only when the online rating group is in restrict-access mode (OCS sends FUA AVP set to `RESTRICT_ACCESS`). When a grant is received for the online rating group in restrict-access mode, the restrict-access rating group is automatically deactivated.
- The online rating group is no longer in use, such as when all the service rules associated with the online rating group are removed from the PCRF.
- The PCRF disables online charging for all service rules associated with the online rating group.

Modifying Default Diameter AVP Flags

Operators can modify the default AVP flag values for Diameter AVPs on a specific Diameter interface (Gx, Gy, Gz, SWm, STa, and S6b). Diameter AVPs have flags that indicate to the receiver how each AVP must be handled in Diameter messages. An AVP can have one or both of these flags set.

- Mandatory “M” flag – Indicates that support for this AVP is required. If either the AVP or its value is not recognized, then the message is not processed.
- Vendor-specific “V” flag – Indicates that this AVP has the optional Vendor ID field and when set, the AVP belongs to the specific vendor code address space.

For example, if the RAT-Type AVP has the Mandatory flag and Vendor-Specific flag set by default, all Diameter messages must have this AVP, otherwise the message is not processed. Also, when this AVP is sent, the Vendor ID field will be populated with the value associated with the specific Vendor.

With this feature, if the Operator changes the RAT-Type AVP Flag setting so that only the Vendor-Specific flag is set for the Gx interface, then subsequent Gx messages may or may not have this AVP. However, the Vendor ID field will still be populated with the value associated with the specific Vendor when this AVP is sent.

Use the following CLI command to modify the default Diameter AVP flag values.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter avp-flags <entry-name> attribute-id
<attribute-id> vendor-id <vendor-id> [gx|gy|gz|swm|sta|s6b] flag-value
[none|mandatory|vendor-specific|mandatory-and-vendor-specific]
```

where

Parameter	Description
avp-flags	Specifies the configured name for the AVP flag entry. The entry name is a custom name used for the AVP flag entry and is not necessarily the name of the Diameter AVP.
attribute-id	Specifies the Diameter AVP attribute ID. <i>Note: The attribute ID and vendor ID are mandatory fields. This feature will only modify the AVP flag value if the attribute ID and vendor ID are correctly provided. If there is no AVP found with the values provided, the configuration will have no effect.</i>
vendor-id	Specifies the Diameter AVP vendor ID. <i>Note: The attribute ID and vendor ID are mandatory fields. This feature will only modify the AVP flag value if the attribute ID and vendor ID are correctly provided. If there is no AVP found with the values provided, the configuration will have no effect.</i>
[gx gy gz swm sta s6b]	Specifies the Diameter interface.

Parameter	Description
<code>flag-value</code>	Specifies the flag value for the Diameter AVP. ■ If an AVP has a non-zero Vendor ID, the possible flag values are <code>vendor-specific</code> and <code>mandatory-and-vendor-specific</code>. ■ If an AVP has a Vendor ID of zero, the possible flag values are <code>none</code> and <code>mandatory</code>.

Configuring an SCTP Profile

To use Stream Control Transmission Protocol (SCTP) as a transport protocol on the Gy interface, you must configure an SCTP profile and associate the profile with the local Diameter node.

You also have the option of configuring a secondary loopback IP address for the local Diameter node and Diameter peer to support SCTP Multi-homing. SCTP multi-homing enables the MCC to switchover to an alternate path in the case of a failure on the primary path. The MCC will revert back to the primary path once the primary connection becomes available. The primary and secondary loopback IP addresses can be IPv4 or IPv6. MCC supports socket creation to be associated to a single network context. SCTP multi-homing will be supported in a single network context.

Note: If using a mix of IPv4/IPv6, ensure the primary loopback IP address is IPv4 and the secondary is IPv6. SCTP Multi-homing will not work if the primary loopback IP address is IPv6 and the secondary is IPv4. For this reason, Affirmed Networks recommends not using a mix of IPv4/IPv6 addresses for SCTP Multi-homing.

1 Create an SCTP profile.

```
an3000-mcm-slot7cpu1(config)# service-construct sctp-profile
<profile-name>
```

The following table describes the configurable parameters.

Parameter	Description
<code>cookie-life</code>	Specifies the lifetime of the SCTP cookie, in 100 milliseconds.
<code>heartbeat-interval</code>	Specifies the periodic time interval (in milliseconds) at which the SCTP heart beat messages are sent. Value 0 disables SCTP heart beats.
<code>init-retry</code>	Specifies the maximum number of attempts to retransmit the INIT message after which the attempted association is terminated.
<code>path-max-retransmit</code>	Specifies the maximum number of consecutive retransmissions over a destination transport address of a peer endpoint before it is marked as inactive.
<code>max-in-streams</code>	Specifies the maximum number of incoming SCTP streams.
<code>max-out-streams</code>	Specifies the number of outgoing SCTP streams.
<code>path-mtu</code>	Specifies the maximum transmission unit (MTU), in bytes, for SCTP streams. Value 0 enables auto-discovery of the path MTU.

Parameter	Description
rto-initial	Specifies the initial retransmission timeout (RTO) value, in milliseconds. Subsequent SCTP RTO will be recomputed starting from this value.
rto-max	Specifies the maximum RTO value (in milliseconds) that any subsequent recomputed RTO may never exceed.
rto-min	Specifies the minimum RTO value (in milliseconds) that any subsequent recomputed RTO may never go below.
sack-period	Specifies the delay before sending the Selective Acknowledgment (SACK), in milliseconds. Value 0 disables the delay value.

- 2 Specify SCTP as the transport type for the local Diameter node and associate the SCTP profile.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> transport sctp
sctp-profile <profile-name>
```

- 3 (Optional) Configure a secondary loopback IP address for the local Diameter node and Diameter peer to support SCTP Multi-homing. The secondary loopback IP address is only configurable for the SCTP transport type.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name>
secondary-loopback-ip <loopback-name>

an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name> secondary-loopback-ip <loopback-name>
```

Note: The `secondary-loopback-ip` parameter is only visible when the local Diameter node transport type is set to SCTP.

Redirecting the First HTTP Request of a Session Based on the FUI AVP

This feature adds the ability to redirect the first HTTP request of a session based on the Final-Unit-Indication (FUI) AVP received from the OCS within the Service-Information → PS-Information AVP via CCA-I or CCA-U (or both) messages.

When the PGW receives the FUI AVP within the Service-Information → PS-Information AVP in the CCA message from the OCS, the PGW performs a one-time HTTP redirection to the URL specified in the FUI AVP. The redirection is only valid for the first HTTP request of the session and is not valid for subsequent HTTP transactions. Non-HTTP traffic should pass as usual.

The PGW uses the following order of priority for handling the FUI AVP received in the CCA message.

1	Command-level FUI AVP
2	Service-Information → PS-Information-level FUI AVP
3	MSCC-level FUI AVP

The Service-Information → PS-Information → FUI AVP is sent via the CCA message from the OCS and is configurable in CCA data record templates. The default `presentFlag` setting is `no`, indicating the PGW ignores the AVP when received in the CCA response.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|ggsn] ocs diameter cca <template-name>
psInformationFinalUnitIndication presentFlag [yes|no|conditional]
```

Customizing the APN Reported on External Interfaces

Operators can customize the APN reported on external interfaces. The customized APN is used in the Called-Station-ID AVP and sent in messages on the following interfaces:

- Access-Request on RADIUS
- Accounting-Request on Gz and RADIUS
- CCR on Gx and Gy interfaces

For Ga (GTPP) and local interfaces, the customized APN-reporting value is filled in the Access Point Name Network Identifier CDR field.

Set parameters for this feature using the following commands:

```
an3000-mcm-slot7cpu1(config)# workflow control-profile <name>  
apn-reporting custom custom-reporting-apn-name <apn-name>
```

To enable this feature, set the `apn-reporting` parameter to `custom`. The value set in the `custom-reporting-apn-name` parameter is used to fill the Called-Station-Id value in messages to external interfaces.

Note: The `custom-reporting-apn-name` parameter is visible only if `apn-reporting` is set to `custom`.

Use the following command to display the reported APN name (`reporting-apn-name`) sent as the Called-Station-Id AVP on external interfaces.

```
an3000-mcm-slot7cpu1# show subscriber pdn-session <session-id>
```


Configuring APN Reporting at the Interface Level

You can configure apn-reporting at the interface level to allow for different APNs to be reported over different external interfaces. The `apn-reporting` value configured at the interface level is used to fill the Called-Station-ID AVP and User-Name AVP sent in messages on the following interfaces:

- Access-Request on RADIUS
- Accounting-Request on Gz and RADIUS
- CCR on Gx and Gy interfaces

For Ga (GTPP) and local interfaces, the `apn-reporting` value configured at the interface level is filled in the Access Point Name Network Identifier CDR field.

The `apn-reporting` value configured at the interface level overrides the `apn-reporting` value configured for the Control-Profile. If the `apn-reporting` value is not configured at the interface level, the Called-Station-ID AVP and User-Name AVP will be filled based on the `apn-reporting` value configured for the Control Profile.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <name> apn-reporting [original|mapped|prior-mapped]
```

Note: The interface-level `apn-reporting` parameter does not support the `custom` option and does not apply to the S6b interface.

Note: This feature applies to the PGW only.

Weighted Round-Robin Distribution to Multiple Diameter Peers

Operators can configure the MCC to use a weighted round-robin algorithm to distribute initial session requests among the available Diameter peers within a Diameter server group based on the configured weight assigned to each peer. The Diameter peer with the highest weight (with 10 being the highest and 1 being the lowest) receives the most distribution of new session requests. The MCC only uses the configured weight for the initial peer selection (for example, CCR-I).

For example, if Peer A is set to 4 and Peer B is set 2, the MCC will distribute the initial session requests as ABABAA. In this example, the MCC alternates between both peers starting with Peer A because it has the highest weight.

When a Diameter peer is configured with a weight of 0, the MCC places the peer in “maintenance mode” and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-U/CCR-T) for existing sessions.

If the Diameter peer with the highest weight is unavailable, the request is sent to the peer with the next highest weight in the same Diameter server group. If multiple Diameter peers are assigned the highest weight (10), the MCC uses a round-robin approach to select from among those peers for new requests.

Operators can configure the MCC to use a weighted round-robin algorithm to select a Diameter peer from among a list of primary peers or from among a list of secondary peers, but not *across* primary and secondary peers.

Use the following command to assign a weight to a Diameter peer:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name> weight <value>
```

- **weight** – Specifies the weight allocated to this Diameter peer. The Diameter peer with the highest weight (10) receives the most distribution of new session requests. The default is 1 (lowest weight). When set to 0, the MCC places the peer in “maintenance mode” and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-U/CCR-T) for existing sessions. *Applies to Gy/Gx/Gz/Sta/Swm/S6b interfaces.*

Using the CRBN Profile to Select an Online-Charging Interface

You can configure the MCC to select an online-charging interface using the CRBN profile if that CRBN profile has an online-charging interface configured.

```
an3000-mcm-slot7cpu1(config)# services crbn-profile <profile-name>  
online-interface <interface-name>
```

Note the following guidelines when using this feature:

- The online-charging interface configured for the CRBN profile overrides the online-charging interface configured for the workflow control profile.
- If the MCC gateway receives multiple CRBN profiles from the PCRF or RADIUS server, the MCC uses the first CRBN profile in the list to select the online-charging interface.

Monitoring

This section describes how to monitor the operational status of the OCS via the Gy interface and how to display statistical information for OCS communication messages. This section also describes the alarm notification that occurs when the TCP connection between the GGSN/PGW and the OCS fails. For more information on statistics, see *Acuitas Performance Statistics*.

Node Status

Use the following CLI command to display the operational status of the local diameter node.

```
an3000-mcm-slot7cpu1# show service-construct interface  
aaa-interface-group diameter node-status <diameter-node-name>
```

The system displays output similar to the following:

```
origin-host          host.ocs.affirmed.net  
origin-realm         affirmed.net  
hostIPAddress        10.48.53.4  
transport            tcp  
port                 3868  
geo-redundant-group ""  
status               inservice
```

Peer Status

Use the following CLI command to display the operational status of the TCP connection between the system and the specified OCS. Use the following CLI command to display the flow-off status and override the peer host name:

```
an3000-mcm-slot7cpu1# show service-construct interface  
aaa-interface-group diameter peer-status <diameter-node-name>  
<peer-host-name>
```

The system displays output similar to the following:

```
realm                affirmed.local  
ip-address           10.6.1.217  
port                 9119  
transport            tcp  
role                 primary  
backup-host-name     ""  
status               okay  
last-state-change-timestamp 2015-06-23T12:44:08-0400  
override-hostname    ""
```

Peer-cmd-Stats

Use the following CLI command to display the number of request/response messages sent per command-code between the system and the specified OCS.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter peer-cmd-stats <diameter-node-name>
<peer-host-name>
```

The system displays output similar to the following:

NODE NAME	HOST NAME	COMMAND CODE	REALM	NUM OUT REQ	NUM OUT RSP	NUM IN REQ	NUM IN RSP
OCS	ocs-server.affirmed.com	257	affirmed.com	98	0	0	98
OCS	ocs-server.affirmed.com	271	affirmed.com	8559	0	0	8559
OCS	ocs-server.affirmed.com	280	affirmed.com	5744	5831	5831	5743
OCS	ocs-server.affirmed.com	282	affirmed.com	0	20	20	0

Peer Stats

Use the following CLI command to display the number of inbound/outbound requests/responses, protocol errors and failures between the system and the specified OCS.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter peer-stats <diameter-node-name>
<peer-host-name>
```

The system displays output similar to the following:

```
realm                                affirmed.com
num-protocol-error                  0
num-permanent-failure               0
num-routing-failure                 0
num-out-request                     5844
num-out-response                    5850
num-in-request                      5850
num-in-response                     14399
```

Diameter Message-Level Statistics

Use the following CLI command to display statistics about Diameter messages exchanged between the local diameter node and diameter peer via the Gy interface.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter statistics cluster-level peer
interface-type GY <diameter-node-name> <peer-host-name>
```

Note: The *num-ccr-initial-failure*, *num-ccr-update-failure* and *num-ccr-terminate-failure* statistics counters reflect the totals of the 3xxx, 4xxx, and 5xxx counts combined.

Note: For more information on Gy interface performance statistics, see *Acuitas Performance Statistics*.

Use the following CLI command to display procedure level success/failure statistics for the OCS Re-Auth message processing:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
[cluster-level|grid-level] procedure pgw
```

Data Usage Cumulative Statistics

Use the following CLI commands to display cumulative statistics for data usage at the interface, service rule, and bearer level.

Interface Level

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
data-usage-stats interface-usage-stats <gw-type> <gw-name>
<interface-type> <interface-name>
```

```
total-used-bytes
```

```
total-uplink-bytes/total-downlink-bytes
```

Service Rule Level

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
data-usage-stats service-data-container-usage-stats <gw-type>
<gw-name> <rating-group> <service-id>
```

Note: The service-id can be the configured service-data-flow-id or the service-id received from the PCRF.

```
total-uplink-bytes/total-downlink-bytes
```

```
total-uplink-packets/total-downlink-packets
```

```
total-unreported-uplink-bytes/total-unreported-downlink-bytes
```

```
total-unreported-uplink-packets/total-unreported-downlink-packets
```

Bearer Level

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
data-usage-stats bearer-usage-stats <gw-type> <gw-name> <qci>
```

```
total-uplink-bytes/total-downlink-bytes
```

```
total-uplink-packets/total-downlink-packets
```

Alarms

A major-level trap similar to the following occurs when the TCP connection between the system and the OCS goes down. The alarm clears after the connection is restored.

```
anDiameterPeerLinkDown
```

```
Description:Transport connection to the diameter peer is down.
```

```
Trap ID:89
```

```
Status:Current
```

```
Severity:Major
```

```
Auto Clearing:No
```

```
Potential Impact:The Diameter Links are Down towards the OFCS server.
This might result in Gz Interface not able to send
the Call Data Records to Billing system.
```

```
Corrective Action:The alarm clears when the interface becomes active.
Check the routing/connectivity towards the
Diameter peer. The alarm clears after the
connectivity is restored. Once you have restored
the IP connectivity make sure the Diameter Peer is
up and routes are correctly configured and
Diameter configuration is correct.
```

Failure Handling

The system uses the watchdog request-based mechanism defined in RFC 3588 and RFC 3539 for failure detection. By exchanging Device-Watchdog-Request (DWR) and Device-Watchdog-Answer (DWA) messages between the GGSN/PGW and OCS, the system can detect OCS or network-related failures and migrate sessions to a secondary OCS, if available.

This section describes failure handling on the Gy interface and includes the following sections.

- [Peer Selection \(page 1-69\)](#)
- [Failover Handling \(page 1-71\)](#)
- [Default Gy Quota \(page 1-72\)](#)
- [Gy Rejoin and Gy Replay \(page 1-77\)](#)
- [Configuring the Diameter Result Code List \(page 1-78\)](#)
- [Storing Gy CCR-T Records in a Local File \(page 1-84\)](#)
- [Configuring Multiple Pairs of Primary and Secondary Peers \(page 1-94\)](#)
- [Failure Handling Message Flows \(page 1-96\)](#)

Peer Selection

Diameter Node Peer List

The diameter node (GGSN/PGW) contains a list of peers (diameter servers) with which it communicates. Each peer must be assigned one of two roles; primary or secondary. Primary peers are active peers used for diameter sessions and secondary peers are used for diameter sessions only when the active peers are unavailable or the OCS returns a temporary failure (result code 3xxx).

For new session requests, the system uses a round-robin approach to select peers from the list of primary peers. If there are no active primary peers, the system selects from the list of active secondary peers.

When the system detects that a peer is down, it removes the peer from the active primary or secondary peer list, depending on its role. When the peer comes back up (CER/CEA exchanges successfully complete), the system adds it back to the active peer list.

Backup Server

You can configure a secondary peer to act as the backup server for a primary peer. The backup configuration is a way to establish a redundancy relationship between two peers so that if the primary peer is unavailable, the system automatically sends all new and pending requests to the configured peer (backup server) instead of the next peer in the list. If the backup server fails, the system automatically fails back to the primary server. Only primary peers can be configured with a backup server. When configuring a primary peer with a backup server, enter the secondary peer host name in the `backup-peer-name` parameter. See [Configuring the Gy Online Charging Interface \(page 1-11\)](#) for more information.

If the system determines that the peer to which the CCR message was sent is unavailable, subsequent messages are sent to the backup server. Existing sessions that were redirected to the backup server, stay on the backup server even when the primary peer recovers. Once connectivity to the primary peer is restored, the GGSN/PGW uses the primary peer for new sessions.

For the CCR-I, if no backup server is found, then there is another lookup to send the message to a different primary/secondary peer. For all other messages, if there is no backup server found, messages are retransmitted to the *same* peer.

RADIUS Server Peer List

The system also supports external OCS peer selection using the RADIUS server. If the RADIUS server sends the system a list of OCS peers using the Nokia-OCS-ID1 and Nokia-OCS-ID2 AVPs in the RADIUS Access-Accept message, the system selects an active OCS from the peer list in the Nokia-OCS-IDs AVP instead of the diameter node peer list.

The MCC treats the peer selected from the Nokia-OCS-ID1 AVP as the primary OCS and the peer selected from the Nokia-OCS-ID2 AVP as the secondary OCS for this subscriber. MCC sends all credit-control messages for this subscriber to the primary OCS (Nokia-OCS-ID1 peer). If the system detects that the primary OCS is down, the system sends credit-control messages for this subscriber to the secondary OCS (Nokia-OCS-ID2 peer).

Failover Handling

Failover to an alternate OCS can be triggered if one of the following events occurs:

- The OCS returns a temporary failure (result code 3xxx).
- The OCS does not respond to DWRs.
- The CCR timer (`retransmit-timeout`) expires. You can configure the `retransmit-timeout` parameter when you configure the online charging interface. See [Configuring the Gy Online Charging Interface \(page 1-11\)](#) for more information.
- The system detects a broken connection (packet send fails).

You can configure a secondary peer to act as the backup server for a primary peer so that when a failover event occurs, the system automatically sends all new and pending requests to the backup server. If the backup server fails, the system automatically fails over to the primary server. When configuring a secondary peer as a backup server, enter the host name of the secondary server in the `backup-peer-name` parameter. See [Configuring the Gy Online Charging Interface \(page 1-11\)](#) for more information.

The system handles failovers for new and existing sessions differently:

New Sessions – When a failover event occurs, the system sends new sessions to the backup server. If no backup server is found, the system sends new session requests to another active peer from the list of primary servers. If there are no active primary servers, the system selects from the list of active secondary servers. When the primary OCS recovers, the system sends new session requests to the primary OCS.

Existing Sessions – When a failover event occurs, the system redirects all messages for existing sessions to the backup server. If no backup server is found, the system retransmits the CCR messages to the *same* peer. Existing sessions that were redirected to the backup OCS, stay on the backup server even when the primary OCS recovers.

Note: If the OCS returns a CCA error message (result code 4xxx or 5xxx), the system terminates the session. See [Configuring the Diameter Result Code List \(page 1-78\)](#) for information on how to configure the MCC to take specific action based on the result code received in the CCA message over the Gy interface.

Note: For failover information when using the Geographic Redundancy Manager, see the *Acuitas User's Guide*.

Default Gy Quota

The Default Gy Quota feature allows the GGSN/PGW to set up an online charging session and to ensure session continuity by allocating a predefined default quota for a session when one of the following events occurs.

- No OCS is available.
- The OCS returns a CCR-I or CCR-U temporary failure (result code 3xxx).
- The CCR `retransmit-count` is reached. The `retransmit-count` specifies the maximum number of times the gateway transmits the CCR. You can configure this value when you configure the online charging interface. See [Configuring the Gy Online Charging Interface \(page 1-11\)](#).

Note: While the default Gy quota (`ocs-down-default-gy-quota`) is used to set up an online charging session when a failure event occurs, the configurable initial quota (`rating-group-initial-quota`) is used with the Gy Async feature to set up an online charging session and prevent user disruption while the PGW is requesting grants from the OCS, especially when the OCS is slow to respond. For more information on the Gy Async feature, see [Setting up the PDN Session and Gy Session Asynchronously \(page 1-43\)](#).

When there is a primary and backup OCS configured and the primary server is unavailable or responds with a temporary failure (result code 3xxx), the GGSN/PGW retransmits the CCR-I/CCR-U request to the backup OCS. If the backup OCS is unavailable or responds with a temporary failure (result code 3xxx), the GGSN/PGW applies the default quota, if enabled.

If no backup server is configured, the system retransmits the CCR-I/CCR-U request to another active peer from the list of primary servers. If there are no active primary servers, the system selects from the list of active secondary servers. If there are no active primary or secondary servers or the configured CCR `retransmit-count` is reached, the GGSN/PGW applies the default quota, if enabled.

When the OCS recovers, the GGSN/PGW sends new session requests to the recovered OCS and continues to process the sessions using the locally configured default quota until the sessions are complete. Existing sessions using the default quota do not rejoin the OCS unless the Gy Rejoin feature is enabled. See [Gy Rejoin and Gy Replay \(page 1-77\)](#) for more information. For Default Gy Quota message flow information, see [Failure Handling Message Flows \(page 1-96\)](#).

The default Gy quota only applies when the OCS returns a CCA error message with a temporary failure (result code 3xxx). To apply the default quota to other CCA error messages (result code 4xxx or 5xxx), you can configure the Diameter result code list. See [Configuring the Diameter Result Code List \(page 1-78\)](#) for configuration information.

However, if you enable the Diameter result code list, the default Gy quota only applies to the ranges configured in the list. Therefore, you will need to configure result code 3xxx as a range in the Diameter result code list to ensure the default Gy quota is applied when the OCS returns a temporary failure (result code 3xxx). To apply the default quota and rejoin capabilities to Diameter result code ranges configured in the list, set the command-level result code `action` parameter to `default-action`.

If the Default Gy Quota feature is *disabled*, the system terminates the session when one of the following events occurs:

- The OCS returns a temporary failure (result code 3xxx).
- The CCR `retransmit-count` is reached. You can configure this value when you configure the online charging interface. See [Configuring the Gy Online Charging Interface \(page 1-11\)](#).

Use the following CLI command to configure the Default Gy Quota feature:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name> ocs-down-default-gy-call-setup
enabled ocs-down-default-gy-quota <value> ocs-down-default-time-quota
<value>
```

The following table describes the configurable parameters.

Parameter	Description
<code>ocs-down-default-gy-call-setup</code>	Specifies whether to enable or disable the Default Gy Quota feature. ■ enabled – When no OCS is available or the OCS returns a temporary failure (result code 3xxx), the GGSN/PGW sets up an online charging session and applies the default quota to the session. ■ disabled – (default) The GGSN/PGW does not set up an online charging session when no OCS is available or the OCS returns a temporary failure (result code 3xxx).
<code>ocs-down-default-gy-quota</code>	Specifies the maximum number of credits, in bytes, that the PGW allocates to a new or existing online charging session in the event of an OCS failure. The default is 10000000.
<code>ocs-down-default-time-quota</code>	Specifies the default time threshold, in seconds, allocated during an OCS failure when the rating group <code>measurement-method</code> is set to <code>time</code> or <code>timeAndVolume</code>. The default is 90 seconds.

Limiting the Number Default Gy Quota Allocations

There is a configuration option to limit the number of times a session can be allocated a default Gy quota while the OCS is down. If the user exceeds the maximum number of default quota allocations, the session is terminated. The default value of 0 allows for an unlimited number of default quota allocations.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name> ocs-down-max-num-default-quota
<value>
```

Reporting Default Gy Quota Subscriber Usage

The subscriber usage consumed while a session is using the default quota is reported to the OCS if the Gy Rejoin feature is enabled (see [Gy Rejoin and Gy Replay \(page 1-77\)](#) for more information). In this case, when the subscriber “rejoins” the OCS, the usage consumed while a session is using the default quota is counted against the quota allocated upon rejoining the OCS. For example, if a subscriber uses up to 1 million bytes while using the default quota and later rejoins the OCS, which allocates a quota of 5 million bytes, the PGW starts counting towards the 5 million units starting at 1 million.

There is also a configuration option to report all interim usage (including all default quota allocations) in the CCR-U when the PGW rejoins the OCS in the event of an OCS failure. By default, this configuration parameter is set to disabled.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name> report-all-usage-on-rejoin
[enabled|disabled]
```

Terminating a Session Upon Depletion of Default Gy Quota

You can configure the PGW/GGSN to terminate a session instead of going to the OCS again when the default Gy quota is exhausted. To activate this feature, configure a session terminate timer for the online-charging interface. By default, this parameter is set to 0 (disabled).

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name> ocs-down-session-terminate-timer
<value-in-seconds>
```

When the session terminate timer is configured and the OCS returns a temporary failure (result code 3xxx), the CCR `retransmit-count` is reached, or no OCS is available, the PGW/GGSN applies the default quota (if enabled) and adds the Failure-Handling-Continue field to GTPP service data containers (SDCs) in Interim CDRs (for rating groups using the default Gy quota). When the session terminate timer expires or the default Gy quota is exhausted, the PGW/GGSN:

- Terminates the session instead of attempting to rejoin the OCS but does not send a CCR-T to the OCS.

- Adds the Failure-Handling-Continue field to SDCs in GTPP Stop CDRs for rating groups using the default Quota.

In the case of multiple rating groups, if one rating group is using the default quota and a rating group using the OCS granted GSU exhausts its quota, the default Gy quota is applied without sending a CCR-U to the OCS to request additional quota.

***Note:** To use this feature, Gy Rejoin (`ocs-down-gy-rejoin`) must be disabled and the Default Gy Quota (`ocs-down-default-gy-call-setup`) must be enabled for the online-charging interface.*

By default, the session terminate timer only applies when the OCS returns a temporary failure (result code 3xxx). To apply the session terminate timer to other CCA error messages (result code 4xxx or 5xxx), you can configure the Diameter result code list.

However, if you enable the Diameter result code list (`gy-action-per-result-code`), the default Gy quota and session terminate timer only apply to the ranges configured in the list. Therefore, you will need to configure result code 3xxx as a range in the Diameter result code list to ensure the session terminate timer and default Gy quota are used when the OCS returns a temporary failure (result code 3xxx). See [Configuring the Diameter Result Code List \(page 1-78\)](#) for configuration information.

See the following sections for message flow information.

- [Failure Handling – Session Terminate Timer Enabled \(Session Setup Failure\) \(page 1-105\)](#)
- [Failure Handling – Session Terminate Timer Enabled \(Mid-Session Failure\) \(page 1-106\)](#)

Default Gy Quota for CUPS

The MCC supports handling of default Gy quota allocations and buffering of data usage messages on the User Plane Function (UPF) if there is a Packet Forwarding Control Protocol (PFCP) link failure or the charging information messages time out.

The Control Plane Function (CPF) sends the default Gy quota configuration to the UPF upon initial session setup. When default Gy quota is enabled and the PFCP link is down when the current default quota is exhausted or the quota request to the CPF times out, the UPF allocates the default quota up to the maximum configured value received from the CPF. If the PFCP link is still down or the quota request times out when the maximum default quota is exhausted, the UPF closes the session. All default quota allocations are reset to 0 if the PFCP link is restored and CPF response messages are successfully received.

If default Gy quota is disabled and the PFCP link is down or the request message times out when the quota is exhausted, the UPF closes the session.

Use the following command to buffer data usage messages on the UPF if the PFCP link is down between the UPF and CPF.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway user-plane-service
pfcf-link-down-queue-messages [disabled|enabled]
```

- **disabled** – (default) If the PFCP link is down or the charging information message times out, the UPF drops the messages and data usage is not reported to the OCS.
- **enabled** – If the PFCP link is down or the charging information message times out, the UPF aggregates and buffers data usage messages for all rating groups and sends those messages (as Sx Session Report messages) to the CPF when the PFCP link is restored.

Note: *If sessions are closed on the UPF before the PFCP link is restored, user data is lost and not reported to the OCS.*

Note: *If the default Gy quota configuration is modified on the CPF, the new configuration will only be replicated on the UPF for new session setups or if the CPF sends any messages to the UPF for existing sessions.*

Gy Rejoin and Gy Replay

The Gy Rejoin feature allows the PGW to rejoin the OCS in the event of an OCS failure. If Gy Rejoin is enabled, the PGW tries to rejoin the OCS if the default quota is consumed or if the `ocs-down-rejoin-timer` expires.

Gy Replay is used if all OCS servers go down after the Gy session is already established and the session using the default quota is terminated before the OCS comes back up. In this case, if Gy Replay is enabled, the GGSN/PGW caches the USU based on the OCS GSU and replays the cached session information via a CCR-T message. For message flow information, see [Failure Handling Message Flows \(page 1-96\)](#).

Note: *If Gy Replay is enabled and the OCS returns a temporary failure (result code 3xxx) in response to the CCR-T message, the system retransmits the CCR-T every 30 seconds until the OCS sends a success code.*

To avoid overloading the OCS, the GGSN/PGW replays all stored session information before marking the OCS as active for new sessions.

Use the following command to enable Gy Rejoin and Gy Replay:

```
an3000-mcm-slot7cpu1(config)# service-construct interface  
online-charging <interface-name> ocs-down-gy-rejoin enabled  
ocs-down-rejoin-timer <value> ocs-down-gy-replay enabled
```

Configuring the Diameter Result Code List

Operators can configure the Mobile Content Cloud (MCC) to take specific action based on the result code received in the CCA message over the Gy interface.

Operators have the ability to configure actions per subscriber-type (home/roaming/both) and to separate actions for Diameter result codes received in the CCA message at the Multiple-Services-Credit-Control (MSCC) level and at the command level. Operators can also specify whether the PGW/GGSN retransmits the request to a secondary server before applying the configured action.

***Note:** To apply default quota and rejoin capabilities to Diameter result code ranges configured in the list, set the command-level result code `action` parameter to `default-action`.*

Use the following CLI commands to configure this feature.

- 1 Configure the Gy Diameter result code list. You can configure a Diameter result code list for the command level and a separate Diameter result code list for the MSCC level.

```
an3000-mcm-slot7cpu1(config)# service-construct  
diameter-result-code-policy-list <list-name> policy <policy-name>
```

The following table describes the configurable parameters.

Parameter	Description
diameter-result-code-policy-list	Specifies the name of result code list.
policy	Specifies the name of the result code policy.
result-code-range	Specifies the low and high end of the result code range. ■ low ■ high <i>Note: Ensure the result code range does not overlap with any previously configured ranges.</i> <i>Note: To configure a range with a single result code, configure the low and high value with the same code.</i>

Parameter	Description
action <i>(Applies to the Gy interface only)</i>	Specifies the action that the PGW/GGSN takes when the result code received in the CCA message is within the configured range. Actions for command-level result codes. ■ block – Block all online rating groups for the subscriber that are in active/blocked states. Online rating groups that have not been activated will not be set to blocked. ■ default-action – (default) The system treats all CCA error result codes as temporary failures. If the OCS returns a CCA error message (result code other than 2001) at the command level, the message is retransmitted and the default quota may be applied and the Gy rejoin timer started based on the configuration. In this case, the PGW/GGSN does not process the OCS response. ■ redirect <i><redirect-url></i> – Redirect traffic to the configured URL for all online rating groups that are in active/blocked states. Online rating groups that have not been activated will not be set to redirect. ■ terminate – Terminate the session immediately. If the PGW/GGSN receives a CCA error message (result code other than 2001) at the command level and if the <i>command-level-result-code-list</i> is configured but the result code is not in the list, the PGW terminates the session. The <i>command-level-result-code-list</i> overrides the <i>mscc-level-result-code-list</i>.

Parameter	Description
action <i>(Continued)</i> <i>(Applies to the Gy interface only)</i>	Specifies the action that the PGW/GGSN takes when the result code received in the CCA message is within the configured range. Note: If the PGW receives a non-zero GSU AVP within the CCA message from the OCS, the MSCC-level result-code configured action is applied after the quota consumption. Actions for MSCC-level result codes. ■ block – Block all traffic for a specific rating group immediately regardless of the quota allocated for the rating group. The Final-Unit-Action (FUA) AVP within the MSCC AVP is ignored. ■ default-action – (default) If the OCS returns a CCA error message (result code other than 2001) at the MSCC level, the gateway executes the action indicated in the FUA AVP if set to REDIRECT or RESTRICT-ACCESS. If the FUA is set to TERMINATE or the CCA does not contain the FUA AVP, then all traffic for the rating group is blocked. Also, the Used-Service-Units received in the MSCC AVP are ignored. ■ redirect <redirect-url> – Redirect traffic to the configured URL for a specific rating group immediately regardless of the quota allocated for the rating group. The FUA AVP within the MSCC AVP is ignored. ■ terminate – Terminate the session immediately regardless of the quota allocated for the rating group. The FUA AVP within the MSCC AVP is ignored. If the PGW/GGSN receives a CCA error message (result code other than 2001) at the MSCC-level and if the <code>mscc-level-result-code-list</code> is configured but the result code is not in the list, the PGW processes subscriber traffic for the specific rating group using the <code>default-action</code> option. The <code>mscc-level-result-code-list</code> is only considered when the command-level result code is 2001.
retry-secondary <i>(Applies to command-level result codes only)</i>	Specifies whether the PGW/GGSN retransmits the request to a secondary server before applying the configured action. ■ enabled – (default) When the primary server returns a command-level result code within the configured range, the PGW retransmits the request to a secondary server before applying the configured action. ■ disabled – When the primary server returns a command-level result code within the configured range, the PGW applies the configured action without retransmitting the request to the secondary server.

Parameter	Description
subscriber-type (Applies to the Gy interface only)	Specifies the type of subscriber to which the configured action is applied. ■ home ■ roaming ■ both (default) The home/roaming subscriber status is based on the home/roaming PLMN ID lists configured on the PGW/GGSN. A subscriber is classified as “roaming” if the serving node PLMN ID is part of the roaming PLMN ID list. If the serving PLMN ID is not found in either the roaming or home PLMN ID lists, the PGW/GGSN uses the value configured in the <code>default-gtp-peer-classification</code> parameter under the <code>gateway-common</code> profile to determine the home/roaming subscriber status.

2 Associate the result code list with the Gy interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name> command-level-result-code-list
<list-name> msccl-level-result-code-list <list-name>
gy-action-per-result-code [enabled|disabled]
```

where

Parameter	Description
online-charging	Specifies the charging interface name.
command-level-result-code-list	Specifies the name of command-level result code list. Note: The <code>command-level-result-code-list</code> overrides the <code>mscc-level-result-code-list</code>.
mscc-level-result-code-list	Specifies the name of MSCC-level result code list. Note: The <code>mscc-level-result-code-list</code> is only considered when the command-level result code is 2001.

Parameter	Description
gy-action-per-result-code	Specifies whether to enable or disable the command-level or MSCC-level result code policy lists. ■ enabled – The result code returned in the CCA message is compared to the configured result code list, and if there is a match, the PGW/GGSN applies the configured action. ■ disabled – (default) The result code list functionality is not used. At the command level, if the OCS responds with a temporary failure (result code 3xxx), the message is retransmitted and the default quota may be applied and the Gy rejoin timer started based on the configuration. For all other CCA error messages (result code other than 2001), the system terminates the session. At the MSCC level, if the OCS returns a CCA error message (result code other than 2001), the gateway executes the action indicated in the FUA AVP if set to REDIRECT or RESTRICT-ACCESS. If the FUA is set to TERMINATE or the CCA does not contain the FUA AVP, then all traffic for the rating group is blocked.

Example 1

The following configuration terminates the session for roaming subscribers and blocks the session for home subscribers when the OCS returns a CCA message with result code `DIAMETER_CREDIT_LIMIT_REACHED` (4012) at the MSCC level and `DIAMETER_SUCCESS` (2001) at the command level.

```
an3000-mcm-slot7cpu1(config)# service-construct
diameter-result-code-policy-list mscc-list policy policy 1
result-code-range low 4012 result-code-range high 4012 retry-secondary
disabled action block subscriber-type home
```

```
an3000-mcm-slot7cpu1(config)# service-construct
diameter-result-code-policy-list mscc-list policy policy 2
result-code-range low 4012 result-code-range high 4012 retry-secondary
disabled action terminate subscriber-type roaming
```

Example 2

The following configuration terminates the session for both home and roaming subscribers when the OCS returns a CCA message with result code `DIAMETER_USER_UNKNOWN` (5030) at the MSCC level and `DIAMETER_SUCCESS` (2001) at the command level.

```
an3000-mcm-slot7cpu1(config)# service-construct
diameter-result-code-policy-list mscc-list policy policy 1
result-code-range low 5030 result-code-range high 5030 retry-secondary
disabled action terminate subscriber-type both
```

Example 3

The following configuration blocks all rating groups for home subscribers when the OCS returns a CCA message with result code 4xxx at the command level.

```
an3000-mcm-slot7cpu1(config)# service-construct
diameter-result-code-policy-list command-list policy policy 1
result-code-range low 4000 result-code-range high 4999 retry-secondary
disabled action block subscriber-type home
```

Diameter Result Code Statistics

Use the following commands to display cluster-level Diameter result code statistics for each message type (for example, CCA-I/CCA-U/CCA-T) for Gy peers in each node. The result code statistics for each peer will be registered dynamically as they are received from the OCS.

To display Diameter result code statistics:

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter statistics cluster-level peer
result-code-statistics <interface-type> <node-name> <peer-name>
<result-code>
```

where:

- **interface-type** – Specifies the interface type.
- **node-name** – Specifies the name of the node.
- **peer-name** – Specifies the peer name.
- **result-code** – Specifies the result code received from the OCS.

To clear Diameter result code statistics:

```
an3000-mcm-slot7cpu1# service-construct interface aaa-interface-group
diameter statistics cluster-level peer result-code-statistics
<interface-type> <node-name> <peer-name> <result-code> clear
```

Note: If a non-redundant Control/Content Services Module (CSM) reboots, the PGW may delete the result code statistics table.

Storing Gy CCR-T Records in a Local File

This feature adds support for storing session usage information in a predefined file and format when one of the following events occur:

- The OCS fails to respond to a CCR-T message or responds with a temporary failure (result code 3xxx).
- The OCS is unavailable and the session using the default quota is terminated before the OCS comes back up. When the PDN session using the default quota is terminated, the PGW tries to initiate a Gy session by sending the CCR-I message. If the OCS is available and the CCR-I succeeds, the PGW sends all usage information in the CCR-T message to the OCS. Otherwise, the PGW stores all session usage information in a predefined file and format.
- If a session that is using the default quota due to a CCR-U failure/timeout is terminated, the PGW sends all usage information in the CCR-T message to the OCS, if available. Otherwise, the PGW stores all session usage information in a predefined file and format.

This feature enables Operators to configure the MCC to automatically store the contents of the CCR-T messages in a predefined file and format. The stored CCR-T records are available for retrieval using SFTP. The amount of time in which they are stored and the size of the file is configurable. When the maximum size is reached, the system creates a new file. File rotation is also configurable based on the number of records and the age of the file. If the file fails to write to disk, the system generates an alarm notification.

This section includes:

- [Export Rules \(page 1-85\)](#)
- [File Naming Conventions \(page 1-86\)](#)
- [custom1 File Format \(page 1-87\)](#)
- [Configuring Gy CCR-T File Storage Functionality \(page 1-89\)](#)

Export Rules

Gy CCR-T records are stored in either the *gy/export* or *gy/export/AP_primary* directory according to the export rules and user permissions configured for the file management profile. The MCC supports several different export rules for Gy CCR-T records configurable using the following CLI command:

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
gy-record <filename-prefix> export-rule
```

- **exportrule-default** – The system exports files to the */public/logs/datarecord.lnk/gy/export* directory according to the default format.
 - Uses the file naming convention:
`<filename-prefix><YYYYMMDDHHMMSS>_<123>.extension`
 - Applies purge rules for this file profile to all files in the directory.
- **exportrule-gy-custom1** – The system exports files to the */public/logs/datarecord.lnk/gy/export/AP_primary* directory according to the custom1 format configured for Gy CCR-T records.
 - Uses the file naming convention:
`<filename-prefix>_<RC>.<date>_<time>`
 - Updates the filename with the last modified timestamp when exporting the file to the *gy/export/AP_primary* directory.
 - Once the stored files have been successfully retrieved, the export user (user configured with read/write access to the files in the *gy/export/AP_primary* directory) should move the files to the *gy/export/AP_secondary* directory and the purge rules configured for the file profile apply to the files in the *gy/export/AP_secondary* directory. Alternatively, the user can delete the files after a successful SFTP file transfer.

If neither of these options are performed, files will remain in the *gy/export/AP_primary* directory and will only be purged when storage becomes full. They will not be cleaned up based on the purge rules configured for this file profile.
- **exportrule-none** – Specifies that there is no export rule for this file profile. In this case, the MCC does not store the Gy CCR-T records.

File Naming Conventions

The MCC uses the following file naming conventions for storing Gy CCR-T records if the Gy CCR-T file storage functionality is enabled.

- *Default* – The default file naming convention is used when the `export-rule` is set to `exportrule-default`.

`<filename-prefix><YYYYMMDDHHMMSS>_<123>.extension`

where `filename-prefix` indicates the prefix of the filename in which to store Gy CCR-T records, `YYYYMMDDHHMMSS` represents the timestamp and `123` represents the file sequence number.

The sequence number runs from 1 to 2147483647, so that at one record per second, it would take a little over 68 years to rollover. The sequence number resets to 1 after a system reboot, MCM switchover or system failure. Depending on the frequency of these events, CDR records could potentially have the same sequence number.

- *custom1* – The custom1 file naming convention is used when the `export-rule` is set to `exportrule-gy-custom1`.

`<filename-prefix>_<RC>.<date>_<time>`

where

- `filename-prefix` – Indicates the prefix of the filename in which to store Gy CCR-T records.
- `RC` – Indicates the file running counter value, which starts with 1 and is incremented by 1 for every file added. When it reaches 65535, it is reset to 1.
- `date` – Indicates the date (YYYYMMDD) when the file is closed.
- `time` – Indicates the time in UTC format when the file is closed.

Note: For more information on export rules, see [Export Rules \(page 1-85\)](#).

custom1 File Format

You can configure the encoding format for Gy CCR-T records as part of the online charging object shown below.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name> pgw diameter
write-records-to-disk-on-failure enabled record-encode-format custom1
```

Currently, the MCC only supports the custom1 encoding format. The Gy CCR-T custom1 file format is illustrated below. The file contains one file header, and zero or more CCR data records as shown below.

File Header
CCR Header
Diameter Header
AVP 1
AVP 2
CCR Header
Diameter Header
AVP 1
AVP 2

Table 1-2 illustrates the Gy CCR-T custom1 File header format. The *length of the CCR routing filter* field is always set to a value of 1 and the *CCR routing filter* field is always set to “ ”.

Table 1-2 Gy CCR-T custom1 File Header Format

Bits								
Octets	8	7	6	5	4	3	2	1
1..4	File Length							
5..8	Header Length							
9	High Release Identifier			High Version Identifier				
10	Low Release Identifier			Low Version Identifier				
11..14	File opening timestamp							
15..18	Timestamp when last CCR was appended to file							
19..22	Number of CCRs in file							
23..26	File sequence number							
27	File closure trigger reason							
28..47	IP address of node that generated file							
48	Lost CCR indicator							
49..50	Length of CCR routing filter							
51..xy	CCR routing filter							

Table 1-3 illustrates the Gy CCR-T custom1 CCR header format.

Table 1-3 Gy CCR-T custom1 CCR Header Format

Bits								
Octets	8	7	6	5	4	3	2	1
1..2	Record Length							
3	Release Identifier			Version Identifier				
4	Data Record Format			TS Number				

Configuring Gy CCR-T File Storage Functionality

Use the following sequence to configure the system to store Gy CCR-T records in a file in the event of a failure.

- 1 Issue the following command to set the default SFTP directory for the export-user:

```
an3000-mcm-slot7cpu1# user-access change-sftp-directory
```

The system prompts:

```
Enter sftp directory:
```

- 2 Type `/public/logs/datarecord.lnk/gy/export` if using the default export rule or type `/public/logs/datarecord.lnk/gy/export/AP_primary` if using the Gy custom1 export rule.

- 3 Create a local file in which to store Gy CCR-T records in the event of a failure.

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
gy-record <filename-prefix>
```

where `<filename-prefix>` specifies the prefix of the filename in which to store Gy CCR-T records.

The following table describes the configurable parameters.

Parameter	Description
<code>compress-enabled</code>	Specifies whether this file should be compressed (zipped) when it is closed. ■ <code>enabled</code> ■ <code>disabled</code>
<code>compress-format</code>	If compress is enabled, specify the format for compressing files: ■ <code>gzip</code> – (default) The data records are written to a file. The file is gzipped after it is closed.
<code>export-rule</code>	Specifies the rules/actions to apply when exporting CCR-T records to a directory for SFTP access. ■ <code>exportrule-default</code> – The system exports files to the <code>/public/logs/datarecord.lnk/gy/export</code> directory and applies purge rules for this file profile to all files in the directory. ■ <code>exportrule-gy-custom1</code> – The system exports files to the <code>/public/logs/datarecord.lnk/gy/export/AP_primary</code> directory according to the custom1 format configured for Gy CCR-T records. For more information, see Export Rules (page 1-85). ■ <code>exportrule-none</code> – Specifies that there is no export rule for this file profile. In this case, the MCC does not store Gy CCR-T records.
<code>export-user</code>	Specifies the user that has read/write access to the files in the directory to which CCR-T records are exported.

Parameter	Description
<code>file-rotate-empty</code>	Specifies whether empty CDR files are rotated. ■ enabled – Empty CDR files are rotated based on the configured <code>file-rotate-frequency</code> interval. ■ disabled – (default) Empty CDR/HTDR files are not rotated.
<code>file-rotate-frequency</code>	Specifies the frequency at which files should be rotated if the current file has not reached the maximum size (<code>file-rotate-size-bytes</code>) or max number of records (<code>file-rotate-num-records</code>). For example, MCC rotates the file on the hour if 1hour is selected, or at midnight, if 1day is selected, etc. Midnight is determined by the timezone and is not based on UTC.
<code>file-rotate-num-records</code>	Specifies the number of records at which to rotate the files.
<code>file-rotate-size-bytes</code>	Specifies the size (in bytes) at which files should be rotated.
<code>purge-enabled</code>	Specifies whether to enable or disable purge/cleanup for this file. ■ enabled ■ disabled
<code>purge-file-older-than</code>	Specifies the frequency at which files should be purged.
<code>purge-low-thr</code>	Specifies the percentage utilization at which to stop transferring/purging files.
<code>purge-thr</code>	Specifies the percentage utilization at which to transfer/purge files.

- 4 Configure the Gy CCR-T file storage functionality on the Gy online charging interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name>
write-records-to-disk-file-on-call-termination [enabled|disabled]
gy-cdr-encode-format custom1 gy-cdr-file <filename> gy-cdr-file-type
gy-record
```

The following table describes the configurable parameters.

Parameter	Description
<code>write-records-to-disk-file-on-call-termination</code>	Specifies whether to enable or disable the Gy CCR-T file storage functionality used to store session usage information when the OCS is unavailable. ■ enabled – When the OCS is unavailable, Gy CCR-T records are stored in a predefined file and format based on the configuration. The stored CCR-T records are available for retrieval using SFTP. ■ disabled – (default) When the remote server is unavailable, the system does not store Gy CCR-T records in a file.
<code>gy-cdr-encode-format</code>	Specifies the format in which to encode the Gy CCR-T records in the file. custom1 – Gy CCR-T records are stored according to the custom1 format. See custom1 File Format (page 1-87) for more information. <i>Note: Visible in the CLI only when the <code>write-records-to-disk-file-on-call-termination</code> parameter is set to enabled.</i>
<code>gy-cdr-file</code>	Specifies the filename used to store Gy CCR-T records in the event of a failure. <i>Note: Visible in the CLI only when the <code>write-records-to-disk-file-on-call-termination</code> parameter is set to enabled.</i>
<code>gy-cdr-file-type</code>	Specifies the type of record file. In this case, select <code>gy-record</code>. <i>Note: Visible in the CLI only when the <code>write-records-to-disk-file-on-call-termination</code> parameter is set to enabled.</i>

Transferring Gy CCR-T Records Through North/South Interface

You can use a separate network-context loopback-ip to enable file-management export users to access Gy CCR-T records without having to connect through the standard management-context and mgmt-ip address.

You can enable and disable the network-context to act as an SFTP login port. Connections are made through the loopback-ip address on the north-south subnet using port 2221, for example:

```
sftp -P 2221 admin@<loopback-ip>
```

***Note:** Some export transfers may experience lower performance rates using this port.*

The infrastructure file-management rule requires that both an `export-rule` and `export-user` be defined for export users to access the network-context address through SFTP.

***Note:** Depending on the loopback-ip address you choose for the export-ip network context, you may need to create a default route on the client machine to forward traffic intended for the loopback-ip to the network-context's interface-ip address. The system will recognize this address as the correct route.*

Issue the following commands:

- 1 Create a loopback-ip on the network context:

```
an3000-mcm-slot7cpu1(config)# network-context <network-context-name>
loopback-ip <loopback-ip-name> ip-address <ip-address>
```

- 2 Associate the loopback-ip with the export object:

```
an3000-mcm-slot7cpu1(config)# infrastructure export-ip mgmtip
network-context <network-context-name> loopback-ip <loopback-ip-name>
```

- 3 Configure the export rule to apply when exporting the file to the `/export` directory.

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
<type> <filename-prefix> export-rule
[exportrule-default|exportrule-gy-custom1|exportrule-none]
```

Where:

- `exportrule-default` – The system exports files to the `/public/logs/datarecord.lnk/gy/export` directory and applies purge rules for this file profile to all files in the directory.

- `exportrule-gy-custom1` – The system exports files to the `/public/logs/datarecord.lnk/gy/export/AP_primary` directory according to the `custom1` format configured for CCR-T records. See the [Storing Gy CCR-T Records in a Local File \(page 1-84\)](#) for more information on this option.
 - `exportrule-none` – Specifies that there is no export rule for this file profile. In this case, the MCC does not store the Gy CCR-T records.
- 4 Specify the export user that has read/write access to the files in the `/export` directory.

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
<type> <filename-prefix> export-user <string>
```

Show Command

```
an3000-mcm-slot7cpu1# show running-config infrastructure export-ip
mgmtip network-context GN loopback-ip fmExportLoopback
```

Configuring Multiple Pairs of Primary and Secondary Peers

The MCC allows for the configuration of multiple primary active peers and secondary peers on the Gy interface within a Diameter server group. In this case, all of the primary active and secondary peers within a Diameter server group must share the same origin-host name and origin-realm (configured at the Diameter server-group level) and each peer is identified by a unique peer host name (configured at the peer level).

This feature adds two new CLI parameters that allow the origin-host and origin-realm to be configured at the peer level thereby allowing Operators to configure up to four pairs of peers (four primary peers and four secondary peers) on the Gy interface within a Diameter server group.

Figure 1-5 shows an example of the configuration used to configure multiple pairs of primary and secondary peers on the Gy interface. The new CLI parameters added at the peer level appear as red text.

Figure 1-5 Example Configuration for Multiple Pairs of Primary/Secondary Peers

Local Diameter Node Configuration

```
an3000-mcm-slot7cpu1(config)# service-construct interface aaa-interface-group diameter node-admin OCS
node-role client
origin-host defaultpeer.ocs.gwgroup
origin-realm ocs.gwgroup
network-context GI
loopback-ip OCSLOOP
transport tcp
port 3700
dscp-value dscp_0_be
connection-attempt-count 0
connection-attempt-timeout 3
cea-timeout 5
dwr-timer 15
dwr-retry-timer 1
dwr-retry-count 3
product-name AN-3000
vendor-id 10415
send-destination-host-avp true
```

Diameter Peer Configuration

Pair 1 <pre>peer OCS-Sim1.ocs.gwgroup peer-identifier 7 origin-host primarypeer1.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.148.3 dest-port 3868 transport tcp role primary backup-peer-name OCS-Sim2.ocs.gwgroup admin-state enabled override-dest-hostname PrimaryOCS.ocs.gwgroup</pre>	Pair 2 <pre>peer OCS-Sim3.ocs.gwgroup peer-identifier 8 origin-host primarypeer2.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.148.3 dest-port 3868 transport tcp role primary backup-peer-name OCS-Sim4.ocs.gwgroup admin-state enabled override-dest-hostname PrimaryOCS.ocs.gwgroup</pre>	Pair 3 <pre>peer OCS-Sim5.ocs.gwgroup peer-identifier 9 origin-host primarypeer3.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.148.3 dest-port 3868 transport tcp role primary backup-peer-name OCS-Sim6.ocs.gwgroup admin-state enabled override-dest-hostname PrimaryOCS.ocs.gwgroup</pre>	Pair 4 <pre>peer OCS-Sim7.ocs.gwgroup peer-identifier 10 origin-host primarypeer4.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.148.3 dest-port 3868 transport tcp role primary backup-peer-name OCS-Sim8.ocs.gwgroup admin-state enabled override-dest-hostname PrimaryOCS.ocs.gwgroup</pre>
<pre>peer OCS-Sim2.ocs.gwgroup peer-identifier 7 origin-host primarypeer1.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.158.3 dest-port 3869 transport tcp role secondary admin-state enabled override-dest-hostname SecondaryOCS.ocs.gwgroup</pre>	<pre>peer OCS-Sim4.ocs.gwgroup peer-identifier 8 origin-host primarypeer2.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.158.3 dest-port 3869 transport tcp role secondary admin-state enabled override-dest-hostname SecondaryOCS.ocs.gwgroup</pre>	<pre>peer OCS-Sim6.ocs.gwgroup peer-identifier 9 origin-host primarypeer3.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.158.3 dest-port 3869 transport tcp role secondary admin-state enabled override-dest-hostname SecondaryOCS.ocs.gwgroup</pre>	<pre>peer OCS-Sim8.ocs.gwgroup peer-identifier 10 origin-host primarypeer4.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.158.3 dest-port 3869 transport tcp role secondary admin-state enabled override-dest-hostname SecondaryOCS.ocs.gwgroup</pre>

Figure 1-5 uses the following configuration:

- All of the primary peers are configured with the same:
 - Destination host name (`override-dest-hostname`)
 - Destination IP address (`dest-ip-address`)
 - Destination port (`dest-port`)
- All of the secondary peers are configured with the same:
 - Destination host name (`override-dest-hostname`)
 - Destination IP address (`dest-ip-address`)
 - Destination port (`dest-port`)
- Each peer is configured with a unique peer host name (`peer`). The peer host name is used by the MCC for load balancing between peers and for statistics.
- The value configured for the destination host name (`override-dest-hostname`) for each peer overrides the configured peer host name (`peer`) in the Destination-Host AVP within CCR-I messages only.
- In CCR-U/CCR-T, the gateway uses the value from the Origin-Host AVP received in the CCA-I/CCA-U message to fill the Destination-Host AVP. In RAA messages, the gateway uses the value from the Origin-Host AVP received in the RAR message to fill the Destination-Host AVP.
- Each primary and secondary peer in a pair is configured with the same:
 - Origin-host name (`origin-host`)
 - Origin-realm (`origin-realm`)

The origin-host and origin-realm configured at the peer level overrides the origin-host and origin-realm configured at the Diameter server-group level.
- Each primary peer is configured with the secondary peer host name (`peer`) in the `backup-peer-name` parameter.
- In this configuration scenario, the CCR-I messages should include the Destination-Realm AVP but not the Destination-Host AVP. Only the CCR-U and CCR-T messages will have both the Destination-Realm and Destination-Host AVPs. To ensure the Destination-Host AVP is not included in Gx/Gy CCR-I messages, set the `presentFlag` setting to `no` in the CCR start data record templates.

```
an3000-mcm-slot7cpu1 (config)# service-construct
data-record-template [pgw|ggsn] [ocs|pcrf] diameter start
<template-name> destinationhost presentFlag no
```

Failure Handling Message Flows

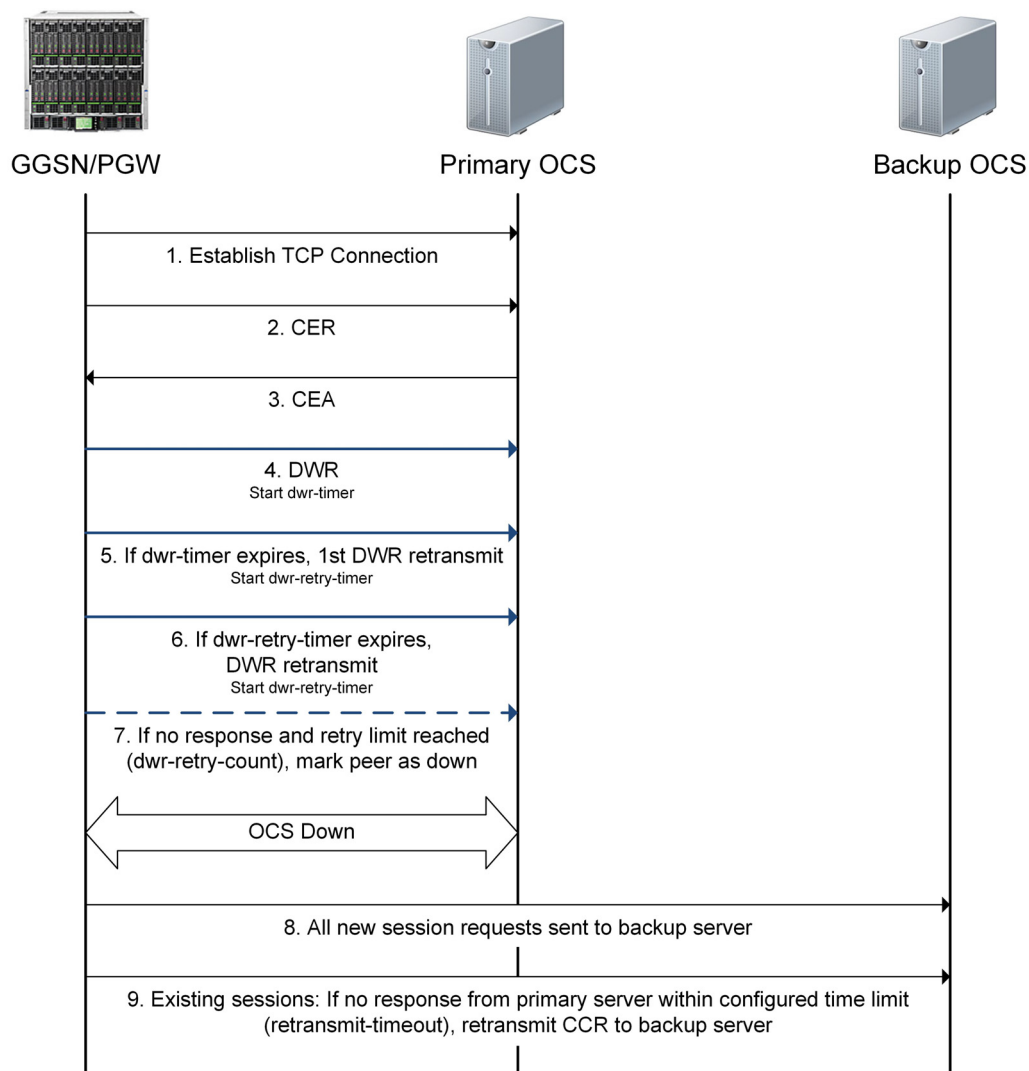
This section provides examples of the following OCS message flows:

- [Failover to Backup OCS \(Transport Failure\) \(page 1-97\)](#)
- [Failover to Backup OCS \(Application Failure\) \(page 1-99\)](#)
- [Failure Handling – Default Gy Quota Enabled \(page 1-101\)](#)
- [Failure Handling – Default Gy Quota and Gy Rejoin Enabled \(page 1-102\)](#)
- [Failure Handling – Gy Replay \(page 1-103\)](#)
- [Failure Handling – Session Terminate Timer Enabled \(Session Setup Failure\) \(page 1-105\)](#)
- [Failure Handling – Session Terminate Timer Enabled \(Mid-Session Failure\) \(page 1-106\)](#)

Failover to Backup OCS (Transport Failure)

Figure 1-6 demonstrates the message flow during a failover from the primary to the configured backup OCS when the OCS does not respond to DWRs. A failure at the transport level affects all sessions.

Figure 1-6 Failover to Backup OCS (Transport Failure)



- 1 The GGSN/PGW establishes a TCP Connection with the OCS.
- 2 The GGSN/PGW sends a Capabilities-Exchange-Request (CER) to the OCS.
- 3 Assuming both diameter peers share the same capabilities, such as diameter applications and security mechanisms, the OCS returns with a successful Capabilities-Exchange-Answer (CEA). The TCP connection is successfully established and the peers can begin servicing sessions.
- 4 Periodically, the GGSN/PGW sends DWRs to the OCS to monitor the TCP connection.

- 5 If the primary OCS does not respond to the first DWR with a DWA within the configured time limit (`dwr-timer`), the system retransmits the DWR and starts the DWR retry timer (`dwr-retry-timer`) with each retransmission.

Note: The `dwr-retry-timer` is typically set at a shorter interval than the `dwr-timer`.

- 6 If the primary OCS does not respond within the time limit (`dwr-retry-timer`), the GGSN/PGW retransmits the DWR. This cycle continues until the GGSN/PGW receives a valid DWA response from the primary OCS or the configured retry limit (`dwr-retry-count`) is reached.
- 7 If the configured retry limit (`dwr-retry-count`) is reached, the GGSN/PGW marks the primary OCS as down.
- 8 The GGSN/PGW sends all new CC session requests to the backup OCS.
- 9 For existing CC sessions, when the GGSN/PGW does not receive a CCA response from the primary OCS and the CCR timer (`retransmit-timeout`) expires, it retransmits the CCR to the backup server, if configured. The TCP connection with the backup OCS is pre-configured.

Note: If a CCR comes in before the `dwr-retry-timer` expires, the CCR timer (`retransmit-timeout`) itself can expire before the watchdog mechanism detects a failure.

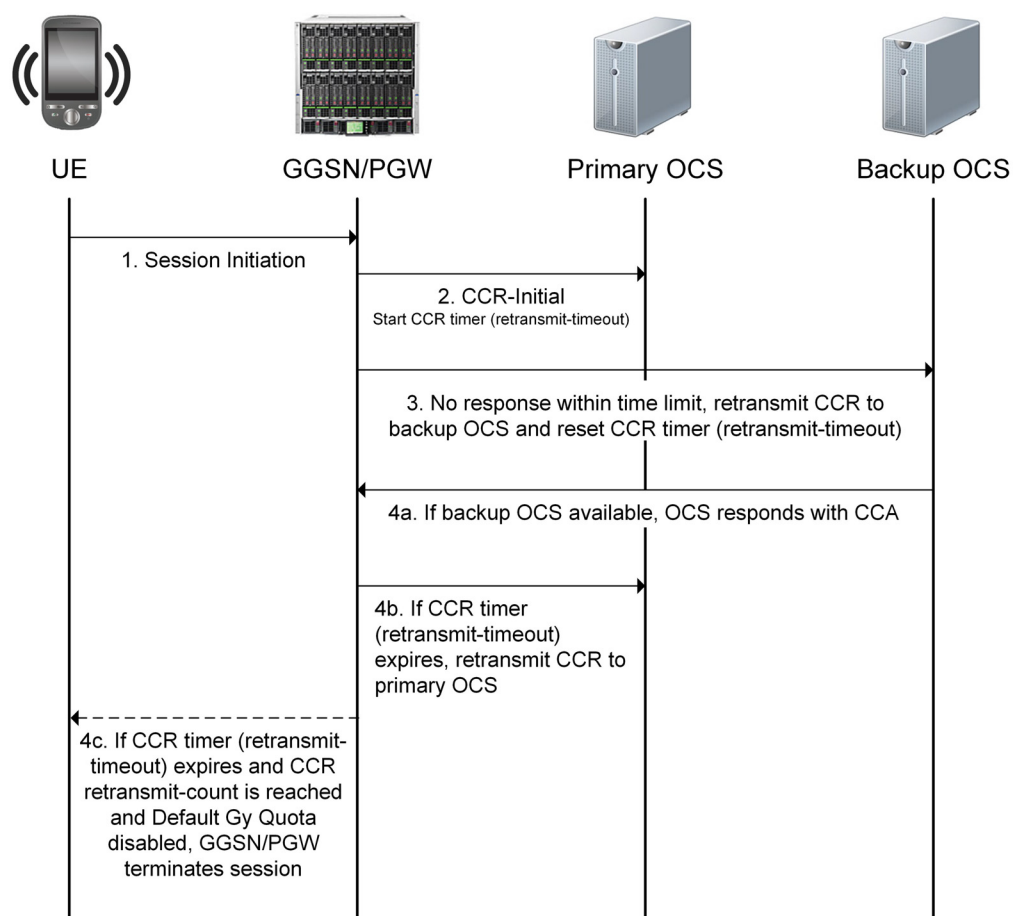
Note: If there is no available backup server and the failure occurs because the Watchdog mechanism expired, all new sessions fail to establish and all existing sessions are terminated.

Failover to Backup OCS (Application Failure)

Figure 1-7 demonstrates the message flow during a failover from the primary to the backup OCS when the CCR timer (`retransmit-timeout`) expires and the retransmit-count is reached before the GGSN/PGW receives the CCA message. In this case, the Default Gy Quota is disabled.

While a failure at the transport level affects all sessions, a failure at the application level only affects the session on which the CCR expiration occurs.

Figure 1-7 Failover to Backup OCS (Application Failure)

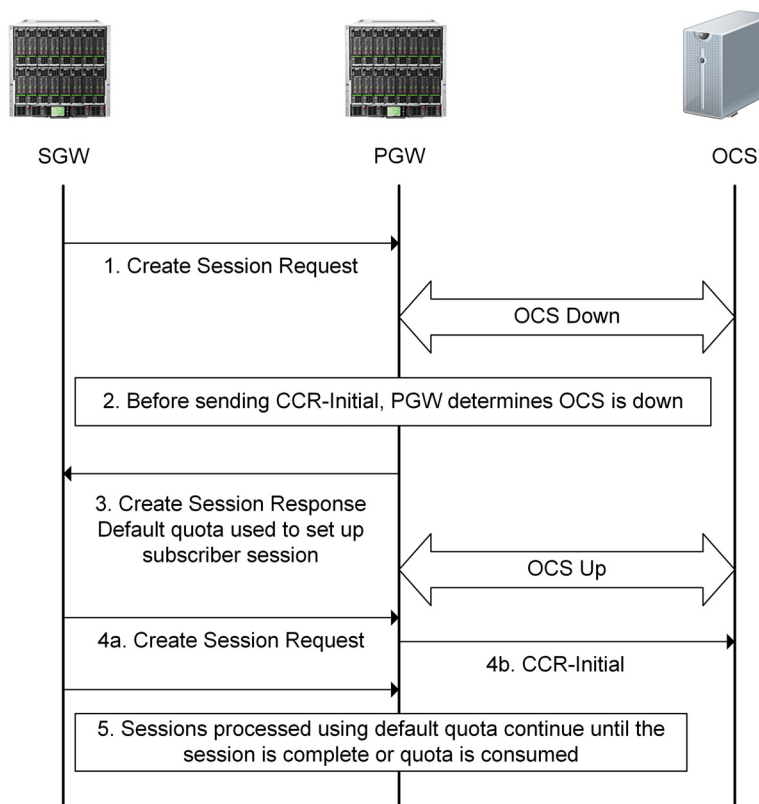


- 1 The User Equipment (UE) initiates a session with the GGSN/PGW.
- 2 The GGSN/PGW sends a CCR to the OCS requesting service units required for the session.
- 3 If the OCS does not respond to the first CCR within the configured time limit (`retransmit-timeout`), the GGSN/PGW retransmits the CCR to the backup OCS.
- 4 Depending on the response from the OCS, one of the following actions can occur:
 - a The backup OCS responds with a CCA message, if available.
 - b If there is no response from the backup OCS within the configured time limit (`retransmit-timeout`), the GGSN/PGW retransmits the CCR to the primary server once again. This cycle continues until the OCS responds with a CCA message or the CCR `retransmit-count` is reached.
 - c If there is no response from the backup OCS within the configured time limit and the CCR `retransmit-count` is reached and the Default Gy Quota feature is disabled, the GGSN/PGW terminates the session.

Failure Handling – Default Gy Quota Enabled

Figure 1-8 demonstrates the message flow for setting up a session when no OCS is available and the Default Gy Quota feature is enabled. In this case, the Gy Rejoin and Gy Replay features are disabled.

Figure 1-8 Failure Handling – Default Gy Quota Enabled



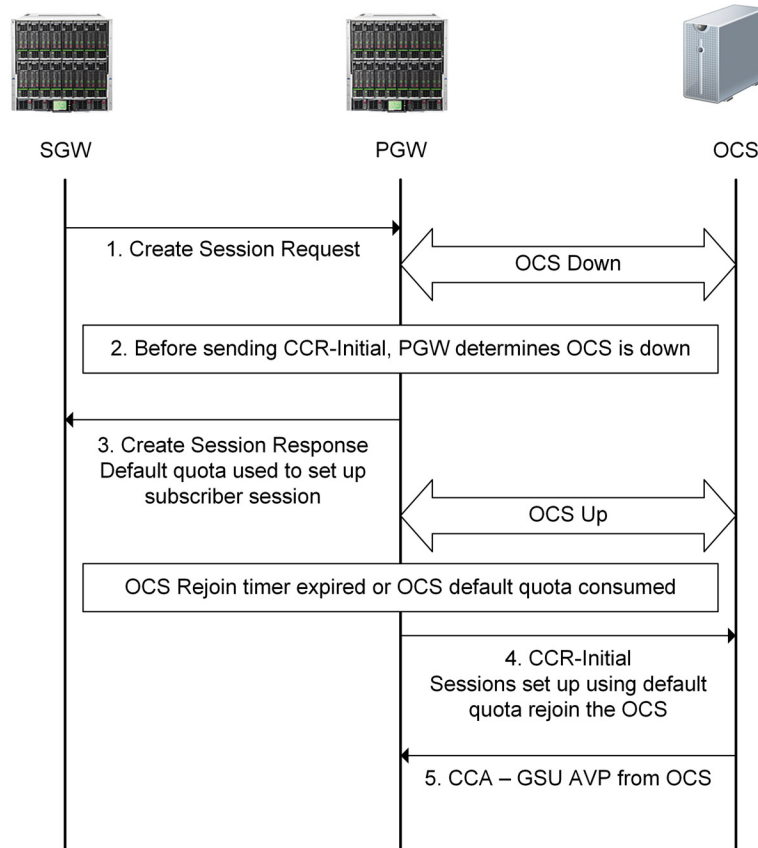
- 1 The SGW sends a Create Session Request to the PGW.
- 2 The PGW determines no OCS is available.
- 3 The PGW uses the locally configured default quota to set up subscriber sessions.
- 4 If the PGW receives a Create Session Request from the SGW and the OCS is available, the PGW sets up an online charging session by sending the OCS a CCR-I message.
- 5 Sessions processed using the default quota continue until the session is complete or the quota is consumed.

Note: Note that the subscriber usage consumed while a session is processed using the default quota is reported to the OCS when the subscriber “rejoins” the OCS. See [Reporting Default Gy Quota Subscriber Usage \(page 1-74\)](#) for more information.

Failure Handling – Default Gy Quota and Gy Rejoin Enabled

Figure 1-9 demonstrates the message flow for setting up a session when no OCS is available and Default Gy Quota and Gy Rejoin are enabled.

Figure 1-9 Failure Handling – Default Gy Quota and Gy Rejoin Enabled



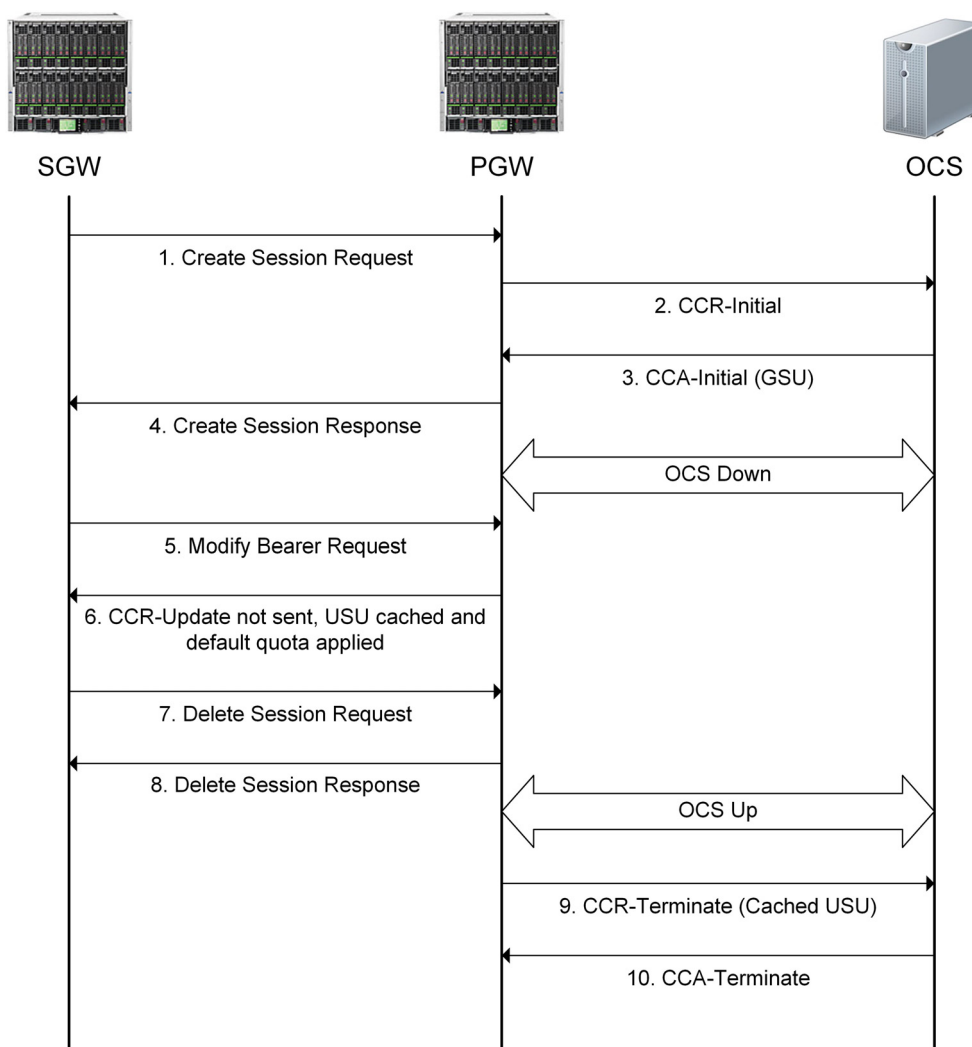
- 1 The SGW sends a Create Session Request to the PGW.
- 2 The PGW determines no OCS is available.
- 3 The PGW uses the locally configured default quota to set up subscriber sessions.
- 4 The PGW determines that the OCS is available and sessions are set up using the default quota rejoin the OCS via CCR-I message.
- 5 The OCS sends a CCA message to the PGW containing the Granted-Service-Unit (GSU) AVP.

Note: The subscriber usage consumed while a session is processed using the default quota is reported to the OCS when the subscriber “rejoins” the OCS. See [Reporting Default Gy Quota Subscriber Usage \(page 1-74\)](#) for more information.

Failure Handling – Gy Replay

Figure 1-10 demonstrates the message flow during an OCS failure on an established Gy session when Default Gy Quota, Gy Rejoin, and Gy Replay are enabled and the session is terminated before the OCS comes back up.

Figure 1-10 Failure Handling – Gy Replay Enabled



- 1 The SGW sends a Create Session Request to the PGW.
- 2 The PGW starts the session setup process by sending a CCR-I message to the OCS.
- 3 The OCS responds with a CCA message containing the GSU.
- 4 The PGW completes the session setup by sending the Create Session Response to the SGW.
- 5 During the session, the SGW sends a Modify Bearer Request to the PGW requesting more charging credits.

- 6 The PGW determines that the OCS is down and does not send the CCR-U, applies the default quota to the session and caches the USU.
- 7 The SGW sends a Delete Session Request to the PGW to terminate the session.
- 8 The PGW sends a Delete Session Response and the session is terminated.
- 9 Once the PGW determines that the OCS is back up, it sends a CCR-T message containing the cached USU based on the OCS grant.
- 10 The OCS responds with a CCA-T message.

Failure Handling – Session Terminate Timer Enabled (Session Setup Failure)

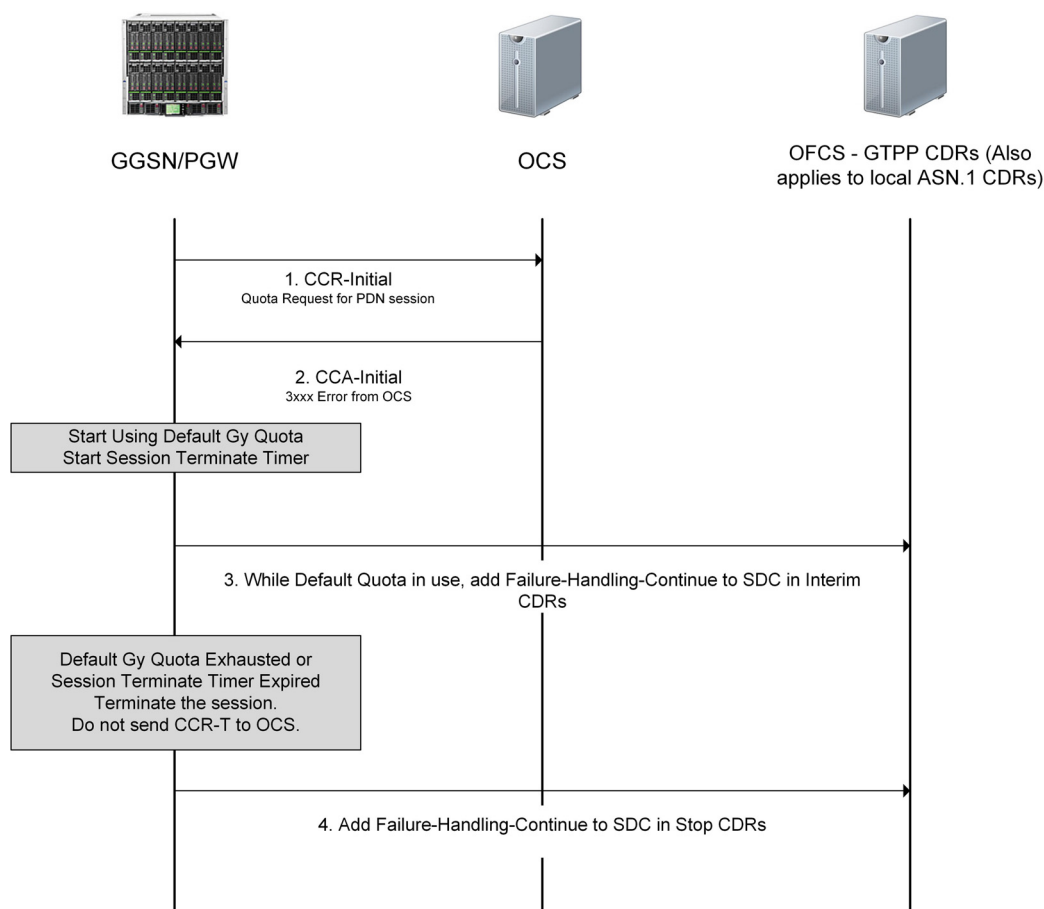
Figure 1-11 demonstrates the message flow for setting up a session using the default Gy quota when the session terminate timer (`ocs-down-session-terminate-timer`) is configured and the OCS returns a temporary failure (result code 3xxx).

Note the following:

- The Default Gy Quota (`ocs-down-default-gy-call-setup`) is enabled.
- The Diameter result code list (`gy-action-per-result-code`) is disabled.
- The Gy Rejoin feature (`ocs-down-gy-rejoin`) is disabled.

For configuration information, see [Terminating a Session Upon Depletion of Default Gy Quota \(page 1-74\)](#).

Figure 1-11 Failure Handling – Session Terminate Timer Enabled (Session Setup Failure)



Failure Handling – Session Terminate Timer Enabled (Mid-Session Failure)

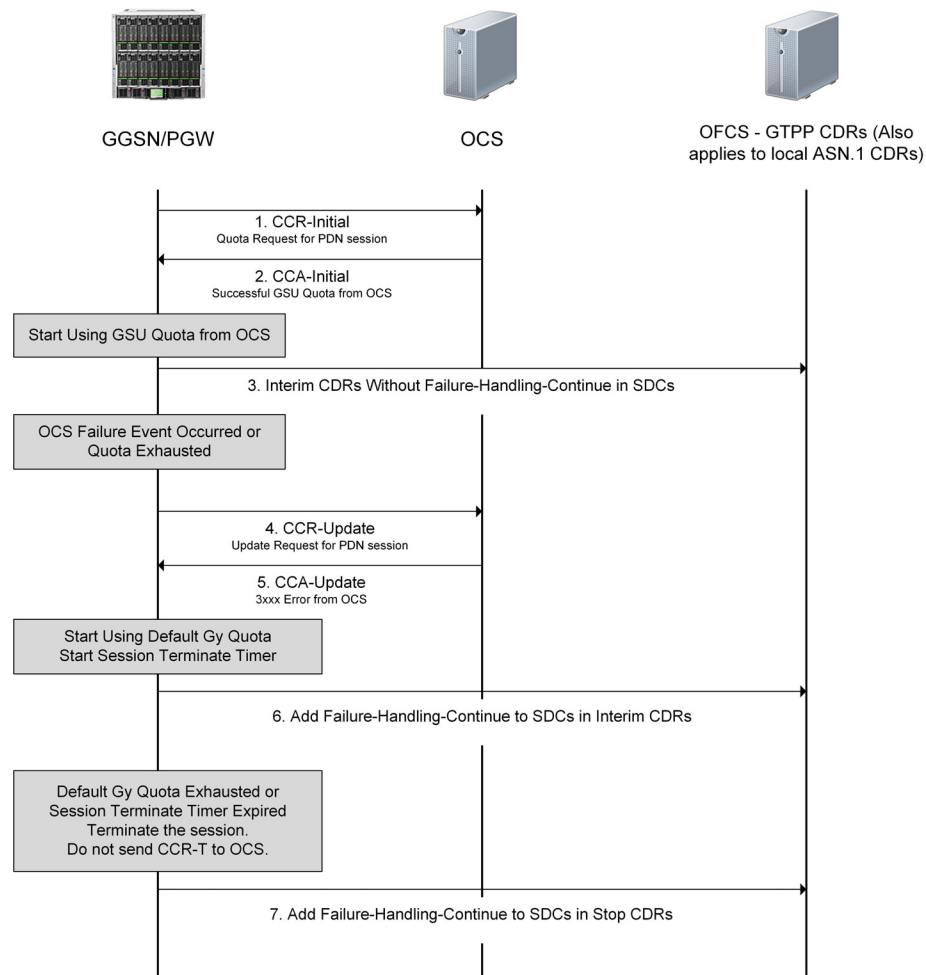
Figure 1-12 demonstrates the message flow when a temporary failure (result code 3xxx) occurs mid-session and the default Gy quota is used to complete the session and the session terminate timer (`ocs-down-session-terminate-timer`) is configured.

Note the following:

- The Default Gy Quota (`ocs-down-default-gy-call-setup`) is enabled.
- The Diameter result code list (`gy-action-per-result-code`) is disabled.
- The Gy Rejoin feature (`ocs-down-gy-rejoin`) is disabled.

For configuration information, see [Terminating a Session Upon Depletion of Default Gy Quota \(page 1-74\)](#).

Figure 1-12 Failure Handling – Session Terminate Timer Enabled (Mid-Session Failure)



Message Definitions

Table 1-4 describes the message definitions for the Gy interface.

Table 1-4 Gy Interface Message Definitions

Command	Acronym	Command Code	Reference
Abort-Session-Request (page 1-109)	ASR	274	RFC 3588
Abort-Session-Answer (page 1-110)	ASA	274	RFC 3588
Capabilities-Exchange-Request (page 1-111)	CER	257	RFC 3588
Capabilities-Exchange-Answer (page 1-112)	CEA	257	RFC 3588
Credit-Control-Request (page 1-113)	CCR	272	RFC 4006, 3GPP 32.299 V10.7.0, Vodafone Specifications
Credit-Control-Answer (page 1-123)	CCA	272	RFC 4006, RFC 3588, 3GPP 32.299 V10.7.0, Vodafone Specifications
Device-Watchdog-Request (page 1-127)	DWR	280	RFC 3588
Device-Watchdog-Answer (page 1-127)	DWA	280	RFC 3588
Re-Auth-Request (page 1-128)	RAR	258	RFC 4006, 3GPP 32.299 V10.7.0, Vodafone Specifications
Re-Auth-Answer (page 1-128)	RAA	258	RFC 4006, 3GPP 32.299 V10.7.0, Vodafone Specifications

Table 1-5 describes the special symbols in the AVP name for each message definition.

Table 1-5 AVP Name Symbols

Symbol	Description
<>	AVP is mandatory and the AVP with this symbol is at the beginning of a message.
{ }	AVP is mandatory.
[]	AVP is optional.
* []	AVP is optional and is repeated.
M	Mandatory item.
C	Optional item.
Om	Mandatory item defined by the carrier.
Oc	Optional item defined by the carrier.
–	Not supported.

Diameter AVPs have flags that indicate to the receiver how each AVP must be handled in Diameter messages. [Table 1-6](#) describes the AVP flag symbols.

Table 1-6 AVP Flag Symbols

Symbol	Description
M	The Mandatory “M” flag indicates that support for this AVP is required. If either the AVP or its value is not recognized, the message is not processed.
V	The Vendor-specific “V” flag indicates that this AVP has the optional Vendor ID field and when set, the AVP belongs to the specific vendor code address space.
M, V	Indicates that this AVP has both the mandatory “M” flag and vendor-specific “V” flag set.

Note: Operators can modify the default AVP flag values for Diameter AVPs on a specific Diameter interface (Gx, Gy, Gz, SWm, STa, and S6b). See [Modifying Default Diameter AVP Flags \(page 1-55\)](#) for more information.

Note: To control the sending (or not sending) of an optional Diameter AVP in the CCR and CCA messages, configure the OCS data record templates. See [Configuring the Gy Online Charging Interface \(page 1-11\)](#) for more information.

Abort-Session-Request

The OCS initiates session termination by sending an Abort-Session-Request (ASR).

Message Format:

```
<ASR> ::= < Diameter Header: 274, REQ, PXY >  
    <Session-Id>  
    {Origin-Host}  
    {Origin-Realm}  
    {Destination-Realm}  
    {Destination-Host}  
    {Auth-Application-Id}  
    [User-Name]  
    [Origin-State-Id]  
    *[Proxy-Info]  
    *[Route-Record]  
    *[AVP]
```

Abort-Session-Answer

The system sends an Abort-Session-Answer (ASA) in response to an ASR.

- The result code is set to DIAMETER_SUCCESS if the session corresponding to the Session-Id in an ASR message is successfully terminated.
- The system sends a result code DIAMETER_UNKOWN_SESSION_ID if the session corresponding to Session-Id in an ASR message is not found.
- The system sends a result code DIAMETER_UNABLE_TO_COMPLY if it does not terminate the session due to a certain reason, for example the Destination-Host in ASR does not match the system Local-Host.

Message format:

```
<ASA> ::= < Diameter Header: 274, PXY >
    <Session-Id>
    {Result-Code}
    {Origin-Host}
    {Origin-Realm}
    [User-Name]
    [Origin-State-Id]
    [Error-Message]
    [Error-Reporting-Host]
    *[Failed-AVP]
    *[Redirect-Host]
    [Redirect-Host-Usage]
    [Redirect-Max-Cache-Time]
    *[Proxy-Info]
    *[AVP]
```

Capabilities-Exchange-Request

The Capabilities-Exchange-Request (CER), indicated by the Command-Code set to 257 and the Command Flags 'R' bit set, is sent to exchange local capabilities. Upon detection of a transport failure, this message MUST NOT be sent to an alternate peer.

Message Format:

```
<CER> ::= < Diameter Header: 257, REQ >
    {Origin-Host}
    {Origin-Realm}
    1*{Host-IP-Address}
    {Vendor-Id}
    {Product-Name}
    [Origin-State-Id]
    *[Supported-Vendor-Id]
    *[Auth-Application-Id]
    *[Inband-Security-Id]
    *[Acct-Application-Id]
    *[Vendor-Specific-Application-Id]
    [Firmware-Revision]
    *[AVP]
```

Capabilities-Exchange-Answer

The Capabilities-Exchange-Answer (CEA), indicated by the Command-Code set to 257 and the Command Flags' 'R' bit cleared, is sent in response to a CER message.

Message Format:

```
<CEA> ::= < Diameter Header: 257 >
    {Result-Code}
    {Origin-Host}
    {Origin-Realm}
    1*{Host-IP-Address}
    {Vendor-Id}
    {Product-Name}
    [Origin-State-Id]
    [Error-Message]
    *[Failed-AVP]
    *[Supported-Vendor-Id]
    *[Auth-Application-Id]
    *[Inband-Security-Id]
    *[Acct-Application-Id]
    *[Vendor-Specific-Application-Id]
    [Firmware-Revision]
    *[AVP]
```

Credit-Control-Request

The Credit-Control-Request (CCR) is used for the DCCA application sent to the OCS.

Vodafone Variants

Message format:

```
<Credit-Control-Request> ::= <Diameter Header: 272, REQ, PXY>
    <Session-Id>
    {Origin-Host}
    {Origin-Realm}
    {Destination-Realm}
    {Auth-Application-Id}
    {Service-Context-Id}
    {CC-Request-Type}
    {CC-Request-Number}
    {Called-Station-ID}
    {3GPP-Charging-ID}
    {3GPP-Charging-Characteristics}
    {3GPP-GPRS-QoS-Negotiated-Profile}
    {3GPP-IMSI}
    {3GPP-NSAPI}
    {3GPP-Selection-Mode}
    {vf-Rulebase-ID}
    [Destination-Host]
    [Event-Timestamp]
    *[Subscription-Id]
    [Multiple-Services-Indicator]
    *[Multiple-Services-Credit-Control]
    [3GPP-SGSN-Address]
    [3GPP-SGSN-IPv6-Address]
    [3GPP-MS-TimeZone]
    [3GPP-SGSN-MCC-MNC]
    [3GPP-IMEISV]
    [3GPP-IMSI-MCC-MNC]
    [3GPP-GGSN-MCC-MNC]
    [3GPP-PDP-Type]
    [vf-Context-Type]
    [vf-Radio-Access-Technology]
    [vf-User-Location-Information]
    [User-Name]
    [Origin-State-Id]
    [Termination-Cause]
```

[Requested-Service-Unit]
 [Framed-IP-Address]
 * [Framed-IPv6-Prefix]
 [Framed-Interface-ID]
 [3GPP-GGSN-Address]
 [3GPP-GGSN-IPv6-Address]
 [3GPP-Session-Stop-Indicator]
 [Dynamic-Address-Flag]
 [PDN-Connection-Charging-ID]
 [QoS-Information]
 [IMSI-Unauthenticated-Flag]
 [Bearer-Usage]
 * [AVP]

Table 1-7 lists the Vodafone variant CCR message format. The AVPs not listed in the table are not supported. The system discards the unsupported AVPs and they are not sent. For a description of each AVP, see [Appendix A, Diameter AVP Descriptions](#).

Table 1-7 Vodafone Variant CCR Message Format

AVP Name	AVP Code	Initial	Update	Terminate	AVP Flag
<Session-ID>	263	M	M	M	M
{Origin-Host}	264	M	M	M	M
{Origin-Realm}	296	M	M	M	M
{Destination-Realm}	283	M	M	M	M
{Auth-Application-Id}	258	M	M	M	M
{Route-Record}	461	M	M	M	M
{CC-Request-Type}	416	M	M	M	M
{CC-Request-Number}	415	M	M	M	M
{Called-Station-Id}	30	Om	Om	Om	M
{3GPP-Charging-Id}	2	Om	Om	Om	M,V
{3GPP-Charging-Characteristics}	13	Om	Om	Om	M,V
{3GPP-GPRS-Negotiated-QoS-Profile}	5	Om	Om	Om	M,V
{3GPP-IMSI}	1	Om	Om	Om	M,V
{3GPP-NSAPI}	10	Om	Om	Om	M,V
{3GPP-Selection-Mode}	12	Om	Om	Om	M,V
{vf-Rulebase-Id}	262	Om	Om	Om	M,V

Table 1-7 Vodafone Variant CCR Message Format

AVP Name	AVP Code	Initial	Update	Terminate	AVP Flag
[Destination-Host]	293	Oc	Oc	Oc	M
[Event-Timestamp]	55	Oc	Oc	Oc	M
*[Subscription-Id]	443	Om	Om	Om	M
{Subscription-Id-Type}	450	M	M	M	M
{Subscription-Id-Data}	444	M	M	M	M
[Multiple-Services-Indicator]	455	Oc	Oc	Oc	M
*[Multiple-Services-Credit-Control]	456	Oc	Oc	Oc	M
[Requested-Service-Unit]	437	-	Oc	-	M
[CC-Time]	420	-	Oc	-	M
[CC-Total-Octets]	421	-	Oc	-	M
[CC-Service-Specific-Units]	417	-	Oc	-	M
*[Used-Service-Unit]	446	-	Oc	Oc	M
[Tariff-Change-Usage]	452	-	Oc	Oc	M
[CC-Time]	420	-	Oc	Oc	M
[CC-Total-Octets]	421	-	Oc	Oc	M
[CC-Input-Octets]	412	-	Oc	Oc	M
[CC-Output-Octets]	414	-	Oc	Oc	M
[CC-Service-Specific-Units]	417	-	Oc	Oc	M
[vf-Reporting-Reason]	261	-	Oc	Oc	M,V
[Rating-Group]	432	Oc	Oc	Oc	M
[vf-Time-Of-First-Usage]	263	-	Oc	Oc	M,V
[vf-Time-Of-Last-Usage]	264	-	Oc	Oc	M,V
*[vf-Envelopes]	269	-	Oc	Oc	M,V
[vf-Envelope-Start-Time]	270	-	Oc	Oc	M,V
[vf-Envelope-End-Time]	271	-	Oc	Oc	M,V
[CC-Total-Octets]	421	-	Oc	Oc	M
[CC-Input-Octets]	412	-	Oc	Oc	M
[CC-Output-Octets]	414	-	Oc	Oc	M

Table 1-7 Vodafone Variant CCR Message Format

AVP Name	AVP Code	Initial	Update	Terminate	AVP Flag
[CC-Service-Specific-Units]	417	-	Oc	Oc	M
[3GPP-SGSN-Address]	6	Oc	Oc	Oc	M,V
[3GPP-SGSN-IPv6-Address]	15	Oc	Oc	Oc	M,V
[3GPP-MS-TimeZone]	23	Oc	Oc	Oc	M,V
[3GPP-SGSN-MCC-MNC]	18	Oc	Oc	Oc	M,V
[3GPP-IMEISV]	20	Oc	Oc	Oc	M,V
[3GPP-IMSI-MCC-MNC]	8	Oc	Oc	Oc	M,V
[3GPP-GGSN-MCC-MNC]	9	Oc	Oc	Oc	M,V
[3GPP-PDP-Type]	3	Oc	Oc	Oc	M,V
[vf-Context-Type]	256	Oc	Oc	Oc	M,V
[vf-Radio-Access-Technology]	260	Oc	Oc	Oc	M,V
[vf-User-Location-Information]	267	Oc	Oc	Oc	M,V
[User-Name]	1	Oc	Oc	Oc	M
[Origin-State-Id]	278	Oc	Oc	Oc	M
[Termination-Cause]	295	-	-	Oc	M
[Requested-Service-Unit]	437	Oc	-	-	M
[Framed-IP-Address]	8	Oc	Oc	Oc	M
*[Framed-IPv6-Prefix]	97	Oc	Oc	Oc	M
[Framed-Interface-Id]	96	Oc	Oc	Oc	M
[3GPP-GGSN-Address]	7	Oc	Oc	Oc	M,V
[3GPP-GGSN-IPv6-Address]	16	Oc	Oc	Oc	M,V
[3GPP-Session-Stop-Indicator]	11	-	-	Oc	M,V
[Dynamic-Address-Flag]	2051	Oc	Oc	Oc	M,V
[PDN-Connection-Charging-ID]	2050	Oc	Oc	Oc	M,V
[QoS-Information]	1016	Oc	Oc	Oc	M,V
[IMSI-Unauthenticated-Flag]	2308	Oc	Oc	Oc	M,V
[Bearer-Usage]	1000	Oc	Oc	Oc	M,V

3GPP Variant

Message format:

```
<CCR> ::= < Diameter Header: 272, REQ, PXY >  
    <Session-Id>  
    {Origin-Host}  
    {Origin-Realm}  
    {Destination-Realm}  
    {Auth-Application-Id}  
    {Service-Context-Id}  
    {CC-Request-Type}  
    {CC-Request-Number}  
    [Destination-Host]  
    [User-Name]  
    [Origin-State-Id]  
    [Event-Timestamp]  
    *[Subscription-Id]  
    [Termination-Cause]  
    [Requested-Action]  
    [Multiple-Services-Indicator]  
    *[Multiple-Services-Credit-Control]  
    [CC-Correlation-Id]  
    [User-Equipment-Info]  
    *[Route-Record]  
    [Service-Information]  
    *[AVP]
```

Table 1-8 lists the 3GPP CCR message format in accordance with 3GPP TS 32.299 V10.7.0. The AVPs not listed in the table are not supported. The system discards the unsupported AVPs and they are not sent. For a description of each AVP, see [Appendix A, Diameter AVP Descriptions](#). The AVPs with the vendor-specific “V” flag set include a Vendor-ID of 10415 (3GPP).

Table 1-8 3GPP CCR message format

AVP Name	AVP Code	Initial	Update	Terminate	AVP Flag
<Session-ID>	263	M	M	M	M
{Origin-Host}	264	M	M	M	M
{Origin-Realm}	296	M	M	M	M
{Destination-Realm}	283	M	M	M	M
{Auth-Application-Id}	258	M	M	M	M
{Route-Record}	461	M	M	M	M
{CC-Request-Type}	416	M	M	M	M
{CC-Request-Number}	415	M	M	M	M
[Destination-Host]	293	Oc	Oc	Oc	M
[User-Name]	1	Oc	Oc	Oc	M
[Origin-State-Id]	278	Oc	Oc	Oc	M
[Event-Timestamp]	55	Oc	Oc	Oc	M
*[Subscription-Id]	443	Om	Om	Om	M
{Subscription-Id-Type}	450	M	M	M	M
{Subscription-Id-Data}	444	M	M	M	M
[Termination-Cause]	295	-	-	Oc	M
[Requested-Action]	436	Oc	Oc	Oc	M
[Multiple-Services-Indicator]	455	Oc	Oc	Oc	M
[Multiple-Services-Credit-Control]	456	Om	Om	Om	M
[Requested-Service-Unit]	437	-	Oc	-	M
[CC-Time]	420	-	Oc	-	M
[CC-Total-Octets]	421	-	Oc	-	M
[CC-Service-Specific-Units]	417	-	Oc	-	M
*[Used-Service-Unit]	446	-	Oc	Oc	M

Table 1-8 3GPP CCR message format

AVP Name	AVP Code	Initial	Update	Terminate	AVP Flag
[Tariff-Change-Usage]	452	-	Oc	Oc	M
[CC-Time]	420	-	Oc	Oc	M
[CC-Total-Octets]	421	-	Oc	Oc	M
[CC-Input-Octets]	412	-	Oc	Oc	M
[CC-Output-Octets]	414	-	Oc	Oc	M
[CC-Service-Specific-Units]	417	-	Oc	Oc	M
[Reporting-Reason]	872	-	Oc	Oc	M,V
*[Service-Identifier]	439	-	Oc	Oc	M
[Rating-Group]	432	Oc	Oc	Oc	M
[Reporting-Reason]	872	-	Oc	Oc	M,V
*[AF-Correlation-Information]	1276	Oc	Oc	Oc	M,V
[AF-Charging-Identifier]	505	Oc	Oc	Oc	M,V
*[Flows]	510	Oc	Oc	Oc	M,V
*[Envelope]	1266	-	Oc	Oc	M,V
[Envelope-Start-Time]	1269	-	Oc	Oc	M,V
[Envelope-End-Time]	1267	-	Oc	Oc	M,V
[CC-Total-Octets]	421	-	Oc	Oc	M
[CC-Input-Octets]	412	-	Oc	Oc	M
[CC-Output-Octets]	414	-	Oc	Oc	M
[CC-Service-Specific-Units]	417	-	Oc	Oc	M
[User-Equipment-Info]	458	Oc	Oc	Oc	M
{User-Equipment-Info-Type}	459	Oc	Oc	Oc	M
{User-Equipment-Info-Value}	460	Oc	Oc	Oc	M
*[Rule-Activation-Time]	282	Oc	Oc	Oc	M,V
[Service-Information]	873	C	C	C	M,V
[PS-Information]	874	C	C	C	M,V
[3GPP-Charging-Id]	2	Om	Om	Om	M,V
[PDN-Connection-Charging-ID]	2050	Oc	Oc	Oc	M,V

Table 1-8 3GPP CCR message format

AVP Name	AVP Code	Initial	Update	Terminate	AVP Flag
[3GPP-PDP-Type]	3	Oc	Oc	Oc	M,V
*[PDP-Address]	1227	Oc	Oc	Oc	M,V
[Dynamic-Address-Flag]	2051	Oc	Oc	Oc	M,V
[Session-Release-Cause]	1228	Oc	Oc	Oc	M,V
[GGSN-Address]	847	Oc	Oc	Oc	M,V
[CG-Address]	846	Oc	Oc	Oc	M,V
[Serving-Node-Type]	2047	Oc	Oc	Oc	M,V
[3GPP-IMSI-MCC-MNC]	18	Oc	Oc	Oc	M,V
[IMSI-Unauthenticated-Flag]	2308	Oc	Oc	Oc	M,V
[3GPP-GGSN-MCC-MNC]	9	Oc	Oc	Oc	M,V
[3GPP-NSAPI]	10	Oc	Oc	Oc	M,V
[Called-Station-Id]	30	Oc	Oc	Oc	M
[3GPP-Session-Stop-Indicator]	11	-	-	Oc	M,V
[3GPP-Selection-Mode]	12	Oc	Oc	Oc	M,V
[3GPP-Charging-Characteristics]	13	Oc	Oc	Oc	M,V
[3GPP-SGSN-MCC-MNC]	18	Oc	Oc	Oc	M,V
[3GPP-MS-TimeZone]	23	Oc	Oc	Oc	M,V
[Charging-Rule-Base-Name]	1004	Oc	Oc	Oc	M,V
[3GPP-User-Location-Info]	22	Oc	Oc	Oc	M,V
[TWAN-User-Location-Info]	2714	Oc	-	Oc	M,V
[QoS-Information]	1016	Oc	Oc	Oc	M,V
[Max-Requested-Bandwidth-DL]	515	Oc	Oc	Oc	M,V
[Max-Requested-Bandwidth-UL]	516	Oc	Oc	Oc	M,V
[Guaranteed-Bitrates-DL]	1025	Oc	Oc	Oc	M,V
[Guaranteed-Bitrates-UL]	1026	Oc	Oc	Oc	M,V
[APN-Aggregate-Max-Bitrates-DL]	1040	Oc	Oc	Oc	V
[APN-Aggregate-Max-Bitrates-UL]	1041	Oc	Oc	Oc	V
[Extended-Max-Requested-BW-DL]	554	Oc	Oc	Oc	V

Table 1-8 3GPP CCR message format

AVP Name	AVP Code	Initial	Update	Terminate	AVP Flag
[Extended-Max-Requested-BW-UL]	555	Oc	Oc	Oc	V
[Extended-GBR-DL]	2850	Oc	Oc	Oc	V
[Extended-GBR-UL]	2851	Oc	Oc	Oc	V
[Extended-APN-AMBR-DL]	2848	Oc	Oc	Oc	V
[Extended-APN-AMBR-UL]	2849	Oc	Oc	Oc	V
[3GPP-RAT-Type]	21	Oc	Oc	Oc	M,V
[PDP-Context-Type]	1247	Oc	Oc	Oc	M,V
[SGi-PtP-Tunnelling-Method]	3931	Oc	Oc	Oc	M,V
[UNI-PDU-CP-Only-Flag]	3932	Oc	Oc	Oc	M,V
[Serving-PLMN-Rate-Control]	4310	Oc	Oc	-	M,V
[APN-Rate-Control]	3933	Oc	Oc	-	M,V
[SCS-AS-Address]	3940	Oc	Oc	Oc	M,V

[Table 1-9](#) lists the vendor-specific attributes (VSAs) for CCR messages. For a description of each attribute, see [Appendix A, Diameter AVP Descriptions](#).

Table 1-9 CCR Vendor-Specific Attributes

AVP Name	Vendor ID	AVP Code	Initial	Update	Terminate
[Origination-Time-Stamp]	12951	7102	-	Oc	-
[Max-Wait-Time]	12951	7103	-	Oc	-
[MVNO-Reseller-Id]	9	131507	Oc	Oc	Oc
[MVNO-Sub-Class-Id]	9	131508	Oc	Oc	Oc
[TMO-Transparent-Data]	29168	1	Oc	Oc	Oc
[TMO-Virtual-Gi-ID]	29168	15	Oc	-	-

Credit-Control-Answer

The OCS sends a Credit-Control-Answer (CCA) in response to a CCR.

Vodafone Variants

Message format:

```
<Credit-Control-Answer> ::= <Diameter Header: 272, PXY>
    <Session-ID>
    {Result-Code}
    {Origin-Host}
    {Origin-Realm}
    {Auth-Application-Id}
    {CC-Request-Type}
    {CC-Request-Number}
    [CC-Session-Failover]
    *[Subscription-Id]
    *[Multiple-Services-Credit-Control]
    [Credit-Control-Failure-Handling]
    [vf-Rulebase-ID]
    *[AVP]
```

Table 1-10 lists the Vodafone variant CCA message format. For a description of each AVP, see [Appendix A, Diameter AVP Descriptions](#).

Table 1-10 Vodafone Variant CCA Message Format

AVP Name	AVP Code	Initial	Update	Terminate
<Session-ID>	263	M	M	M
{Result-Code}	268	M	M	M
{Origin-Host}	264	M	M	M
{Origin-Realm}	296	M	M	M
{Auth-Application-Id}	258	M	M	M
{CC-Request-Type}	416	M	M	M
{CC-Request-Number}	415	M	M	M
[CC-Session-Failover]	418	Oc	-	-
*[Subscription-Id]	443	Om	Om	Om
{Subscription-Id-Type}	450	M	M	M
{Subscription-Id-Data}	444	M	M	M

Table 1-10 Vodafone Variant CCA Message Format

AVP Name	AVP Code	Initial	Update	Terminate
*[Multiple-Services-Credit-Control]	456	-	Oc	-
[Granted-Service-Unit]	431	-	Oc	-
[Tariff-Time-Change]	451	-	Oc	-
[CC-Time]	420	-	Om	-
[CC-Total-Octets]	421	-	Oc	-
[CC-Service-Specific-Units]	417	-	Oc	-
[Result-Code]	268	-	Oc	-
[Final-Unit-Indication]	430	-	Oc	-
{Final-Unit-Action}	449	-	Oc	-
*[Filter-ID]	11	-	Oc	-
[Redirect-Server]	434	-	Oc	-
[Rating-Group]	432	-	Oc	-
[Validity-Time]	448	-	Oc	-
[Tariff-Change-Usage]	452	-	Oc	-
[vf-Quota-Consumption-Time]	257	-	Oc	-
[vf-Quota-Holding-Time]	258	-	Oc	-
[vf-Time-Quota-Threshold]	259	-	Oc	-
[vf-Trigger]	265	-	Oc	-
[vf-Trigger-Type]	266	-	Oc	-
[vf-Volume-Quota-Threshold]	268	-	Oc	-
[vf-Envelope-Reporting]	272	-	Oc	-
[vf-Unit-Quota-Threshold]	273	-	Oc	-
[vf-Time-Quota-Mechanism]	274	-	Oc	-
[vf-Time-Quota-Type]	275	-	Oc	-
[vf-Base-Time-Interval]	276	-	Oc	-
[Credit-Control-Failure-Handling]	427	Oc	-	-
{vf-Rulebase-Id}	262	Om	-	-

3GPP Variants

Message format:

```
<CCA> ::= < Diameter Header: 272, PXY >
    <Session-Id>
    {Result-Code}
    {Origin-Host}
    {Origin-Realm}
    {Auth-Application-Id}
    {CC-Request-Type}
    {CC-Request-Number}
    [CC-Session-Failover]
    *[Multiple-Services-Credit-Control]
    [Credit-Control-Failure-Handling]
    [PS-Furnish-Charging-Information]
    [Service-Information]
    *[AVP]
```

Table 1-11 lists the 3GPP CCA message format in accordance with 3GPP TS 32.299 V10.7.0. For a description of each AVP, see [Appendix A, Diameter AVP Descriptions](#).

Table 1-11 3GPP CCA Message Format

AVP Name	AVP Code	Initial	Update	Terminate
<Session-ID>	263	M	M	M
{Result-Code}	268	M	M	M
{Origin-Host}	264	M	M	M
{Origin-Realm}	296	M	M	M
{Acct-Application-Id}	258	M	M	M
{CC-Request-Type}	416	M	M	M
{CC-Request-Number}	415	M	M	M
[CC-Session-Failover]	418	Oc	-	-
*[Multiple-Services-Credit-Control]	456	-	Oc	-
[Granted-Service-Unit]	431	-	Oc	-
[Tariff-Time-Change]	451	-	Oc	-
[CC-Time]	420	-	Om	-
[CC-Total-Octets]	421	-	Oc	-
[CC-Service-Specific-Units]	417	-	Oc	-

Table 1-11 3GPP CCA Message Format

AVP Name	AVP Code	Initial	Update	Terminate
[Result-Code]	268	-	Oc	-
[Final-Unit-Indication]	430	Oc	Oc	-
{Final-Unit-Action}	449	Oc	Oc	-
[Redirect-Server]	434	Oc	Oc	-
[Rating-Group]	432	-	Oc	-
[Validity-Time]	448	-	Oc	-
[Tariff-Change-Usage]	452	-	Oc	-
[Quota-Consumption-Time]	881	Oc	Oc	-
[Quota-Holding-Time]	871	Oc	Oc	-
[Time-Quota-Threshold]	868	Oc	Oc	-
[Trigger]	1264	Oc	Oc	-
[Trigger-Type]	870	Oc	Oc	-
[PS-Furnish-Charging-Information]	865	Oc	Oc	-
[Volume-Quota-Threshold]	869	Oc	Oc	-
[Envelope-Reporting]	1268	Oc	Oc	-
[Unit-Quota-Threshold]	1226	Oc	Oc	-
[Time-Quota-Mechanism]	1270	Oc	Oc	-
[Time-Quota-Type]	1271	Oc	Oc	-
[Base-Time-Interval]	1265	Oc	Oc	-
[Credit-Control-Failure-Handling]	427	Oc	-	-
[PS-Furnish-Charging-Information]	865	Oc	Oc	-
[Service-Information]	873	Oc	Oc	-
[PS-Information]	874	Oc	Oc	-
[Final-Unit-Indication]	430	Oc	Oc	-

Disconnect-Peer-Request

The system sends a Disconnect-Peer-Request (DPR) to gracefully bring down the connection with the OCS when the online charging interface is taken down administratively.

Message format:

```
< Disconnect-Peer-Request > ::= <Diameter Header: 282, REQ>
    {Origin-Host}
    {Origin-Realm}
    {Disconnect-Cause}
```

Disconnect-Peer-Answer

The OCS is expected to send a Disconnect-Peer-Answer (DPA) in response to a DPR. The system brings down the connection on receipt of the DPA.

Message format:

```
< Disconnect-Peer -Answer> ::= <Diameter Header: 282>
    {Result-Code}
    {Origin-Host}
    {Origin-Realm}
    [Error-Message]
    *[Failed-AVP]
```

Device-Watchdog-Request

The system sends a Device-Watchdog-Request (DWR) to the OCS at configured intervals.

Message format:

```
<Device-Watchdog-Request> ::= <Diameter Header: 280, REQ>
    {Origin-Host}
    {Origin-Realm}
    [Origin-State-Id]
```

Device-Watchdog-Answer

The system expects to receive a Device-Watchdog-Answer (DWA) to complete the DWR request it sends out to the OCS.

Message format:

```
<Device-Watchdog-Answer> ::= <Diameter Header: 280>
    {Result-Code}
    {Origin-Host}
    {Origin-Realm}
    [Error-Message]
    *[Failed-AVP]
    [Original-State-Id]
```

Re-Auth-Request

The OCS initiates the credit re-authorization using a Re-Auth-Request (RAR) with the Re-Auth-Request-Type set to AUTHORIZE_ONLY (0).

Message format:

```
<Re-Auth-Request> ::= <Diameter Header: 258, REQ, PXY>
    <Session-Id>
    {Origin-Host}
    {Origin-Realm}
    {Destination-Realm}
    {Destination-Host}
    {Auth-Application-Id}
    {Re-Auth-Request-Type}
    [Origin-State-Id]
    [Rating-Group]
    *[AVP]
```

Re-Auth-Answer

The system sends a Re-Auth-Answer (RAA) in response to an RAR. The result code is set to DIAMETER_SUCCESS if the system can successfully process the RAR.

- By default, the system sends result code DIAMETER_SUCCESS (2001) if it can successfully process the RAR. However, Operators can configure the PGW to instead send result code DIAMETER_LIMITED_SUCCESS (2002) in the RAA message to indicate success.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name> reauth-response-success-
result-code [success_2001|limited_success_2002]
```

- The system sends result code DIAMETER_UNKOWN_SESSION_ID if the session corresponding to Session-Id in an RAR message is not found.
- The system sends result code DIAMETER_UNABLE_TO_COMPLY if it does not terminate the session for a certain reason, for example the Destination-Host in ASR does not match the system Local-Host.

Message format:

```
<Re-Auth-Answer> ::= <Diameter Header: 258, PXY>
    <Session-Id>
    {Result-Code}
    {Origin-Host}
    {Origin-Realm}
    [Origin-State-Id]
    *[AVP]
```

Result Codes

The Result-Code AVP specifies whether a request has successfully completed or failed. The system interprets an online-charging request as successful if a diameter peer returns a request with the Result-Code AVP value of 2001. However, a result code other than 2001 is considered unsuccessful. [Table 1-12](#) summarizes the supported result codes that indicate a request has failed.

Note: Operators can configure the MCC to take specific action based on the result code received in the CCA message over the Gy interface. See [Configuring the Diameter Result Code List \(page 1-78\)](#) for more information.

Note: By default, the system sends result code `DIAMETER_SUCCESS (2001)` if it can successfully process the RAR. However, Operators can configure the PGW to instead send result code `DIAMETER_LIMITED_SUCCESS (2002)` in the RAA message to indicate success. See [Re-Auth-Answer \(page 1-128\)](#).

Table 1-12 Supported Result Codes

Procedure	Value	Name	Description	Action
Initial	3002	DIAMETER_UNABLE_TO_DELIVER	The diameter device cannot deliver the message to the desired destination because the host described in the Destination-Host AVP does not match a host within the associated realm.	See Failure Handling (page 1-68) for more information.
Initial	3004	DIAMETER_TOO_BUSY	The diameter device cannot deliver the message to the desired destination.	See Failure Handling (page 1-68) for more information.
Initial	3005	DIAMETER_LOOP_DETECTED	A diameter agent detected a loop while trying to get the message to the desired destination.	See Failure Handling (page 1-68) for more information.
Initial	4011	DIAMETER_CREDIT_CONTROL_NOT_APPLICABLE	The diameter server determines that the service can proceed without OCS processing. For example, the requested service is free of charge.	The system overrides online charging and sets up the call without using OCS procedures.
Update	3002	DIAMETER_UNABLE_TO_DELIVER	The diameter device cannot deliver the message to the desired destination because the host described in the Destination-Host AVP does not match a host within the associated realm.	See Failure Handling (page 1-68) for more information.
Update	3004	DIAMETER_TOO_BUSY	The diameter device cannot deliver the message to the desired destination.	See Failure Handling (page 1-68) for more information.

Table 1-12 Supported Result Codes

Procedure	Value	Name	Description	Action
Update	3005	DIAMETER_LOOP_DETECTED	A diameter agent detected a loop while trying to get the message to the desired destination.	See Failure Handling (page 1-68) for more information.
Update	4001	DIAMETER_AUTHENTICATION_REJECTED	The user authentication process failed, most likely because the user provided an invalid password.	The system logs the event, updates statistics, sets the GPRS Tunneling Protocol (GTP) result to failure and terminates the session without OCS processing.
Update	4010	DIAMETER_END_USER_SERVICE_DENIED	The OCS denied the service request due to service restrictions. If the CCR contains Used-Service-Units, they are deducted from the account.	The system logs the event, updates statistics, sets the GTP result to failure and terminates the session.
Update	4012	DIAMETER_CREDIT_LIMIT_REACHED	The OCS denied the request because the subscriber's account does not contain sufficient credit. If the CCR contains Used-Service-Units, they are deducted from the account.	The system logs the event, updates statistics, sets the GTP result to failure and terminates the session. If this occurs while processing MSCC AVPs, the system requests a final report from the data plane (fastpath application).
Update	5003	DIAMETER_AUTHORIZATION_REJECTED	The OCS could not authorize the user's request.	The system blacklists the rating group and starts a timer. This rating group cannot be used for the duration of the timer.
Update	5030	DIAMETER_USER_UNKNOWN	The OCS does not recognize this user.	The system logs the event, updates statistics, sets the GTP result to failure and terminates the session.
Update	Other non-successful codes	N/A	N/A	The system logs the event, updates statistics, and sets the GTP result to failure. If this occurs while processing event triggers and the quota list is not empty, the system moves the charging groups.
Terminate	3002	DIAMETER_UNABLE_TO_DELIVER	The diameter device cannot deliver the message to the desired destination because the host described in the Destination-Host AVP does not match a host within the associated realm.	See Failure Handling (page 1-68) for more information.
Terminate	3004	DIAMETER_TOO_BUSY	The diameter device cannot deliver the message to the destination.	See Failure Handling (page 1-68) for more information.

Table 1-12 Supported Result Codes

Procedure	Value	Name	Description	Action
Terminate	3005	DIAMETER_LOOP_DETECTED	A diameter agent detected a loop while trying to get the message to the desired destination.	See Failure Handling (page 1-68) for more information.
Terminate	4001	DIAMETER_AUTHENTICATON_REJECTED	The user authentication process failed, most likely because the user provided an invalid password.	The system logs the event, updates statistics, sets the GTP result to failure and terminates the session without OCS processing.
Terminate	4010	DIAMETER_END_USER_SERVICE_DENIED	The OCS denied the service request due to service restrictions. If the CCR contains Used-Service-Units, they are deducted from the account.	The system logs the event, updates statistics, sets the GTP result to failure and terminates the session.
Terminate	4012	DIAMETER_CREDIT_LIMIT_REACHED	The OCS denied the request because the subscriber's account does not contain sufficient credit. If the CCR contains Used-Service-Units, they are deducted from the account.	The system logs the event, updates statistics, sets the GTP result to failure and terminates the session. If this occurs while processing MSCC attributes, the system requests a final report from the data plane (fastpath application).
Terminate	5030	DIAMETER_USER_UNKNOWN	The OCS does not recognize this user.	The system logs the event, updates statistics, sets the GTP result to failure and terminates the session.

Diameter AVP Descriptions

This section describes the supported Attribute Value Pair (AVP) definitions for the Gy, Gx, and Gz/Rf Diameter interfaces and therefore the AVPs described in this section do not apply to all interfaces.

- [Standard AVP Definitions \(page A-2\)](#)
- [3GPP AVP Definitions \(page A-28\)](#)
- [NASREQ/RADIUS AVP Definitions \(page A-102\)](#)
- [Vodafone AVP Definitions \(page A-104\)](#)
- [Vendor-Specific AVP Definitions \(page A-112\)](#)

Note: Refer to the 3GPP specifications and RFCs listed in References and Specifications for more details.

Standard AVP Definitions

This section describes the following standard AVPs used in the messages for DCCA.

- Accounting-Input-Octets (page A-3)
- Accounting-Input-Packets (page A-3)
- Accounting-Output-Octets (page A-3)
- Accounting-Output-Packets (page A-3)
- Accounting-Record-Number (page A-4)
- Accounting-Record-Type (page A-4)
- Acct-Application-Id (page A-4)
- Acct-Interim-Interval (page A-4)
- Auth-Application-Id (page A-5)
- CC-Input-Octets (page A-5)
- CC-Output-Octets (page A-5)
- CC-Request-Number (page A-6)
- CC-Request-Type (page A-6)
- CC-Session-Failover (page A-6)
- CC-Service-Specific-Units (page A-7)
- CC-Time (page A-7)
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- Credit-Control-Failure-Handling (page A-7)
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- User-Equipment-Info-Value (page A-25)
- User-Name (page A-25)
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- Validity-Time (page A-26)
- Vendor-Id (page A-26)
- Vendor-Specific-Application-Id (page A-27)

Accounting-Input-Octets

AVP Name	Accounting-Input-Octets
AVP Code	363
AVP Data Type	Integer64
Description	Together with the Accounting-Input-Packets AVP, this AVP contains the number of octets (resp packets), transmitted during the data container recording interval, reflecting the volume counts for uplink traffic for a data flow.

Accounting-Input-Packets

AVP Name	Accounting-Input-Packets
AVP Code	365
AVP Data Type	Integer64
Description	Together with the Accounting-Input-Octets AVP, this AVP contains the number of packets (resp octets), transmitted during the data container recording interval, reflecting the volume counts for uplink traffic for a data flow.

Accounting-Output-Octets

AVP Name	Accounting-Output-Octets
AVP Code	364
AVP Data Type	Integer64
Description	Together with the Accounting-Output-Packets AVP, this AVP contains the number of octets (resp packets), transmitted during the data container recording interval, reflecting the volume counts for downlink traffic for a data flow.

Accounting-Output-Packets

AVP Name	Accounting-Output-Packets
AVP Code	366
AVP Data Type	Integer64
Description	Together with the Accounting-Output-Octets AVP, this AVP contains the number of packets (resp octets), transmitted during the data container recording interval, reflecting the volume counts for downlink traffic for a data flow.

Accounting-Record-Number

AVP Name	Accounting-Record-Number
AVP Code	485
AVP Data Type	Unsigned32
Description	Contains the sequence number of the transferred messages.

Accounting-Record-Type

AVP Name	Accounting-Record-Type
AVP Code	480
AVP Data Type	Enumerated
Description	Defines the type of accounting record: ■ START – Starts an accounting session. ■ INTERIM – Updates an accounting session. ■ STOP – Ends an accounting session. ■ EVENT – Indicates a one-time accounting event.

Acct-Application-Id

AVP Name	Acct-Application-Id
AVP Code	259
AVP Data Type	Unsigned32
Description	Indicates the unique ID of the Diameter Accounting Application and is defined with the value 3.

Acct-Interim-Interval

AVP Name	Acct-Interim-Interval
AVP Code	85
AVP Data Type	Unsigned32
Description	Indicates the number of seconds between each interim update for this session.

Auth-Application-Id

AVP Name	Auth-Application-Id
AVP Code	258
AVP Data Type	Unsigned32
Description	Indicates the unique ID that is used for re-authentication or re-authorization. The AVP contains the following values: ■ Gy interface – 4 (DCCA) ■ Gx interface – 16777238 ■ S6b interface – You can specify the application ID (standard or non-standard) on the S6b interface using the following command. This is any value up to 32 bits. The default is 16777272 (default - 3GPP S6b). <pre>an3000-mcm-slot7cpu1(config)# service-construct interface s6b-interface <name> auth-application-id <value></pre>

CC-Input-Octets

AVP Name	CC-Input-Octets
AVP Code	412
AVP Data Type	Unsigned64
Description	The amount of octets to be received.

CC-Output-Octets

AVP Name	CC-Output-Octets
AVP Code	414
AVP Data Type	Unsigned64
Description	The amount of octets to be sent.

CC-Request-Number

AVP Name	CC-Request-Number
AVP Code	415
AVP Data Type	Unsigned32
Description	Indicates the identity of a certain request in a session, starting from 0.

CC-Request-Type

AVP Name	CC-Request-Type
AVP Code	416
AVP Data Type	Enumerated
Description	Indicates the reason for sending a CCR message. This AVP must be included in the CCR message. The CC-Request-Type AVP contains the following values: ■ INITIAL_REQUEST (1) – Indicates that the initial CCR is used to initiate a credit control session. The session contains the credit control information about initiating a session. ■ UPDATE_REQUEST (2) – Indicates that the update CCR contains the credit control information about the sessions of which credit control is set up. When the re-authorization needs to be initiated, an update CCR needs to be initiated. ■ TERMINATION_REQUEST (3) – Indicates an interruption to a credit control session by terminating a credit request.

CC-Session-Failover

AVP Name	CC-Session-Failover
AVP Code	418
AVP Data Type	Enumerated
Description	Indicates whether the OCS supports moving the CCR messages to a secondary server for a CC session. If the OCS supports session failover, the system sends CCR messages to the secondary server when the transport connection with the primary server goes down. The following values are defined: ■ FAILOVER_NOT_SUPPORTED (0) ■ FAILOVER_SUPPORTED (1)

CC-Service-Specific-Units

AVP Name	CC-Service-Specific-Units
AVP Code	417
AVP Data Type	Unsigned64
Description	The amount of service specific units. For example, the number of events for a rating-group.

CC-Time

AVP Name	CC-Time
AVP Code	420
AVP Data Type	Unsigned32
Description	The requested, distributed, or used time of the charging party, in seconds.

CC-Total-Octets

AVP Name	CC-Total-Octets
AVP Code	421
AVP Data Type	Unsigned64
Description	The amount of octets to be sent and received.

Credit-Control-Failure-Handling

AVP Name	Credit-Control-Failure-Handling
AVP Code	427
AVP Data Type	Enumerated
Description	The OCS uses this AVP to instruct clients how to handle the PDN session during a failure condition (transport failure etc.). The following values are defined: ■ TERMINATE (0) ■ CONTINUE (1) ■ RETRY_AND_TERMINATE (2)

Destination-Host

AVP Name	Destination-Host
AVP Code	293
AVP Data Type	DiameterIdentity
Description	Indicates the identity of the destination device. This AVP must be contained in messages that are initiated by an agent. This AVP can be contained in request messages but cannot be contained in response messages.

Destination-Realm

AVP Name	Destination-Realm
AVP Code	283
AVP Data Type	DiameterIdentity
Description	Indicates the home realm of the destination device.

Event-Timestamp

AVP Name	Event-Timestamp
AVP Code	55
AVP Data Type	Time
Description	Indicates the time of an event. The value of this AVP is generated by the DCC client. The time is converted from January 1, 1900, 00:00 UTC, in seconds. <i>Note: UTC: Coordinated Universal Time.</i> Gx Interface – Indicates the time when the Create Session Request was received. Gy Interface – Indicates the time when the quota was requested.

Experimental-Result

AVP Name	Experimental-Result
AVP Code	297
AVP Data Type	Grouped
Description	Indicates whether a particular vendor-specific request was completed successfully or whether an error occurred. The detailed syntax is as follows: <pre>Experimental-Result ::= <AVP Header: 297> {Vendor-Id} {Experimental-Result-Code}</pre>

Failed-AVP

AVP Name	Failed-AVP
AVP Code	279
AVP Data Type	Grouped
Description	Provides debugging information in cases where a request is rejected or not fully processed due to erroneous information in a specific AVP. The value of the Result-Code AVP will provide information on the reason for the Failed-AVP.

Filter-ID

AVP Name	Filter-ID
AVP Code	11
AVP Data Type	UTF8String
Description	Indicates the name of the filter list for this user. Used to restrict user access when the FUI AVP (embedded within the MSCC AVP), received in the CCA message from the OCS, contains the Final-Unit-Action AVP set to RESTRICT_ACCESS and one or more Filter-ID AVPs referencing a locally configured Rating Group with an IP address filter list. Multiple Filter-IDs are supported by configuring a Rating Group for each Filter-ID. See “Mapping a Filter-ID to a Rating-Group” in the <i>Affirmed Networks Gy Interface Specification</i>.

Final-Unit-Action

AVP Name	Final-Unit-Action
AVP Code	449
AVP Data Type	Enumerated
Description	Indicates the action to be taken once the subscriber has consumed all of the final granted units. The following values are defined: ■ TERMINATE (0) – Upon consumption of the final granted units, the session is terminated. ■ REDIRECT (1) – Upon consumption of the final granted units, the subscriber is redirected to the address specified in the Redirect-Server AVP. ■ RESTRICT-ACCESS (2) – Gy Interface – Upon consumption of the final granted units, user access is restricted according to the IP packet filters identified by the Filter-ID AVP. All of the packets that do not match the filters are dropped. For more information, see “Mapping a Filter-ID to a Rating-Group” in the <i>Affirmed Networks Gy Interface Specification</i>. – Gx Interface – If a flow is out of charging credits and when a packet for that flow arrives, the packet is dropped except when an Advice of Charge (AoC) ID is configured for the associated Rating Group on the MCC device. In this case, the subscriber is redirected to the redirect URL configured for the AoC ID.

Final-Unit-Indication

AVP Name	Final-Unit-Indication
AVP Code	430
AVP Data Type	Grouped
Description	The Final-Unit-Indication (FUI) AVP indicates that the Granted-Service-Unit AVP in the CCA contains the final units for the service. After these units have expired, the gateway executes the action indicated in the Final-Unit-Action AVP. Note the following: Gy Interface The FUI AVP is embedded within the Multiple-Services-Credit-Control (MSCC) AVP in the CCA message. The MCC supports FUI-based redirection when the MSCC AVP received in the CCA message from the OCS contains: ■ MSCC result code = 2001 (DIAMETER_SUCCESS) or 4012 (CREDIT_LIMIT_REACHED) + FUI AVP ■ MSCC result code = 2001 (DIAMETER_SUCCESS) + FUI AVP, no grant (no GSU AVP) ■ MSCC result code = 2001 (DIAMETER_SUCCESS) + FUI AVP + zero Grant (GSU AVP with zero grant / no grant info) The detailed syntax is as follows: <pre>Final-Unit-Indication ::= < AVP Header: 430 > {Final-Unit-Action} *[Filter-Id] [Redirect-Server]</pre> Gx Interface The FUI AVP is included in the CCA message at the command level. The detailed syntax is as follows: <pre>Final-Unit-Indication ::= < AVP Header: 430 > {Final-Unit-Action} [Redirect-Server]</pre> The FUI AVP is also included in the CCR message within the Charging-Rule-Report AVP to report the status of the PCC rules for which credit is no longer available or credit has been reallocated after the former out-of-credit indication. When reporting an out-of-credit condition, the Final-Unit-Indication AVP indicates the termination action the PCEF applies to the PCC rules as instructed by the OCS. The detailed syntax is as follows: <pre>Final-Unit-Indication ::= < AVP Header: 430 > {Final-Unit-Action} *[Filter-Id] [Redirect-Server]</pre>

Firmware-Revision

AVP Name	Firmware-Revision
AVP Code	267
AVP Data Type	Unsigned32
Description	Used to inform a diameter peer of the firmware revision of the issuing device.

Granted-Service-Unit

AVP Name	Granted-Service-Unit
AVP Code	431
AVP Data Type	Grouped
Description	The total service units that the DCC client can provide for end users.

Host-IP-Address

AVP Name	Host-IP-Address
AVP Code	257
AVP Data Type	Address
Description	Informs a diameter peer of the sender's IP address.

Inband-Security-Id

AVP Name	Inband-Security-Id
AVP Code	299
AVP Data Type	Unsigned32
Description	Indicates support of the security portion of the application.

Multiple-Services-Credit-Control

AVP Name	Multiple-Services-Credit-Control
AVP Code	456
AVP Data Type	Grouped
Description	Indicates all parameters for the CTF quota management and defines the quotas to allow traffic to flow. Each Multiple-Services-Credit-Control (MSCC) carries units for a single rating group.

Multiple-Services-Indicator

AVP Name	Multiple-Services-Indicator
AVP Code	455
AVP Data Type	Enumerated
Description	Indicates whether the system is capable of reporting multiple rating groups within a CC session. It is included in the CCR-Initial message. The following values are defined: ■ MULTIPLE_SERVICES_NOT_SUPPORTED (0) ■ MULTIPLE_SERVICES_SUPPORTED (1)

Origin-Host

AVP Name	Origin-Host
AVP Code	264
AVP Data Type	DiameterIdentity
Description	Contains the identification of the source point of the operation and the realm of the operation originator. Relay agents must not modify this AVP. The value of the Origin-Host AVP is guaranteed to be unique within a single host. <i>Note: The Origin-Host AVP may resolve to more than one address as the diameter peer may support more than one address (for example: host1.com).</i>

Origin-Realm

AVP Name	Origin-Realm
AVP Code	296
AVP Data Type	DiameterIdentity
Description	Indicates the home realm of the device that initiates DCC messages.

Origin-State-Id

AVP Name	Origin-State-Id
AVP Code	278
AVP Data Type	Unsigned32
Description	Carries the state associated with an Origin-Host.

Product-Name

AVP Name	Product-Name
AVP Code	269
AVP Data Type	UTF8String
Description	Contains the vendor-assigned name for the product. The Product-Name AVP should remain constant across firmware revisions for the same product.

Rating-Group

AVP Name	Rating-Group
AVP Code	432
AVP Data Type	Unsigned32
Description	The identifier of a rating group. For information on how the Rating-Group AVP is used on the Gx interface to install/activate PCC rules, see “PCC Rules”, in the <i>Affirmed Networks Gx Interface Specification</i>.

Re-Auth-Request-Type

AVP Name	Re-Auth-Request-Type
AVP Code	285
AVP Data Type	Enumerated
Description	Included in application-specific authorization answers to inform the client of the action expected upon expiration of the Authorization-Lifetime. The following values are defined: ■ AUTHORIZE_ONLY (0) – An authorization only re-authorization is expected upon expiration of the Authorization-Lifetime. This is the default value if the AVP is not present in answer messages that include the Authorization-Lifetime. ■ AUTHORIZE_AUTHENTICATE (1) – An authentication and authorization re-authorization is expected upon expiration of the Authorization-Lifetime.

Redirect-Address-Type

AVP Name	Redirect-Address-Type
AVP Code	433
AVP Data Type	Enumerated
Description	Defines the type of address given in the Redirect-Server-Address AVP. The address type can be one of the following: ■ URL (2) – The address is in the form of Uniform Resource Locator (URL). <i>Note: MCC does not support Redirect-Address-Type SIP URI.</i>

Redirect-Host

AVP Name	Redirect-Host
AVP Code	292
AVP Data Type	DiameterURI
Description	Contains the host to which the request should be forwarded. If the PGW receives a CCA (Initial, Update or Terminate) message from the PCRF and the Result-Code AVP is set to DIAMETER_REDIRECT_INDICATION, the PGW will forward the CCR message for that session to the host contained in the Redirect-Host AVP. The PCRF can send multiple Redirect-Host AVPs in the CCA message and if one fails, the MCC selects the next one in the list. The Redirect-Host AVP is configurable in CCA data record templates. The default <code>presentFlag</code> setting is <code>yes</code>, indicating the AVP is processed when received in the CCA message. <pre>an3000-mcm-slot7cpul (config) # service-construct data-record-template [pgw ggsn] pcrf diameter cca <template-name> redirecthost presentFlag [yes no]</pre>

Redirect-Server

AVP Name	Redirect-Server
AVP Code	434
AVP Data Type	Grouped
Description	Contains the address information of the redirect server (for example, HTTP redirect server) with which the end user is to be connected when the account cannot cover the service cost. This AVP must be present when the Final-Unit-Action AVP is set to REDIRECT. The detailed syntax is as follows: <pre>Redirect-Server ::= < AVP Header: 434 > {Redirect-Address-Type} {Redirect-Server-Address}</pre> Note: The MCC does not support Redirect-Address-Type SIP URI.

Redirect-Server-Address

AVP Name	Redirect-Server-Address
AVP Code	435
AVP Data Type	UTF8String
Description	Defines the address of the redirect server.

Requested-Action

AVP Name	Requested-Action
AVP Code	436
AVP Data Type	Enumerated
Description	Contains the requested action sent in the CCR where the CC-Request-Type is set to EVENT_REQUEST. The following values are defined: ■ DIRECT_DEBITING (0) – Indicates a request to decrease the end user's account according to information specified in the Requested-Service-Unit AVP and/or Service-Identifier AVP (additional rating information may be included in service-specific AVPs or in the Service-Parameter-Info AVP). The Granted-Service-Unit AVP in the CCA contains the debited units. ■ REFUND_ACCOUNT (1) – Indicates a request to increase the end user's account according to information specified in the Requested-Service-Unit AVP and/or Service-Identifier AVP (additional rating information may be included in service-specific AVPs or in the Service-Parameter-Info AVP). The Granted-Service-Unit AVP in the CCA contains the refunded units. ■ CHECK_BALANCE (2) – Indicates a balance check request. In this case, the checking of the account balance is done without any credit reservation from the account. The Check-Balance-Result AVP in the CCA contains the result of the balance check. ■ PRICE_ENQUIRY (3) – Indicates a price inquiry request. In this case, neither checking of the account balance nor reservation from the account is done. Only the price of the service is returned in the Cost-Information AVP in the CCA.

Requested-Service-Unit

AVP Name	Requested-Service-Unit
AVP Code	437
AVP Data Type	Grouped
Description	The amount of requested service units for a particular category, including monetary or non-monetary units.

Result-Code

AVP Name	Result-Code
AVP Code	268
AVP Data Type	Unsigned32
Description	A specified request is successfully fulfilled or fails.

Route-Record

AVP Name	Route-Record
AVP Code	282
AVP Data Type	DiameterIdentity
Description	Contains an identifier inserted by a relay or proxy node to identify the node from which the message is received.

Service-Context-Id

AVP Name	Service-Context-Id
AVP Code	461
AVP Data Type	UTF8String
Description	Indicates the unique ID (may be extended) of a Diameter credit-control (DCC) service-specific document that applies to the request. Use the following CLI parameter to specify the Service-Context-Id AVP sent in ACR messages via the Gz/Rf offline-charging interface to the OFCS. The default service-context-id is 9.32251@3gpp.org. <pre>an3000-mcm-slot7cpul(config)# service-construct interface offline-charging <name> service-context-id <service-context-id></pre>

Service-Identifier

AVP Name	Service-Identifier
AVP Code	439
AVP Data Type	Unsigned32
Description	Contains the identifier of a service. The specific service the request relates to is uniquely identified by the combination of the Service-Context-Id and Service-Identifier AVPs. Note the following: ■ For dynamic PCC rules, the Service-Identifier AVP sent from the PCRF as part of the Charging-Rule-Definition AVP does not need to be unique. Multiple Charging-Rule-Definition AVPs can have the same Service-Identifier AVP value. ■ For predefined PCC rules, if the MCC gateway receives a CCA message from the PCRF with the Service-Identifier AVP under the Charging-Rule-Definition AVP, the value in the Service-Identifier AVP overrides the Service-ID (<code>service-construct->service-rule->service-data-flow-id</code>) value provisioned on the gateway assuming the Charging-Rule-Name (<code>service-rule</code>) is valid. ■ If the <code>reporting-level</code> of a Rating Group is set to <code>service-rule</code>, the PGW sends the Service-Identifier AVP and Rating Group AVP (contained within the MSCC AVP) in CCR messages to the OCS. The Service-Identifier AVP only reports the Service-ID (<code>service-data-flow-id</code>) of the first service rule in the Rating Group. If the service rule list contains different Service-IDs, the MSCC AVP does not report all the Service-IDs.

Session-ID

AVP Name	Session-ID
AVP Code	263
AVP Data Type	UTF8String
Description	Identifies a session-ID that uniquely identifies a DCCA session. The default format is as follows: ■ <code><DiameterIdentity>;<high 32 bits>;<low 32 bits>;<optional value></code>. ■ <code><DiameterIdentity></code>: the same as filled in the Origin-Host AVP. ■ <code><high 32 bits></code>: contains the current time of the system. ■ <code><low 32 bits></code>: contains service instance cookie. ■ <code><optional value></code>: contains subscriber and bearer information. For example: <i>example.com;1450803134;2148532240;8;1450812272;262229</i> where ■ <i>example.com</i>: (Diameter identity) represents the Origin-host. ■ <i>1450803134</i>: (high 32 bits) represents the interface timestamp. ■ <i>2148532240</i>: (low 32 bits) represents the subscriber ID. ■ <i>8</i>: (optional value) represents the static Diameter session counter. ■ <i>1450812272</i>: (optional value) represents the session setup time in decimal format. ■ <i>262229</i>: (optional value) represents the gateway service ID (formed using Geo-ID, Block-ID, Node-ID, Bearer-ID). You can configure the format for the Session-ID AVP sent on the Gz/Rf interface using the following command. <pre>an3000-mcm-slot7cpul(config)# service-construct interface offline-charging <interface-name> session-id-format [v0 v1]</pre> ■ v0 – (default) The Session-ID AVP format represents the optional value for the session setup time in decimal format. For example: <i>example.com;1450803134;2148532240;8;1450812272;262229</i> where 1450812272 represents the session setup time in decimal format. ■ v1 – The Session-ID AVP format represents the optional value for the session setup time in hex format. This format applies to the Gz/Rf interface only. For example: 01-<i>example.com;1450803134;2148532240;8;5679A370-262229</i> where 01 is a prefix indicating that this Session-ID format is using version 1 (v1) and 5679A370 represents the session setup time in hex format. Note that the delimiter after the timestamp uses a dash (-) instead of a semi-colon (;).

Subscription-Id

AVP Name	Subscription-Id
AVP Code	443
AVP Data Type	Grouped
Description	The information used to identify the end user of the service subscription party. The detailed syntax is as follows: <pre>*Subscription-Id ::= <AVP Header: 443> {Subscription-Id-Type} {Subscription-Id-Data}</pre> The diameter proxy or agent must route based on this field. In the case of more than one Subscription-Id, the system handles only the first Subscription-Id. Gz interface Operators can suppress the Subscription-Id AVP in ACR messages based on the Subscription-Id-Type (IMSI/MSISDN) using ACR data record templates. In the ACR/CCR data record templates, the Subscription-Id options appear as <code>subscriptionidimsi</code> and <code>subscriptionidmsisdn</code>. Gx interface Operators can suppress the Subscription-Id AVP in CCR messages (CCR-I/CCR-U/CCR-T) based on the Subscription-Id-Type (IMSI/MSISDN/SIP URI) using CCR data record templates. In the CCR data record templates, the Subscription-Id options appear as the following: ■ <code>subscriptionid</code> ■ <code>subscriptionidimsi</code> ■ <code>subscriptionidmsisdn</code> ■ <code>subscriptionidsipuri</code> <i>Note: The GiGW sends a pseudo IMSI in the Subscription-ID AVP, which needs to be suppressed in PCRF CCR Start data record templates to use the Subscription-Id AVP with type SIP URI. In PCRF CCR Start data record templates, set the <code>subscriptionidimsi presentFlag</code> to <code>no</code>.</i> Gy Interface Operators can suppress the Subscription-Id AVP in CCR messages (CCR-I/CCR-U/CCR-T) based on the Subscription-Id-Type (IMSI/MSISDN/NAI) using CCR data record templates. In the CCR data record templates, the Subscription-Id options appear as the following: ■ <code>subscriptionidimsi</code> ■ <code>subscriptionidmsisdn</code> ■ <code>subscriptionidnai</code> <i>Note: For trusted non-3GPP Wi-Fi access using standard RADIUS/EAP methods for authentication, the end user's Network Access Identifier (NAI), received in the User-Name AVP of the RADIUS Access-Request, will be used in the Subscription-Id AVP in CCR messages to the OCS. In addition to the NAI, if the IMSI/MSISDN are available as part of the end user's identity, the MCC TWAG will encode multiple Subscription-Id AVPs in CCR messages on the Gy interface.</i>

Subscription-Id-Data

AVP Name	Subscription-Id-Data
AVP Code	444
AVP Data Type	UTF8String
Description	Used to identify the end user. This AVP indicates the number of the charged party when the Subscription-Id-Type is 0. Supports the following format for the mobile number: Mobile Subscriber Integrated Services Digital Network-Number (MSISDN) – Indicates the IMSI number of the charged party when the Subscription-Id-Type is 1.

Subscription-Id-Type

AVP Name	Subscription-Id-Type
AVP Code	450
AVP Data Type	Enumerated
Description	Indicates the type of user's terminal ID. The following values are defined: ■ END_USER_E164 (0) – The identifier is in international E.164 format defined according to the ITU-T E.164 number scheme. For example, MSISDN. ■ END_USER_IMSI (1) – The identifier is in international IMSI format defined according to the ITU-T E.212 number scheme. ■ END_USER_SIP_URI (2) – The identifier is in the form of a SIP URI, as defined in [SIP]. ■ END_USER_NAI (3) – The identifier is in the form of a Network Access Identifier, as defined in RFC 4282 and formatted as defined in 3GPP TS 23.003, clause 19.3.2. Used for Wi-Fi access. Use the following CLI parameter to configure the IMSI format to be sent in the Subscription-Id AVP for non-3gpp access. The default is <code>imsi format</code>. <i>Applicable to Gx and Gy interfaces.</i> <pre>an3000-mcm-slot7cpul(config)# service-construct interface [online-charging pcrf-charging] <interface-name> subscriptionid-imsi-format-non- 3gpp-access [imsi format imsinaiprefix0format]</pre>

Supported-Vendor-Id

AVP Name	Supported-Vendor-Id
AVP Code	265
AVP Data Type	Unsigned32
Description	Is used in the CER and CEA messages to inform the peer that the sender supports the vendor-specific AVPs defined by the vendor identified in this AVP.

Tariff-Change-Usage

AVP Name	Tariff-Change-Usage
AVP Code	452
AVP Data Type	Enumerated
Description	Identifies the reporting period for the used service unit. That is, before, after, or during tariff change. Omission of this AVP means that no tariff change has occurred. The following values are defined: ■ UNIT_BEFORE_TARIFF_CHANGE (0) ■ UNIT_AFTER_TARIFF_CHANGE (1) ■ UNIT_INDETERMINATE (2)

Tariff-Time-Change

AVP Name	Tariff-Time-Change
AVP Code	451
AVP Data Type	Time
Description	Is used when the OCS expects a tariff change to occur within the validity time of the quota it is granting.

Termination-Cause

AVP Name	Termination-Cause
AVP Code	295
AVP Data Type	Enumerated
Description	Shows the reason why the session of the diameter client terminated. The following values are defined: ■ DIAMETER_LOGOUT (1) – Interruption initiated by users. ■ DIAMETER_SERVICE_NOT_PROVIDED (2) – Session breaks before users receive authentication response messages. ■ DIAMETER_BAD_ANSWER (3) – The authentication answer received by the diameter client is not successfully handled. ■ DIAMETER_ADMINISTRATIVE (4) – Users are not granted the access authority or the session is disconnected due to operation management. For example, receiving an Abort-Session-Request. ■ DIAMETER_LINK_BROKEN (5) – Communication with users is suddenly disconnected. ■ DIAMETER_AUTH_EXPIRED (6) – User access ends because the authorized session time is reached. ■ DIAMETER_USER_MOVED (7) – Users are receiving the services from other diameter clients. ■ DIAMETER_SESSION_TIMEOUT (8) – User session is timed out and the service is stopped.

User-Equipment-Info

AVP Name	User-Equipment-Info
AVP Code	458
AVP Data Type	Grouped
Description	Indicates the identity and capability of the terminal that the subscriber is using for the connection to the network. The detailed syntax is as follows: <pre>User-Equipment-Info ::= < AVP Header: 458 > {User-Equipment-Info-Type} {User-Equipment-Info-Value}</pre> <i>Note: For Wi-Fi access, the User-Equipment-Info AVP contains the MAC address of the device if received.</i>

User-Equipment-Info-Type

AVP Name	User-Equipment-Info-Type
AVP Code	459
AVP Data Type	Enumerated
Description	Defines the type of user equipment information contained in the User-Equipment-Info-Value AVP. Contains the following values: ■ IMEISV (0) – The identifier contains the International Mobile Equipment Identifier and Software Version in the international IMEISV format according to 3GPP TS 23.003 [3GPPIMEI]. ■ MAC (1) – The 48-bit MAC address is formatted as described in [RAD802.1X]. ■ EUI64 (2) – The 64-bit identifier used to identify hardware instance of the product, as defined in [EUI64].

User-Equipment-Info-Value

AVP Name	User-Equipment-Info-Value
AVP Code	460
AVP Data Type	OctetString
Description	Defines which type of identifier is used. <i>Note: When the User-Equipment-Info-Type is set to IMEISV(0), the value within the User-Equipment-Info-Value is a UTF-8 encoded decimal.</i>

User-Name

AVP Name	User-Name
AVP Code	1
AVP Data Type	UTF8String
Description	The user provides the username (extracted from the Protocol Configuration Options field of the Create PDP Context Request message) as provided in the Access-Request message.

Used-Service-Unit

AVP Name	Used-Service-Unit
AVP Code	446
AVP Data Type	Grouped
Description	The amount of used non-monetary service units measured for a rating group. <i>Note: The “M” bit flag for this AVP is set to 0 when present in the Usage-Monitoring-Information AVP.</i> The detailed syntax is as follows: <pre>Used-Service-Unit ::= < AVP Header: 446 > [CC-Time] [CC-Total-Octets] [CC-Input-Octets] [CC-Output-Octets] * [AVP]</pre>

Validity-Time

AVP Name	Validity-Time
AVP Code	448
AVP Data Type	Unsigned32
Description	Sent from the credit-control server to the credit-control client. This AVP contains the validity time of the granted service units. The measurement of the Validity-Time is started upon receipt of the CCA message containing this AVP. <i>Note: The MCC starts the Validity-Time only when the Validity-time AVP is received in the CCA message with the current GSU and does not use Validity-Time (if any) from a previous grant.</i>

Vendor-Id

AVP Name	Vendor-Id
AVP Code	266
AVP Data Type	Unsigned32
Description	Contains the vendor ID of the diameter application.

Vendor-Specific-Application-Id

AVP Name	Vendor-Specific-Application-Id
AVP Code	259
AVP Data Type	Group
Description	Used to advertise support of a vendor-specific diameter application.

3GPP AVP Definitions

This section describes the 3GPP AVPs sent by the system in CCR, CCA, RAR, ACR, and/or ACA messages. The AVPs include a Vendor-ID of 10415 (3GPP).

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3GPP-Charging-Characteristics

AVP Name	3GPP-Charging-Characteristics
AVP Code	13
AVP Data Type	UTF8String
Description	The charging characteristics for this PDP context received in the Create PDP Context Request message (only available in R99 and later releases).

3GPP-Charging-Id

AVP Name	3GPP-Charging-Id
AVP Code	2
AVP Data Type	OctetString
Description	The Charging ID for this PDP Context. The Charging ID together with the GGSN-Address constitutes a unique identifier for the PDP context.

3GPP-GGSN-Address

AVP Name	3GPP-GGSN-Address
AVP Code	7
AVP Data Type	OctetString
Description	The Gateway GPRS Support Node/Packet Data Network Gateway (GGSN/PGW) IP address.

3GPP-GGSN-IPv6-Address

AVP Name	3GPP-GGSN-IPv6-Address
AVP Code	16
AVP Data Type	OctetString
Description	The GGSN IPv6 address that is used by the GTP control plane for the context.

3GPP-GGSN-MCC-MNC

AVP Name	3GPP-GGSN-MCC-MNC
AVP Code	9
AVP Data Type	UTF8String
Description	The Mobile Country Code (MCC) and Mobile Network Code (MNC) of the network to which the GGSN/PGW belongs. When the PLMN ID (MCC + MNC), received in the subscriber's IMSI, matches a PLMN ID configured in the home-plmnid-list, the configured PLMN ID is sent in the 3GPP-GGSN-MCC-MNC AVP on all external interfaces (RADIUS, Gx, Gy, Gz/Rf). If an entry is not found in the list, the first entry in the home-plmnid-list is used in the 3GPP-GGSN-MCC-MNC AVP.

3GPP-GPRS-Negotiated-QoS-Profile

AVP Name	3GPP-GPRS-Negotiated-QoS-Profile
AVP Code	5
AVP Data Type	UTF8String
Description	This AVP contains the QoS profile.

3GPP-IMEISV

AVP Name	3GPP-IMEISV
AVP Code	20
AVP Data Type	UTF8String
Description	The user International Mobile Equipment Identity of the equipment served (IMEISV).

3GPP-IMSI

AVP Name	3GPP-IMSI
AVP Code	1
AVP Data Type	UTF8String
Description	The International Mobile Subscriber Identity (IMSI) for this user.

3GPP-IMSI-MCC-MNC

AVP Name	3GPP-IMSI-MCC-MNC
AVP Code	8
AVP Data Type	UTF8String
Description	The Mobile Country Code (MCC) and Mobile Network Code (MNC) extracted from the user's IMSI (first 5 or 6 digits, as applicable from the presented IMSI).

3GPP-MS-TimeZone

AVP Name	3GPP-MS-TimeZone
AVP Code	23
AVP Data Type	UTF8String
Description	The Time Zone field and the Daylight Saving Time fields are used to indicate the offset between universal time and local time in steps of 15 minutes of where the MS currently resides.

3GPP-NSAPI

AVP Name	3GPP-NSAPI
AVP Code	10
AVP Data Type	UTF8String
Description	Identifies a particular PDP context for the associated PDN and Mobile Subscriber Integrated Services Digital Network-Number (MSISDN)/IMSI from creation to deletion.

3GPP-PDP-Type

AVP Name	3GPP-PDP-Type
AVP Code	3
AVP Data Type	Enumerated
Description	Type of PDP context. For example, IP or PPP. The PDP type may have the following values: ■ 0 = IPv4 ■ 1 = PPP ■ 2 = IPv6 ■ 3 = Ipv4v6 ■ 4 = Non-IP

3GPP-RAT-Type

AVP Name	3GPP-RAT-Type
AVP Code	21
AVP Data Type	OctetString
Description	The Radio Access Technology currently serving the User Equipment (UE).

3GPP-Selection-Mode

AVP Name	3GPP-Selection-Mode
AVP Code	12
AVP Data Type	UTF8String
Description	The selection mode for this PDP context received in the create PDP context request message.

3GPP-Session-Stop-Indicator

AVP Name	3GPP-Session-Stop-Indicator
AVP Code	11
AVP Data Type	UTF8String
Description	Indicates to the AAA server that the last PDP context of a session is released and the IP-CAN session has been terminated.

3GPP-SGSN-Address

AVP Name	3GPP-SGSN-Address
AVP Code	6
AVP Data Type	OctetString
Description	Indicates the IP address of the Serving GPRS Support Node (SGSN). This AVP can be used to identify the Public Land Mobile Network (PLMN) of the SGSN to which a user logs in.

3GPP-SGSN-IPv6-Address

AVP Name	3GPP-SGSN-IPv6-Address
AVP Code	15
AVP Data Type	OctetString
Description	Indicates the SGSN IPv6 address used by the GPRS Tunneling Protocol (GTP) control plane for the handling of control messages. It may be used to identify the PLMN to which the user is attached.

3GPP-SGSN-MCC-MNC

AVP Name	3GPP-SGSN-MCC-MNC
AVP Code	18
AVP Data Type	UTF8String
Description	Indicates the information about the international roam-in area of the charged party. On the Gx interface, the 3GPP-SGSN-MCC-MNC AVP can be used in the CCA message to override the PLMN-ID sent in GTP-C messages. See “Overriding the PLMN-ID sent in GTP-C messages” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

3GPP-User-Location-Info

AVP Name	3GPP-User-Location-Info
AVP Code	22
AVP Data Type	OctetString
Description	Indicates details of where the user equipment is currently located (for example, SAI or CGI).

Access-Network-Charging-Address

AVP Name	Access-Network-Charging-Address
AVP Code	501
AVP Data Type	Address
Description	Indicates the IP Address of the network entity within the access network performing charging (for example, the GGSN IP address).

Access-Network-Charging-Identifier-Gx

AVP Name	Access-Network-Charging-Identifier-Gx
AVP Code	1022
AVP Data Type	Grouped
Description	Contains a charging identifier within the Access-Network-Charging-Identifier-Value AVP and the related PCC rule name(s) within the Charging-Rule-Name AVP(s) and/or within the Charging-Rule-Base-Name AVP(s). If the charging identifier applies to the entire IP CAN session, neither the Charging-Rule-Name AVP or Charging-Rule-Base-Name AVP needs to be provided. The detailed syntax is as follows: <pre>Access-Network-Charging-Identifier-Gx ::= < AVP Header: 1022 > {Access-Network-Charging-Identifier-Value} * [Charging-Rule-Base-Name] * [Charging-Rule-Name]</pre>

Access-Network-Charging-Identifier-Value

AVP Name	Access-Network-Charging-Identifier-Value
AVP Code	503
AVP Data Type	OctetString
Description	Contains a charging identifier.

ADC-Rule-Base-Name

AVP Name	ADC-Rule-Base-Name
AVP Code	1095
AVP Data Type	UTF8String
Description	Indicates the name of a predefined group of Application Detection and Control (ADC) rules. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

ADC-Rule-Definition

AVP Name	ADC-Rule-Definition
AVP Code	1094
AVP Data Type	Grouped
Description	Defines the ADC rule sent by the PCRF. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. Note: The Redirect-Information AVP sent from the PCRF as part of the ADC-Rule-Definition AVP applies to predefined PCC rules only and applies to both online and offline PCC rules. When redirect information is received for an online rule, all HTTP traffic matching that rule's rating group will be redirected. The detailed syntax is as follows: <pre>ADC-Rule-Definition ::= < AVP Header: 1094 > {ADC-Rule-Name} [TDF-Application-Identifier] *[Flow-Information] [Service-Identifier] [Rating-Group] [Reporting-Level] [Online] [Offline] [Metering-Method] [Flow-Status] [QoS-Information] [Monitoring-Key] [Redirect-Information] [Mute-Notification] *[AVP]</pre>

ADC-Rule-Install

AVP Name	ADC-Rule-Install
AVP Code	1092
AVP Data Type	Grouped
Description	Used to activate, install or modify Application Detection and Control (ADC) rules as instructed from the PCRF. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The detailed syntax is as follows: <pre>ADC-Rule-Install ::= < AVP Header: 1092 > * [ADC-Rule-Definition] * [ADC-Rule-Name] * [ADC-Rule-Base-Name] [Monitoring-Flags] [Rule-Activation-Time] [Rule-Deactivation-Time] * [AVP]</pre>

ADC-Rule-Name

AVP Name	ADC-Rule-Name
AVP Code	1096
AVP Data Type	OctetString
Description	Defines a name for an ADC rule. For ADC rules provided by the PCRF, it uniquely identifies an ADC rule within one session. For predefined ADC rules, it uniquely identifies an ADC rule within the PCEF. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

ADC-Rule-Remove

AVP Name	ADC-Rule-Remove
AVP Code	1093
AVP Data Type	Grouped
Description	Used to deactivate or remove ADC rules as instructed from the PCRF. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The detailed syntax is as follows: <pre>ADC-Rule-Remove ::= < AVP Header: 1093 > * [ADC-Rule-Name] * [ADC-Rule-Base-Name] * [AVP]</pre>

ADC-Rule-Report

AVP Name	ADC-Rule-Report
AVP Code	1097
AVP Data Type	Grouped
Description	Used to report the status of the ADC rules which cannot be installed/activated or enforced. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The detailed syntax is as follows: <pre>ADC-Rule-Report ::= < AVP Header: 1097 > * [ADC-Rule-Name] * [ADC-Rule-Base-Name] [PCC-Rule-Status] [Rule-Failure-Code] [Final-Unit-Indication] * [AVP]</pre>

Additional-Exception-Reports

AVP Name	Additional-Exception-Reports
AVP Code	3936
AVP Data Type	Enumerated
Description	Indicates whether additional exception reports are allowed when maximum rate is reached. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The following values are defined: ■ NOT_ALLOWED (0)

AF-Charging-Identifier

AVP Name	AF-Charging-Identifier
AVP Code	505
AVP Data Type	OctetString
Description	Contains the AF Charging Identifier sent by the Application Function (AF). This information may be used for charging correlation with the bearer layer.

AF-Correlation-Information

AVP Name	AF-Correlation-Information
AVP Code	1276
AVP Data Type	Grouped
Description	Contains the AF charging identifier (ICID for IMS) and associated flow identifiers generated by the AF and received by PGW over the Gx interface as defined in TS 29.214 and TS 29.212. The AF-Correlation-Information is defined per Rating Group or per Rating Group and Service Identifier, when Service Identifier level reporting applies. When several AF sessions are conveyed over the same bearer, this AVP may appear several times per MSCC instance. The detailed syntax is as follows: <pre>AF-Correlation-Information:: = < AVP Header: 1276 > {AF-Charging-Identifier} * [Flows]</pre>

Allocation-Retention-Priority

AVP Name	Allocation-Retention-Priority
AVP Code	1034
AVP Data Type	Grouped
Description	Indicates the priority of allocation and retention, the pre-emption capability and pre-emption vulnerability for the SDF if provided within the QoS-Information AVP or for the EPS default bearer if provided within the Default-EPS-Bearer-QoS AVP. The detailed syntax is as follows: <pre>Allocation-Retention-Priority ::= < AVP Header: 1034 > {Priority-Level} [Pre-emption-Capability] [Pre-emption-Vulnerability]</pre>

AN-GW-Address

AVP Name	AN-GW-Address
AVP Code	1050
AVP Data Type	Address
Description	Contains the IPv4 and/or IPv6 (if available) address(es) of the SGW for 3GPP networks and access node gateway (AGW) for non-3GPP networks. For 2G/3G based sessions with an IP-CAN-Type of 3GPP-EPS, the MCC sends the IP address of the SGSN in the AN-GW-Address AVP in the CCR-Initial message to the PCRF. If both IPv4 and IPv6 addresses are provided, the MCC sends two instances of the AN-GW-Address AVP.

AN-Trusted

AVP Name	AN-Trusted
AVP Code	1503
AVP Data Type	Enumerated
Description	Indicates whether the access network is trusted or untrusted for the Non-3GPP access network. For example, if there is an LTE-to-Wi-Fi handover and if the PCRF has requested to be notified of an IP-CAN_CHANGE, the PGW includes the AN-Trusted AVP in the CCR-U along with the IP-CAN_Change event-trigger to indicate whether the access network is trusted or untrusted. The following values are defined: ■ TRUSTED (0) – This value is used when the non-3GPP access network is to be handled as trusted. ■ UNTRUSTED (1) – This value is used when the non-3GPP access network is to be handled as untrusted.

Application-Detection-Information

AVP Name	Application-Detection-Information
AVP Code	1098
AVP Data Type	Grouped
Description	Used to report application Start/Stop notifications when application traffic, defined by the TDF-Application-Identifier, has been detected and when the PCRF has subscribed for APPLICATION_START/APPLICATION_STOP Event-Triggers. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The detailed syntax is as follows: <pre>Application-Detection-Information:: = <AVP Header: 1098> {TDF-Application-Identifier} [TDF-Application-Instance-Identifier] *[Flow-Information] *[AVP]</pre>

Application-Service-Provider-Identity

AVP Name	Application-Service-Provider-Identity
AVP Code	532
AVP Data Type	UTF8String
Description	Used for sponsored data connectivity purposes as an identifier of the application service provider.

APN-Aggregate-Max-Bitrate-DL

AVP Name	APN-Aggregate-Max-Bitrate-DL
AVP Code	1040
AVP Data Type	Unsigned32
Description	Indicates the maximum aggregate bit rate in bits per second for the downlink direction across all non-GBR bearers with the same APN. When provided in a CCR message, it indicates the subscribed maximum bitrate. When provided in a CCA message, it indicates the maximum bandwidth authorized by the PCRF.

APN-Aggregate-Max-Bitrate-UL

AVP Name	APN-Aggregate-Max-Bitrate-UL
AVP Code	1041
AVP Data Type	Unsigned32
Description	Indicates the maximum aggregate bit rate in bits per seconds for the uplink direction across all non-GBR bearers with the same APN. When provided in a CCR message, it indicates the subscribed maximum bitrate. When provided in a CCA message, it indicates the maximum bandwidth authorized by the PCRF.

APN-Rate-Control

AVP Name	APN-Rate-Control
AVP Code	3933
AVP Data Type	Grouped
Description	Indicates the APN Rate Control parameters applicable to data PDU transfers in the uplink and downlink direction for this APN. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The detailed syntax is as follows: <pre>APN-Rate-Control ::= < AVP Header: 3933 > [APN-Rate-Control-Uplink] [APN-Rate-Control-Downlink]</pre>

APN-Rate-Control-Downlink

AVP Name	APN-Rate-Control-Downlink
AVP Code	3934
AVP Data Type	Grouped
Description	Indicates the APN-Rate Control parameters applicable to data PDU transfers in the downlink direction for this APN. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The detailed syntax is as follows: <pre>APN-Rate-Control-Downlink ::= < AVP Header: 3934 > [Rate-Control-Time-Unit] [Rate-Control-Max-Rate] [Rate-Control-Max-Message-Size]</pre>

APN-Rate-Control-Uplink

AVP Name	APN-Rate-Control-Uplink
AVP Code	3935
AVP Data Type	Grouped
Description	Indicates the APN Rate Control parameters applicable to data PDU transfers in the uplink direction for this APN. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The detailed syntax is as follows: <pre> APN-Rate-Control-Uplink ::= < AVP Header: 3935> [Additional-Exception-Reports] [Rate-Control-Time-Unit] [Rate-Control-Max-Rate] </pre>

Base-Time-Interval

AVP Name	Base-Time-Interval
AVP Code	1265
AVP Data Type	Unsigned32
Description	Contains the length of the base time interval for controlling the consumption of time quota, in seconds.

Bearer-Control-Mode

AVP Name	Bearer-Control-Mode
AVP Code	1023
AVP Data Type	Enumerated
Description	Indicates the PCRF selected bearer control mode. The following values are defined: ■ UE_ONLY (0) – Indicates that the UE may request any additional PDP Context establishment. ■ UE_NW (2) – Indicates that both the UE and PCEF may request any additional PDP Context establishment and add own traffic mapping information to a PDP Context.

Bearer-Identifier

AVP Name	Bearer-Identifier
AVP Code	1020
AVP Data Type	OctetString
Description	Indicates the bearer to which specific information refers. When present within the CCR command, subsequent AVPs within the CCR refer to the specific bearer identified by this AVP. The bearer identifier of an IP CAN bearer is unique within the corresponding IP CAN session and is selected by the PCEF.

Bearer-Operation

AVP Name	Bearer-Operation
AVP Code	1021
AVP Data Type	Enumerated
Description	Indicates the bearer event that causes a request for PCC rules. This AVP is supplied if the bearer event relates to an IP CAN bearer initiated by the UE. This AVP is only sent when the IP-CAN-Type AVP is 3GPP-GPRS (2G/3G). The following values are defined: ■ TERMINATION (0) – Indicates that a bearer is being terminated. ■ ESTABLISHMENT (1) – Indicates that a new bearer is being established. ■ MODIFICATION (2) – Indicates that an existing bearer is being modified.

Bearer-Usage

AVP Name	Bearer-Usage
AVP Code	1000
AVP Data Type	Enumerated
Description	Indicates how the bearer is being used. For example, whether it is used as a dedicated IMS signaling context. The following values are defined: ■ GENERAL (0) – Indicates that no specific bearer usage information is available. No TFT filter is associated with the PDP context. ■ IMS_SIGNALLING (1) – Indicates that the bearer is used for IMS signaling only. ■ DEDICATED (2) – Indicates that the bearer is being used for a dedicated purpose and accordingly has an associated TFT filter.

CG-Address

AVP Name	CG-Address
AVP Code	846
AVP Data Type	OctetString
Description	Contains the IP address of the charging gateway.

Change-Condition

AVP Name	Change-Condition
AVP Code	2037
AVP Data Type	Integer32
Description	Indicates the change in charging condition. The following values are defined: ■ 0 = Normal Release – Used to indicate an IP-CAN session termination, an IP-CAN bearer release or a service data flow termination. ■ 1 = Abnormal Release ■ 2 = QoS Change ■ 3 = Volume Limit ■ 4 = Time Limit ■ 8 = RAT Change ■ 14 = CGI-SAI Change ■ 15 = RAI Change ■ 16 = ECGI Change ■ 17 = TAI Change ■ 36 = Change in UP to UE <i>Note: If a single accounting record contains multiple changes (for example, an SGSN change with CGI change, RAI change, QoS change), the MCC sends multiple Change-Condition AVPs containing all of the change conditions.</i>

Change-Time

AVP Name	Change-Time
AVP Code	2038
AVP Data Type	Time
Description	Contains the time in UTC format when the volume counts, associated with the IP-CAN bearer or the service data container, are closed and reported due to a change in charging conditions.

Charging-Characteristics-Selection-Mode

AVP Name	Charging-Characteristics-Selection-Mode
AVP Code	2066
AVP Data Type	Enumerated
Description	Indicates how the applied Charging-Characteristics was selected. The following values are defined: ■ 0 = Serving-Node-Supplied ■ 1 = Subscription-specific ■ 2 = APN-specific ■ 3 = Home-Default ■ 4 = Roaming-Default ■ 5 = Visiting-Default

Charging-Information

AVP Name	Charging-Information
AVP Code	618
AVP Data Type	Grouped
Description	Contains the addresses of the charging functions. <i>Note: You can configure the PGW to extract information from the Charging-Information AVP received from the PCRF and use this information during Gy session setup. See “Using Gx Charging-Information AVP for Gy Session Setup” in the Affirmed Networks Gy Interface specification.</i> The detailed syntax is as follows: <pre>Charging-Information ::= <AVP Header : 618 10415> [Primary-Event-Charging-Function-Name] [Secondary-Event-Charging-Function-Name] * [AVP]</pre>

Charging-Rule-Base-Name

AVP Name	Charging-Rule-Base-Name
AVP Code	1004
AVP Data Type	UTF8String
Description	Indicates the name of a predefined group of PCC rules residing at the PCEF. The value of the Charging-Rule-Base-Name AVP corresponds to the locally configured data-profile on the MCC gateway (workflow → data-profile). For information on the MCC implementation of this AVP, see “PCC Rules” in the <i>Affirmed Networks Gx Interface Specification</i> .

Charging-Rule-Definition

AVP Name	Charging-Rule-Definition
AVP Code	1003
AVP Data Type	Grouped
Description	Defines the dynamic PCC rule for a service data flow sent by the PCRF to the PCEF or used to activate a predefined PCC rule that is pre-configured at the PCEF. For information on the MCC implementation of this AVP, see “PCC Rules” in the <i>Affirmed Networks Gx Interface Specification</i>. Note: <i>The Redirect-Information AVP sent from the PCRF as part of the Charging-Rule-Definition AVP applies to predefined PCC rules only and applies to both online and offline PCC rules. When redirect information is received for an online rule, all HTTP traffic matching that rule's rating group will be redirected.</i> The detailed syntax is as follows: <pre>Charging-Rule-Definition ::= < AVP Header: 1003 > {Charging-Rule-Name} [Service-Identifier] [Rating-Group] *[Flow-Information] [Flow-Status] [QoS-Information] [PS-to-CS-Session-Continuity] [Reporting-Level] [Online] [Offline] [Metering-Method] [Precedence] [Monitoring-Key] [Redirect-Information] [Mute-Notification] [Sponsor-Identity] [Application-Service-Provider-Identity] [Required-Access-Info] *[AVP]</pre>

Charging-Rule-Install

AVP Name	Charging-Rule-Install
AVP Code	1001
AVP Data Type	Grouped
Description	Used to activate, install, or modify PCC rules as instructed from the PCRF to the PCEF. For information on the MCC implementation of this AVP, see “PCC Rules” in the <i>Affirmed Networks Gx Interface Specification</i> .

Charging-Rule-Name

AVP Name	Charging-Rule-Name
AVP Code	1005
AVP Data Type	OctetString
Description	For PCC rules provided by the PCRF, it uniquely identifies a PCC rule within one IP CAN session. For PCC rules predefined at the PCEF, it uniquely identifies a PCC rule within the PCEF. For information on the MCC implementation of this AVP, see “PCC Rules” in the <i>Affirmed Networks Gx Interface Specification</i> .

Charging-Rule-Remove

AVP Name	Charging-Rule-Remove
AVP Code	1002
AVP Data Type	Grouped
Description	Used to deactivate or remove PCC rules from an IP CAN session. For information on the MCC implementation of this AVP, see “PCC Rules” in the <i>Affirmed Networks Gx Interface Specification</i>. The detailed syntax is as follows: <pre>Charging-Rule-Remove ::= < AVP Header: 1002 > * [Charging-Rule-Name] * [Charging-Rule-Base-Name]</pre>

Charging-Rule-Report

AVP Name	Charging-Rule-Report
AVP Code	1018
AVP Data Type	Grouped
Description	Reports the status of the PCC rules that have been successfully installed, modified or removed or that cannot be installed/activated or enforced at the PCEF. The Rule-Failure-Code indicates the reason that the PCC rules cannot be successfully installed/activated or enforced. The detailed syntax is as follows: <pre>Charging-Rule-Report ::= < AVP Header: 1018 > * [Charging-Rule-Name] * [Charging-Rule-Base-Name] [PCC-Rule-Status] [Rule-Failure-Code] [Final-Unit-Indication] * [RAN-NAS-Release-Cause]</pre>

CP-CIoT-EPS-Optimisation-Indicator

AVP Name	CP-CIoT-EPS-Optimisation-Indicator
AVP Code	3930
AVP Data Type	Enumerated
Description	Indicates whether Control Plane CIoT EPS optimisation is used by this PDN connection for data transfer with the UE. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The following values are defined: ■ NOT_APPLY (0) ■ APPLY (1)

CSG-Access-Mode

AVP Name	CSG-Access-Mode
AVP Code	2317
AVP Data Type	Enumerated
Description	Contains the CSG cell access mode. The following values are defined: ■ CLOSED MODE (0) ■ HYBRID MODE (1)

CSG-Id

AVP Name	CSG-Id
AVP Code	1437
AVP Data Type	Unsigned32
Description	Contains the CSG identification information.

CSG-Information-Reporting

AVP Name	CSG-Information-Reporting
AVP Code	1071
AVP Data Type	Enumerated
Description	Sent from the PCRF to the PCEF to request the PCEF to report the user CSG information change to the OCS. The following values are defined: ■ CHANGE_CSG_CELL (0) – Indicates that the PCEF sends a credit re-authorization request to the OCS when the UE enters/leaves/accesses via a CSG cell. ■ CHANGE_CSG_SUBSCRIBED_HYBRID_CELL (1) – Indicates that the PCEF sends a credit re-authorization request to the OCS when the UE enters/leaves/accesses via a hybrid cell in which the subscriber is a CSG member. ■ CHANGE_CSG_UNSUBSCRIBED_HYBRID_CELL (2) – Indicates that the PCEF sends a credit re-authorization request to the OCS when the UE enters/leaves/accesses via a hybrid cell in which the subscriber is not a CSG member.

CSG-Membership-Indication

AVP Name	CSG-Membership-Indication
AVP Code	2318
AVP Data Type	Enumerated
Description	Indicates whether the UE is a member of the accessing CSG cell, if the access mode is Hybrid. The following values are defined: ■ NOT CSG MEMBER (0) ■ CSG MEMBER (1)

Default-EPS-Bearer-QoS

AVP Name	Default-EPS-Bearer-QoS
AVP Code	1049
AVP Data Type	Grouped
Description	Defines the QoS information for the EPS default bearer. When this AVP is sent from the PCEF to the PCRF, it indicates the subscribed QoS for the default EPS bearer. When this AVP is sent from the PCRF to the PCEF, it indicates the authorized QoS for the default EPS bearer.

Diagnostics

AVP Name	Diagnostics
AVP Code	2039
AVP Data Type	Integer32
Description	Provides more information on the reason for sending the ACR Stop message and may contain one of the following values: ■ UNSPECIFIED (0) ■ SESSION_TIMEOUT (1) ■ RESOURCE_LIMITATION (2) ■ ADMIN_DISCONNECT (3) ■ IDLE_TIMEOUT (4) ■ PCRF_UNREACHABLE (5) ■ AAA_UNREACHABLE (6) ■ AAA_INITIATED_SESSION_TERMINATION (7) ■ PCRF_INITIATED_SESSION_TERMINATION (9) ■ PCRF_INITIATED_FLOW_TERMINATION (10) ■ PCRF_ACCOUNTING_PARAMETERS_CHANGED (11) ■ GTP_INITIATED_SESSION_TERMINATION (14) ■ HANDOVER_ERROR (16) ■ REAUTHORIZATION_FAILED (18)

Downlink-Rate-Limit

AVP Name	Downlink-Rate-Limit
AVP Code	4312
AVP Data Type	Unsigned32
Description	Contains the maximum number of NAS Data PDUs per deci hour for this UE for downlink. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i>

Dynamic-Address-Flag

AVP Name	Dynamic-Address-Flag
AVP Code	2051
AVP Data Type	Enumerated
Description	Indicates whether the PDP context/PDN address is statically or dynamically allocated. If this AVP is not present, the address is statically allocated. The following values are defined: ■ STATIC (0) ■ DYNAMIC (1)

Envelope

AVP Name	Envelope
AVP Code	1266
AVP Data Type	Grouped
Description	Reports the start and end time of a one time envelope using the Envelope-Start-Time and Envelope-End-Time AVPs. Further details of its usage are described in 3GPP 32.299. The detailed syntax is as follows: <pre>Envelope ::= < AVP Header: 1266> {Envelope-Start-Time} [Envelope-End-Time] [CC-Total-Octets] [CC-Input-Octets] [CC-Output-Octets] [CC-Service-Specific-Units] * [AVP]</pre>

Envelope-End-Time

AVP Name	Envelope-End-Time
AVP Code	1267
AVP Data Type	Time
Description	This AVP is set to the time of the end of the time envelope.

Envelope-Reporting

AVP Name	Envelope-Reporting
AVP Code	1268
AVP Data Type	Enumerated
Description	Used in the CCA-Initial message to indicate whether the client reports the start and end of each time envelope, in cases where the quota is consumed in envelopes. The following values are defined: ■ DO_NOT_REPORT_ENVELOPES (0) ■ REPORT_ENVELOPES (1) ■ REPORT_ENVELOPES_WITH_VOLUME (2) ■ REPORT_ENVELOPES_WITH_EVENTS (3) ■ REPORT_ENVELOPES_WITH_VOLUME_AND_EVENTS (4)

Envelope-Start-Time

AVP Name	Envelope-Start-Time
AVP Code	1269
AVP Data Type	Time
Description	This AVP is set to the time of the packet of user data which caused the time envelope to start.

Event-Trigger

AVP Name	Event-Trigger
AVP Code	1006
AVP Data Type	Enumerated
Description	When sent from the PCRF to the PCEF, this AVP indicates an event that causes a re-request of PCC rules. When sent from the PCEF to the PCRF, this AVP indicates that the corresponding event has occurred at the gateway (PGW). The system supports the following PCRF event-triggers: ■ SGSN_CHANGE (0) ■ QOS_CHANGE (1) ■ RAT_CHANGE (2) ■ PLMN_CHANGE (4) ■ LOSS_OF_BEARER (5) ■ IP-CAN_CHANGE (7) ■ RAI_CHANGE (12) ■ USER_LOCATION_CHANGE (13) ■ OUT_OF_CREDIT (15) ■ REALLOCATION_OF_CREDIT (16) ■ REVALIDATION_TIMEOUT (17) ■ DEFAULT_EPS_BEARER_QOS_CHANGE (20) ■ SUCCESSFUL_RESOURCE_ALLOCATION (22) ■ UE_TIME_ZONE_CHANGE (25) ■ USAGE_REPORT (26) or USAGE_REPORT (33) – MCC supports the following USAGE_REPORT trigger codes, depending on the PCRF's 3GPP TS 29.212 release compliance: – USAGE_REPORT (26) (3GPP TS 29.212 Release 10 or earlier) – USAGE_REPORT (33) (3GPP TS 29.212 Release 11 or later) Specify the PCRF's 3GPP TS 29.212 release compliance when configuring the PCRF charging interface object. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface pcrf-charging <interface-name> pcrf-3gpp-release-compliance [release-10 release-10-plus]</pre>

AVP Name	Event-Trigger
Description (Continued)	■ TAI_CHANGE (26) ■ ECGI_CHANGE (27) The TAI_CHANGE and ECGI_CHANGE event-triggers are applicable for 3GPP 29.212 V10.0.0 or higher. To ensure the PCEF receives and reports the TAI_CHANGE event-trigger with code 26 and the ECGI_CHANGE event-trigger with code 27, set the PCRF's 3GPP TS 29.212 release compliance to <code>release-10-plus</code>. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface pcrf-charging <interface-name> pcrf-3gpp-release-compliance release-10-plus</pre> ■ APN-AMBR_MODIFICATION_FAILURE (29) ■ DEFAULT-EPS-BEARER-QOS_MODIFICATION_FAILURE (34) ■ APPLICATION_START (39) ■ APPLICATION_STOP (40) ■ CS_TO_PS_HANOVER (42) ■ ACCESS_NETWORK_INFO_REPORT (45)

Extended-APN-AMBR-DL

AVP Name	Extended-APN-AMBR-DL
AVP Code	2848
AVP Data Type	Unsigned32
Description	Indicates the maximum aggregate bit rate in Kpbs for the downlink direction across all non-GBR bearers related with the same APN. For more details on this AVP, refer to 3GPP 29.212 Version 15.0.0. <i>Applies to the 5G NSA solution.</i> See “Enabling 5G NSA QoS AVPs” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Extended-APN-AMBR-UL

AVP Name	Extended-APN-AMBR-UL
AVP Code	2849
AVP Data Type	Unsigned32
Description	Indicates the maximum aggregate bit rate in Kpbs for the uplink direction across all non-GBR bearers related with the same APN. For more details on this AVP, refer to 3GPP 29.212 Version 15.0.0. <i>Applies to the 5G NSA solution.</i> See “Enabling 5G NSA QoS AVPs” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Extended-GBR-DL

AVP Name	Extended-GBR-DL
AVP Code	2850
AVP Data Type	Unsigned32
Description	Indicates the guaranteed bit rate (GBR) in Kpbs for a downlink service data flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above. For example, IP, UDP, RTP and RTP payload. For more details on this AVP, refer to 3GPP 29.212 Version 15.0.0. <i>Applies to the 5G NSA solution.</i> See “Enabling 5G NSA QoS AVPs” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Extended-GBR-UL

AVP Name	Extended-GBR-UL
AVP Code	2851
AVP Data Type	Unsigned32
Description	Indicates the guaranteed bit rate in Kpbs for an uplink service data flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above. For example, IP, UDP, RTP and RTP payload. For more details on this AVP, refer to 3GPP 29.212 Version 15.0.0. <i>Applies to the 5G NSA solution.</i> See “Enabling 5G NSA QoS AVPs” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Extended-Max-Requested-BW-DL

AVP Name	Extended-Max-Requested-BW-DL
AVP Code	554
AVP Data Type	Unsigned32
Description	Defines the maximum authorized bandwidth in Kpbs for downlink. For more details on this AVP, refer to 3GPP 29.214 Version 15.1.0. <i>Applies to the 5G NSA solution.</i> See “Enabling 5G NSA QoS AVPs” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Extended-Max-Requested-BW-UL

AVP Name	Extended-Max-Requested-BW-UL
AVP Code	555
AVP Data Type	Unsigned32
Description	Defines the maximum authorized bandwidth in Kpbs for uplink. For more details on this AVP, refer to 3GPP 29.214 Version 15.1.0. <i>Applies to the 5G NSA solution.</i> See “Enabling 5G NSA QoS AVPs” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Feature-List

AVP Name	Feature-List
AVP Code	630
AVP Data Type	Unsigned32
Description	Contains a list of supported features of the origin host.

Feature-List-ID

AVP Name	Feature-List-ID
AVP Code	629
AVP Data Type	Unsigned32
Description	Contains the identity of a feature list. If there are multiple feature lists defined for the Gx reference point, the Feature-List-ID AVP differentiates those lists from one another.

Fixed-User-Location-Info

AVP Name	Fixed-User-Location-Info
AVP Code	2825
AVP Data Type	Grouped
Description	Indicates the UE location in a Fixed Access Network. It is populated with the Circuit Id from the DHCP Option 82 field. <i>Applies to the Gxa interface only.</i> The detailed syntax is as follows: <pre>Fixed-User-Location-Info ::= < AVP Header: 2825 > [Logical-Access-ID]</pre>

Flows

AVP Name	Flows
AVP Code	510
AVP Data Type	Grouped
Description	Indicates the IP flows via their flow identifiers. The detailed syntax is as follows: <pre>Flows ::= < AVP Header: 510 > {Media-Component-Number} *[Flow-Number]</pre>

Flow-Information

AVP Name	Flow-Information
AVP Code	1058
AVP Data Type	Grouped
Description	Sent from the PCRF to the PCEF and contains the information from a single IP flow packet filter. When included in the Application-Detection-Information AVP, the Flow-Information contains the Flow-Description AVP and Flow-Direction AVP. The detailed syntax is as follows: <pre>Flow-Information ::= < AVP Header: 1058 > [Flow-Description] [Packet-Filter-Identifier] [Packet-Filter-Usage] [ToS-Traffic-Class] [Security-Parameter-Index] [Flow-Label] [Flow-Direction] [AVP]</pre>

Flow-Status

AVP Name	Flow-Status
AVP Code	511
AVP Data Type	Enumerated
Description	Defines whether the service data flow is enabled or disabled. For predefined PCC rules, the Flow-Status AVP overrides whatever is configured statically on the gateway. The following values are defined: ■ ENABLED (2) – Used to enable all associated IP flow(s) in both directions. ■ DISABLED (3) – Used to disable all associated IP flow(s) in both directions.

Guaranteed-Bitrate-DL

AVP Name	Guaranteed-Bitrate-DL
AVP Code	1025
AVP Data Type	Unsigned32
Description	Defines the guaranteed bit rate allowed for the downlink direction, in bps.

Guaranteed-Bitrate-UL

AVP Name	Guaranteed-Bitrate-UL
AVP Code	1026
AVP Data Type	Unsigned32
Description	Defines the guaranteed bit rate allowed for the uplink direction, in bps.

GGSN-Address

AVP Name	GGSN-Address
AVP Code	847
AVP Data Type	Address
Description	The Gateway GPRS Support Node/Packet Data Network Gateway (GGSN/PGW) IP address.

IMS-Information

AVP Name	IMS-Information
AVP Code	876
AVP Data Type	Grouped
Description	Provides additional IMS service-specific information elements. The detailed syntax is as follows: <pre>IMS-Information ::= < AVP Header: 876> {Node-Functionality}</pre>

IMSI-Unauthenticated-Flag

AVP Name	IMSI-Unauthenticated-Flag
AVP Code	2308
AVP Data Type	Enumerated
Description	Indicates whether the served IMSI is authenticated. If this AVP is not present, the served IMSI is authenticated. The following values are defined: ■ AUTHENTICATED (0) ■ UNAUTHENTICATED (1)

IP-CAN-Type

AVP Name	IP-CAN-Type
AVP Code	1027
AVP Data Type	Enumerated
Description	Indicates the type of Connectivity Access Network in which the user is connected. The IP-CAN-Type AVP is always present during the IP-CAN session establishment. During an IP-CAN session modification, this AVP is present when there has been a change in the IP-CAN type and the PCRF requested to be informed of this event. The Event-Trigger AVP with value IP-CAN-CHANGE is provided together with the IP-CAN-Type AVP. The following values are defined: ■ 3GPP-GPRS (0) – Indicates that the IP-CAN is associated with a 3GPP GPRS access that is connected to the GGSN based on the Gn/Gp interfaces and is further detailed by the RAT-Type AVP. The RAT-Type AVP includes applicable 3GPP values, except EUTRAN. ■ DOCSIS (1) – Indicates that the IP-CAN is associated with a DOCSIS access. ■ xDSL (2) – Indicates that the IP-CAN is associated with an xDSL access. ■ WiMAX (3) – Indicates that the IP-CAN is associated with a WiMAX access (IEEE 802.16). ■ 3GPP2 (4) – Indicates that the IP-CAN is associated with a 3GPP2 access connected to the 3GPP2 packet core as specified in 3GPP2 X.S0011 [20] and is further detailed by the RAT-Type AVP. ■ 3GPP-EPS (5) – Indicates that the IP-CAN associated with a 3GPP EPS access and is further detailed by the RAT-Type AVP. ■ Non-3GPP-EPS (6) – Indicates that the IP-CAN associated with an EPC based non-3GPP access and is further detailed by the RAT-Type AVP.

Local-Sequence-Number

AVP Name	Local-Sequence-Number
AVP Code	2063
AVP Data Type	Unsigned32
Description	Contains the service data container sequence number, increased by one for each service data container closed.

Logical-Access-Id

AVP Name	Logical-Access-Id
AVP Code	302
AVP Data Type	OctetString
Description	Contains a Circuit ID from the DHCP Option 82 field. <i>Applies to the Gxa interface only.</i>

Low-Priority-Indicator

AVP Name	Low-Priority-Indicator
AVP Code	2602
AVP Data Type	Enumerated
Description	Indicates if the PDN connection has a low priority (for example, for Machine Type Communications). <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The following values are defined: ■ NO (0) ■ YES (1)

Max-Requested-Bandwidth-DL

AVP Name	Max-Requested-Bandwidth-DL
AVP Code	515
AVP Data Type	Unsigned32
Description	Indicates the maximum bit rate allowed for the downlink direction.

Max-Requested-Bandwidth-UL

AVP Name	Max-Requested-Bandwidth-UL
AVP Code	516
AVP Data Type	Unsigned32
Description	Indicates the maximum bit rate allowed for the uplink direction.

Metering-Method

AVP Name	Metering-Method
AVP Code	1007
AVP Data Type	OctetString
Description	Defines which parameters are metered for offline charging. The following values are defined: ■ VOLUME (1) – The volume of the service data flow traffic is metered. If the Metering-Method AVP is omitted but has been supplied previously, the previous information remains valid. If the Metering-Method AVP is omitted and has not been supplied previously, the metering method pre-configured at the PCEF is used as the default metering method.

Monitoring-Key

AVP Name	Monitoring-Key
AVP Code	1066
AVP Data Type	Enumerated
Description	Contains the monitoring key that may apply to the PCC rule. This AVP is used for usage monitoring control purposes as an identifier to a usage monitoring control instance. For more information, see “Usage Monitoring and Reporting” in the <i>Affirmed Networks Gx Interface Specification</i>.

Mute-Notification

AVP Name	Mute-Notification
AVP Code	2809
AVP Data Type	Enumerated
Description	Used by the PCEF to mute the notification to the PCRF of the detected application's Start/Stop for the specific PCC/ADC rule. The following values are defined: MUTE_REQUIRED (0) – Indicates that the PCEF should not inform the PCRF when the application's Start/Stop for the specific PCC/ADC rule is detected.

Network-Request-Support

AVP Name	Network-Request-Support
AVP Code	1024
AVP Data Type	Enumerated
Description	Indicates whether the UE and access network supports the network requested bearer control mode. The following values are defined: ■ NETWORK_REQUEST NOT SUPPORTED (0) – Indicates that the UE and the access network do not support the network initiated bearer establishment request procedure. ■ NETWORK_REQUEST SUPPORTED (1) – Indicates that the UE and the access network support the network initiated bearer establishment request procedure.

Node-Functionality

AVP Name	Node-Functionality
AVP Code	862
AVP Data Type	Enumerated
Description	Includes the functionality identifier of the node. The functionality identifier can be one of the following: ■ S-CSCF (0) ■ P-CSCF (1) ■ I-CSCF (2) ■ MRFC (3) ■ MGCF (4) ■ BGCF (5) ■ AS (6) ■ IBCF (7) ■ S-GW (8) ■ P-GW (9) ■ HSGW (10) ■ E-CSCF (11) ■ MME (12) ■ TRF (13) ■ TF (14) ■ ATCF (15) ■ Proxy Function (16) ■ ePDG (17)

Node-Id

AVP Name	Node-Id
AVP Code	2064
AVP Data Type	UTF8String
Description	Includes an optional, operator-configurable identifier string for the node that had generated the Accounting-Request (ACR) message.

Offline

AVP Name	Offline
AVP Code	1008
AVP Data Type	Enumerated
Description	When included within the Charging-Rule-definition AVP, this AVP indicates whether the offline charging interface from the PCEF for the associated PCC rule should be enabled. The absence of this AVP within the first provisioning of the Charging-Rule-definition AVP of a new PCC rule indicates that the pre-configured default charging method at the PCEF should be used. When included in the CCR-I message at the command level, this AVP indicates the pre-configured default charging method at the PCEF. The absence of this AVP within the CCR-I indicates that the charging method for offline pre-configured at the PCEF is not available. When included in the CCA-I message at the command level, this AVP controls whether to enable/disable offline charging at the PDN session level. The absence of this AVP within the CCA-I indicates that the pre-configured default charging method at the PCEF should be used. See “Configuring the Default Charging Method” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The default charging method provided by the PCRF takes precedence over any pre-configured default charging method at the PCEF. The following values are defined: ■ DISABLE_OFFLINE (0) – Indicates that the offline charging interface for the associated PCC rule is disabled. ■ ENABLE_OFFLINE (1) – Indicates that the offline charging interface for the associated PCC rule is enabled.

Online

AVP Name	Online
AVP Code	1009
AVP Data Type	Enumerated
Description	When included within the Charging-Rule-Definition AVP, this AVP defines whether the online charging interface from the PCEF for the associated PCC rule should be enabled. <i>Note: If all the dynamic PCC rules associated with the Rating Group have online charging disabled, the Rating Group is moved to a blocked state for online charging and no longer performs online charging. When the last active Rating Group is moved to a blocked state, the PGW terminates the Gy session by sending the CCR-T message to the PCRF.</i> When included in the CCR-I message at the command level, this AVP indicates the pre-configured default charging method at the PCEF. The absence of this AVP within the CCR-I indicates that the charging method for online pre-configured at the PCEF is not available. When included in the CCA-I message at the command level, this AVP controls whether to enable/disable online charging at the PDN session level. The absence of this AVP within the CCA-I indicates that the pre-configured default charging method at the PCEF should be used. The default charging method provided by the PCRF takes precedence over any pre-configured default charging method at the PCEF. See “Configuring the Default Charging Method” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ DISABLE_ONLINE (0) – Indicates that the online charging interface for the associated PCC rule is disabled. ■ ENABLE_ONLINE (1) – Indicates that the online charging interface for the associated PCC rule is enabled.

PCC-Rule-Status

AVP Name	PCC-Rule-Status
AVP Code	1019
AVP Data Type	Enumerated
Description	Indicates the status of one PCC rule or a group of PCC rules. The following values are defined: ■ ACTIVE (0) – Used to indicate that the PCC rule(s) are successfully installed (for those provisioned from PCRF) or activated (for those pre-provisioned in PCEF). ■ INACTIVE (1) – Used to indicate that the PCC rule(s) are removed (for those provisioned from PCRF) or inactive (for those pre-provisioned in PCEF). ■ TEMPORARILY INACTIVE (2) – Used to indicate that, for some reason (for example, loss of bearer), already installed or activated PCC rules are temporarily disabled. ■ ACTIVE_WITHOUT_CREDIT_CONTROL (10)

PDN-Connection-Charging-ID

AVP Name	PDN-Connection-Charging-ID
AVP Code	2050
AVP Data Type	Unsigned32
Description	Contains the charging identifier used to identify different records belonging to the same PDN connection. This AVP includes the Charging ID of the first IP-CAN bearer activated within the PDN connection. The PDN-Connection-Charging-ID together with the PGW address uniquely identifies the PDP connection. Applies to the Gy and Gz interfaces.

PDN-Connection-ID

AVP Name	PDN-Connection-ID
AVP Code	1065
AVP Data Type	OctetString
Description	Indicates the PDN connection to which specific information refers. Applies to the Gx interface.

PDP-Address

AVP Name	PDP-Address
AVP Code	1227
AVP Data Type	Address
Description	Contains the IP address associated with the IP CAN bearer session (PDP context/PDN connection).

PDP-Context-Type

AVP Name	PDP-Context-Type
AVP Code	1247
AVP Data Type	Enumerated
Description	Indicates the type of PDP context. The following values are defined: ■ PRIMARY (0) ■ SECONDARY (1)

Precedence

AVP Name	Precedence
AVP Code	1010
AVP Data Type	Unsigned32
Description	Determines the order in which the service data flow templates are applied at service data flow detection at the PCEF. A PCC rule with the Precedence AVP with a lower value is applied before a PCC rule with the Precedence AVP with a higher value.

Pre-emption-Capability

AVP Name	Pre-emption-Capability
AVP Code	1047
AVP Data Type	Enumerated
Description	Indicates whether a service data flow can get resources that were already assigned to another service data flow with a lower priority level. The following values are defined: ■ PRE-EMPTION_CAPABILITY_ENABLED (0) – Indicates that the service data flow is allowed to get resources that were already assigned to another service data flow with a lower priority level. ■ PRE-EMPTION_CAPABILITY_DISABLED (1) – Indicates that the service data flow is not allowed to get resources that were already assigned to another service data flow with a lower priority level. This is the default value applicable if this AVP is not supplied.

Pre-emption-Vulnerability

AVP Name	Pre-emption-Vulnerability
AVP Code	1048
AVP Data Type	Enumerated
Description	Indicates whether a service data flow can lose the resources assigned to it in order to admit a service data flow with higher priority level. The following values are defined: ■ PRE-EMPTION_VULNERABILITY_ENABLED (0) – Indicates that the resources assigned to the service data flow can be pre-empted and allocated to a service data flow with a higher priority level. This is the default value applicable if this AVP is not supplied. ■ PRE-EMPTION_VULNERABILITY_DISABLED (1) – Indicates that the resources assigned to the service data flow can not be pre-empted and allocated to a service data flow with a higher priority level.

Primary-Event-Charging-Function-Name

AVP Name	Primary-Event-Charging-Function-Name
AVP Code	619
AVP Data Type	DiameterURI
Description	Defines the address of the primary online charging system. The protocol definition in the DiameterURI is either omitted or supplied with a value of “Diameter”.

Priority-Level

AVP Name	Priority-Level
AVP Code	1046
AVP Data Type	Unsigned32
Description	Indicates whether a bearer establishment or modification request can be accepted or needs to be rejected in case of resource limitations (typically used for admission control of GBR traffic). The AVP can also be used to decide which existing bearers to pre-empt during resource limitations. The priority level defines the relative importance of a resource request. ■ Values 1 to 15 are defined, with value 1 as the highest level of priority. ■ Values 1 to 8 should only be assigned for services that are authorized to receive prioritized treatment within an operator domain. ■ Values 9 to 15 may be assigned to resources that are authorized by the home network and thus applicable when a UE is roaming.

PS-Furnish-Charging-Information

AVP Name	PS-Furnish-Charging-Information
AVP Code	865
AVP Data Type	Grouped
Description	Contains online charging session specific information. The detailed syntax is as follows: <pre>PS-Furnish-Charging-Information ::= < AVP Header: 865> { 3GPP-Charging-Id } { PS-Free-Format-Data } [PS-Append-Free-Format-Data]</pre>

PS-Information

AVP Name	PS-Information
AVP Code	874
AVP Data Type	Grouped
Description	Allows the transmission of additional PS service specific information elements. For a list of the supported AVPs within the PS-Information AVP on the Gz/Rf interface, see the <i>Affirmed Networks Charging Data Records Specification</i>.

PS-to-CS-Session-Continuity

AVP Name	PS-to-CS-Session-Continuity
AVP Code	1099
AVP Data Type	Enumerated
Description	Indicates whether the service data flow is a candidate for PS-to-CS session continuity as specified in 3GPP TS 23.216 [40]. The following value is defined: VIDEO_PS2CS_CONT_CANDIDATE (0) – Indicates that the service data flow carries video and is a candidate for PS to CS session continuity.

QoS-Class-Identifier

AVP Name	QoS-Class-Identifier
AVP Code	1028
AVP Data Type	Enumerated
Description	Identifies a set of IP-CAN specific QoS parameters that define the authorized QoS, excluding the applicable bit rates and ARP for the IP-CAN bearer or service flow. The allowed values for the nine standard QoS Class Identifiers (QCI) are defined in Table 6.1.7 of 3GPP TS 23.203 [7].

QoS-Information

AVP Name	QoS-Information
AVP Code	1016
AVP Data Type	Grouped
Description	Defines the QoS information for resources requested by the UE, an IP-CAN bearer, PCC rule, QCI or APN. <i>Gx-Interface</i> – When the PCEF sends this AVP to the PCRF, it indicates the requested QoS information associated with the resources requested by the UE, and IP-CAN bearer or the subscribed QoS information at the Access Point Name (APN) level.

Quota-Consumption-Time

AVP Name	Quota-Consumption-Time
AVP Code	881
AVP Data Type	Unsigned32
Description	Contains an idle traffic threshold time in seconds.

Quota-Holding-Time

AVP Name	Quota-Holding-Time
AVP Code	871
AVP Data Type	Unsigned32
Description	Contains the quota holding time in seconds and is contained in the Multiple-Services-Credit-Control AVP.

RAN-End-Timestamp

AVP Name	RAN-End-Timestamp
AVP Code	1301
AVP Data Type	Time
Description	Indicates the time, in UTC format, of the volume container reported was collected, the end time of the reported usage. For more details on this AVP, refer to 3GPP 32.299 Version 15.2.0. <i>Applies to 5G NSA.</i>

RAN-NAS-Release-Cause

AVP Name	RAN-NAS-Release-Cause
AVP Code	2819
AVP Data Type	OctetString
Description	Indicates the Radio Access Network (RAN) or Non-Access Stratum (NAS) release cause code information. The RAN-NAS-Release-Cause AVP is sent from the PCEF to the PCRF.

RAN-Secondary-RAT-Usage-Report

AVP Name	RAN-Secondary-RAT-Usage-Report
AVP Code	1302
AVP Data Type	Grouped
Description	Contains the volume count as reported by the RAN for the secondary RAT (separated for uplink and downlink) including the time of the report. For more details on this AVP, refer to 3GPP 32.299 Version 15.2.0. <i>Applies to 5G NSA.</i> The detailed syntax is as follows: <pre>RAN-Secondary-RAT-Usage-Report ::= < AVP Header: 1302> [Secondary-RAT-Type] [RAN-Start-Timestamp] [RAN-End-Timestamp] [Accounting-Input-Octets] [Accounting-Output-Octets]</pre>

RAN-Start-Timestamp

AVP Name	RAN-Start-Timestamp AVP
AVP Code	1303
AVP Data Type	Time
Description	Indicates the time, in UTC format, of the volume container reported was collected, the start time of the reported usage. For more details on this AVP, refer to 3GPP 32.299 Version 15.2.0. <i>Applies to 5G NSA.</i>

RAT-Type

AVP Name	RAT-Type
AVP Code	1032
AVP Data Type	Enumerated
Description	The Radio Access Technology (RAT) currently serving the UE. Values 0-999 are used for generic radio access technologies that can apply to different IP-CAN types and are not IP-CAN specific. Notes: ■ Values 1000-1999 are used for 3GPP specific radio access technology types. ■ Values 2000-2999 are used for 3GPP2 specific radio access technology types. ■ The informative Annex C presents a mapping between the code values for different access network types. The following values are defined: ■ WLAN (0) – Indicates that the RAT is WLAN. ■ UTRAN (1000) – Indicates that the RAT is UTRAN. For further details, refer to 3GPP TS 29.060 [18]. ■ GERAN (1001) – Indicates that the RAT is GERAN. For further details, refer to 3GPP TS 29.060 [18]. ■ GAN (1002) – Indicates that the RAT is GAN. For further details, refer to 3GPP TS 29.060 [18] and 3GPP TS 43.318 [29]. ■ HSPA_EVOLUTION (1003) – Indicates that the RAT is HSPA Evolution. For further details, refer to 3GPP TS 29.060 [18]. ■ EUTRAN (1004) – Indicates that the RAT is EUTRAN. For further details, refer to 3GPP TS 29.274 [22]. ■ EUTRAN-NB-IoT (1005) – Indicates that the RAT is NB-IoT. For further details refer to 3GPP TS 29.274 [22]. ■ CDMA2000_1X (2000) – Indicates that the RAT is CDMA2000 1X. For further details, refer to 3GPP2 X.S0011 [20]. ■ HRPD (2001) – Indicates that the RAT is HRPD. For further details, refer to 3GPP2 X.S0011 [20]. ■ UMB (2002) – Indicates that the RAT is UMB. For further details, refer to 3GPP2 X.S0011 [20]. ■ EHRPD (2003) – Indicates that the RAT is eHRPD. For further details, refer to 3GPP2 X.S0057 [24].

Rate-Control-Max-Message-Size

AVP Name	Rate-Control-Max-Message-Size
AVP Code	3937
AVP Data Type	Unsigned32
Description	Indicates the maximum data PDU size in octets. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i>

Rate-Control-Max-Rate

AVP Name	Rate-Control-Max-Rate
AVP Code	3938
AVP Data Type	Unsigned32
Description	Indicates the maximum number of data PDUs per time unit indicated in the Rate-Control-Time-Unit AVP. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i>

Rate-Control-Time-Unit

AVP Name	Rate-Control-Time-Unit
AVP Code	3939
AVP Data Type	Unsigned32
Description	Indicates a time unit for rate control. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The following values are defined: ■ UNRESTRICTED (0) – This value indicates the number of unrestricted data PDUs. ■ MINUTE (1) ■ HOUR (2) ■ DAY (3) ■ WEEK (4)

Redirect-Information

AVP Name	Redirect-Information
AVP Code	1085
AVP Data Type	Grouped
Description	Indicates whether the detected application traffic should be redirected to another controlled address. <i>Note: The Redirect-Information AVP sent from the PCRF as part of the Charging-Rule-Definition/ADC-Rule-Definition AVP applies to predefined PCC rules only and applies to both online and offline PCC rules. When redirect information is received for an online rule, all HTTP traffic matching that rule's rating group will be redirected.</i> The detailed syntax is as follows: <pre>Redirect-Information ::= < AVP Header: 1085 > [Redirect-Support] [Redirect-Address-Type] [Redirect-Server-Address] * [AVP]</pre>

Redirect-Support

AVP Name	Redirect-Support
AVP Code	1086
AVP Data Type	Enumerated
Description	Indicates whether redirection is disabled or enabled for a predefined PCC rule. <i>Note: The Redirect-Support AVP sent from the PCRF as part of the Redirection-Information AVP applies to predefined PCC rules only and is currently not supported for dynamic PCC rules.</i> The following values are defined: ■ REDIRECTION_DISABLED (0) – Indicates that redirection is disabled for a predefined PCC rule. ■ REDIRECTION_ENABLED (1) – Indicates that redirection is enabled for a predefined PCC rule. This is the default value applicable if a Redirect-Information AVP is provided for the first time and if this AVP is not supplied.

Reporting-Level

AVP Name	Reporting-Level
AVP Code	1011
AVP Data Type	Enumerated
Description	Defines on what level the PCEF reports the usage for the related PCC rule. The following values are defined: ■ SERVICE_IDENTIFIER_LEVEL (0) – Indicates that the usage is reported at the service-id and rating-group level and is applicable when the Service-Identifier and Rating-Group are provisioned within the Charging-Rule-Definition AVP. ■ RATING_GROUP_LEVEL (1) – Indicates that the usage is reported at the rating-group level and is applicable when the Rating-Group is provisioned within the Charging-Rule-Definition AVP.

Reporting-Reason

AVP Name	Reporting-Reason
AVP Code	872
AVP Data Type	Enumerated
Description	Specifies the reason for usage reporting for one or more types of quota for a particular category. It can occur directly in the Multiple-Services-Credit-Control AVP or in the Used-Service-Units AVP within a Credit Control Request command reporting credit usage.

Required-Access-Info

AVP Name	Required-Access-Info
AVP Code	536
AVP Data Type	Enumerated
Description	Used by the PCRF to request that the PCEF report the access network information required for that Application Function (AF) session. The following values are defined: ■ USER_LOCATION (0) – Indicates that the PCEF should report the user location information to the PCRF. If the user location information is available, the PCEF reports the user location information within the 3GPP-User-Location-Info AVP and the time when it was last known in the User-Location-Info-Time AVP (if available). If the user location information is not available, the PCEF provides the serving PLMN identifier within the 3GPP-SGSN-MCC-MNC AVP. ■ MS_TIME_ZONE (1) – Indicates that the PCEF should report the user timezone information to the PCRF. The PCEF reports the user timezone information within the 3GPP-MS-TimeZone AVP.

Resource-Allocation-Notification

AVP Name	Resource-Allocation-Notification
AVP Code	1063
AVP Data Type	Enumerated
Description	Included within the Charging-Rule-Install AVP and defines whether the rules included within the Charging-Rule-Install AVP need notification. The following value is defined: ■ ENABLE_NOTIFICATION (0) – Indicates that the allocation of resources for the related PCC rules are confirmed.

Revalidation-Time

AVP Name	Revalidation-Time
AVP Code	1042
AVP Data Type	Time
Description	Sent from the PCRF to the GGSN/PGW in CCA and RAR messages to indicate the NTP time before which the GGSN/PGW should issue a new request for PCC rules from the PCRF. The Revalidation-Time AVP is always sent with the REVALIDATION_TIMEOUT event trigger. <i>Note: The PCRF can disable the revalidation timer on the GGSN/PGW by not including the REVALIDATION_TIMEOUT event-trigger in the CCA or RAR message.</i>

Rule-Activation-Time

AVP Name	Rule-Activation-Time
AVP Code	1043
AVP Data Type	Time
Description	Sent from the PCRF to the PCEF in CCA and RAR messages to indicate the NTP time at which the PCC/ADC rule has to be enforced. The AVP is included in the Charging-Rule-Install AVP/ADC-Rule-Install AVP and is applicable for all the PCC/ADC rules included within the Charging-Rule-Install AVP/ADC-Rule-Install AVP.

Rule-Deactivation-Time

AVP Name	Rule-Deactivation-Time
AVP Code	1044
AVP Data Type	Time
Description	Sent from the PCRF to the PCEF in CCA and RAR messages to indicate the NTP time at which the PCEF has to stop enforcing the PCC/ADC rule. The AVP is included in Charging-Rule-Install AVP/ADC-Rule-Install AVP and is applicable for all the PCC/ADC rules included within the Charging-Rule-Install AVP/ADC-Rule-Install AVP.

Rule-Failure-Code

AVP Name	Rule-Failure-Code
AVP Code	1031
AVP Data Type	Enumerated
Description	Sent by the PCEF to the PCRF within a Charging-Rule-Report AVP to identify the reason a PCC Rule is being reported. The following values are defined: UNKNOWN_RULE_NAME (1) – Indicates that the pre-provisioned PCC rule could not be successfully activated because the Charging-Rule-Name or Charging-Rule-Base-Name is unknown to the PCEF. RATING_GROUP_ERROR (2) – Indicates that the PCC rule could not be successfully installed or enforced because the Rating-Group specified within the Charging-Rule-Definition AVP by the PCRF is unknown or invalid. SERVICE_IDENTIFIER_ERROR (3) – Indicates that the PCC rule could not be successfully installed or enforced because the Service-Identifier specified within the Charging-Rule-Definition AVP by the PCRF is invalid, unknown, or not applicable to the service being charged. GW/PCEF_MALFUNCTION (4) – Indicates that the PCC rule could not be successfully installed (for those provisioned from the PCRF) or activated (for those pre-provisioned in PCEF) or enforced (for those already successfully installed) due to GW/PCEF malfunction. RESOURCES_LIMITATION (5) – Indicates that the PCC rule could not be successfully installed (for those provisioned from PCRF) or activated (for those pre-provisioned in PCEF) or enforced (for those already successfully installed) due to a limitation of resources at the PCEF. MAX_NR_BEARERS_REACHED (6) – Indicates that the PCC rule could not be successfully installed (for those provisioned from PCRF) or activated (for those pre-provisioned in PCEF) or enforced (for those already successfully installed) due to the fact that the maximum number of bearers has been reached for the IP-CAN session. UNKNOWN_BEARER_ID (7) – Indicates that the PCC rule could not be successfully installed or enforced at the PCEF because the Bearer-Id specified within the Charging-Rule-Install AVP by the PCRF is unknown or invalid. Applicable only for GPRS in the case the PCRF performs the bearer binding. MISSING_BEARER_ID (8) – Indicates that the PCC rule could not be successfully installed or enforced at the PCEF because the Bearer-Id is not specified within the Charging-Rule-Install AVP by the PCRF. Applicable only for GPRS in the case the PCRF performs the bearer binding. MISSING_FLOW_DESCRIPTION (9) – Indicates that the PCC rule could not be successfully installed or enforced because the Flow-Information AVP is not specified within the Charging-Rule-Definition AVP by the PCRF during the first install request of the PCC rule. RESOURCE_ALLOCATION_FAILURE (10) – Indicates that the PCC rule could not be successfully installed or maintained since the bearer establishment/modification failed, or the bearer was released.

AVP Name	Rule-Failure-Code
Description (Continued)	UNSUCCESSFUL_QOS_VALIDATION (11) – Indicates that the QoS validation has failed. PS_TO_CS_HANDOVER (13) – Indicates that the PCC rule could not be maintained because of PS to CS handover. Applicable only for 3GPP-GPRS and 3GPP-EPS.

SCS-Address

AVP Name	SCS-Address
AVP Code	3941
AVP Data Type	Address
Description	Contains the IP address of the SCS/AS. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i>

SCS-AS-Address

AVP Name	SCS-AS-Address
AVP Code	3940
AVP Data Type	Grouped
Description	Indicates the address of the Service Capability Server/Application Server (SCS/AS). <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The detailed syntax is as follows: <pre>SCS-AS-Address ::= < AVP Header: 3940> [SCS-Realm] [SCS-Address]</pre>

SCS-Realm

AVP Name	SCS-Realm
AVP Code	3942
AVP Data Type	DiameterIdentity
Description	Indicates the Diameter Realm Identity of the SCS/AS. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i>

Secondary-Event-Charging-Function-Name

AVP Name	Secondary-Event-Charging-Function-Name
AVP Code	620
AVP Data Type	DiameterURI
Description	Defines the address of the secondary online charging system for the bearer. The protocol definition in the DiameterURI is either omitted or supplied with a value of “Diameter”.

Secondary-RAT-Type

AVP Name	Secondary-RAT-Type
AVP Code	1304
AVP Data Type	OctetString
Description	Contains the Secondary RAT Type value, as provided by the RAN. The field is provided by the RAN and transferred to the SGW/PGW in the RAN Traffic Volume element. For more details on this AVP, refer to 3GPP 32.299 Version 15.2.0. <i>Applies to 5G NSA.</i> The following values are defined: ■ Reserved (0) ■ 5G NR (1)

Service-Data-Container

AVP Name	Service-Data-Container
AVP Code	2040
AVP Data Type	Grouped
Description	Provides a list of changes in charging conditions reported for flow-based charging. When there is a change in charging conditions, this container identifies the volume count (separated for uplink and downlink), elapsed time or number of events, per service data flow identified per rating group or combination of the rating group and service ID within an IP-CAN bearer. The detailed syntax is as follows: <pre>Service-Data-Container:: = < AVP Header: 2040> [AF-Correlation-Information] [Charging-Rule-Base-Name] [Accounting-Input-Octets] [Accounting-Output-Octets] [Accounting-Input-Packets] [Accounting-Output-Packets] [Local-Sequence-Number] [QoS-Information] [Rating-Group] [Change-Time] [Service-Identifier] [Service-Specific-Info] [SGSN-Address] [Time-First-Usage] [Time-Last-Usage] [Time-Usage] * [Change-Condition] [3GPP-User-Location-Info] [3GPP-RAT-Type] [Serving-PLMN-Rate-Control] [APN-Rate-Control]</pre>

Service-Information

AVP Name	Service-Information
AVP Code	873
AVP Data Type	Grouped
Description	Contains service-specific parameters. The format and the contents of the fields inside the Service-Information AVP are specified in the middle-tier documents which are applicable for the specific service.

Service-Specific-Info

AVP Name	Service-Specific-Info
AVP Code	1249
AVP Data Type	Grouped
Description	Contains service-specific data if provided by an Application Server or a PCEF only for predefined PCC rules. The detailed syntax is as follows: <pre>Service-Specific-Info ::= < AVP Header: 1249 > [Service-Specific-Data] [Service-Specific-Type]</pre>

Serving-Node-Type

AVP Name	Serving-Node-Type
AVP Code	2047
AVP Data Type	Enumerated
Description	Indicates the type of serving node. The following values are defined: ■ SGSN (0) ■ PMIPSGW (1) ■ GTPSGW (2) ■ ePDG (3) ■ hSGW (4) ■ MME (5)

Serving-PLMN-Rate-Control

AVP Name	Serving-PLMN-Rate-Control
AVP Code	4310
AVP Data Type	Grouped
Description	Indicates the Serving PLMN Rate Control used by the MME. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The detailed syntax is as follows: <pre>Serving-PLMN-Rate-Control ::= <AVP header: 4310 10415> [Uplink-Rate-Limit] [Downlink-Rate-Limit]</pre> A Downlink-Rate-Limit set to 0 indicates that the Serving PLMN Rate Control for downlink messages is deactivated in the MME. If the Serving PLMN Rate Control is activated, the value of the Downlink-Rate-Limit should not be less than 10, see 3GPP TS 23.401 [25]. An Uplink-Rate-Limit set to 0 indicates that the Serving PLMN Rate Control for uplink messages is deactivated in the MME. If Serving PLMN Rate Control is activated, the value of the Uplink-Rate-Limit should not be less than 10, see 3GPP TS 23.401 [25].

SGi-PtP-Tunnelling-Method

AVP Name	SGi-PtP-Tunnelling-Method
AVP Code	3931
AVP Data Type	Enumerated
Description	Indicates whether the SGi PtP tunnelling method is based on UDP/IP or other methods for a non-IP PDN type PDN connection. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The following values are defined: ■ UDP_IP_BASED (0) ■ OTHERS (1)

Session-Release-Cause

AVP Name	Session-Release-Cause
AVP Code	1045
AVP Data Type	Enumerated
Description	Indicates the reason for terminating the IP-CAN session by the PCRF. The following values are defined: ■ UNSPECIFIED_REASON (0) – Used for unspecified reasons. ■ UE_SUBSCRIPTION_REASON (1) – Indicates that the subscription of the UE has changed (for example, has been removed) and the session needs to be terminated. ■ INSUFFICIENT_SERVER_RESOURCES (2) – Indicates that the server is overloaded and needs to abort the session.

SGSN-Address

AVP Name	SGSN-Address
AVP Code	1228
AVP Data Type	Address
Description	Indicates the IP address of the Serving GPRS Support Node (SGSN). This AVP can be used to identify the Public Land Mobile Network (PLMN) of the SGSN to which a user logs in.

SGW-Address

AVP Name	SGW-Address
AVP Code	2067
AVP Data Type	Address
Description	Indicates the IP-address of the SGW Node.

SGW-Change

AVP Name	SGW-Change
AVP Code	2065
AVP Data Type	Enumerated
Description	Indicates this is the first ACR Start message after an SGW change. If this AVP is not present in the ACR Start message, this ACR Start is not due to an SGW change. The following values are defined: ■ ACR START NOT DUE TO SGW CHANGE (0) ■ ACR START DUE TO SGW CHANGE (1)

Sponsor-Identity

AVP Name	Sponsor-Identity
AVP Code	531
AVP Data Type	UTF8String
Description	Used for sponsored data connectivity purposes as an identifier of the sponsor.

Sponsored-Connectivity-Data

AVP Name	Sponsored-Connectivity-Data
AVP Code	530
AVP Data Type	Grouped
Description	Contains the data associated with the sponsored data connectivity. The detailed syntax is as follows: <pre>Sponsored-Connectivity-Data ::= < AVP Header: 530 > [Sponsor-Identity] [Application-Service-Provider-Identity] [Granted-Service-Unit] * [AVP]</pre>

Supported-Features

AVP Name	Supported-Features
AVP Code	628
AVP Data Type	Grouped
Description	Informs the destination host about the features that the origin host requires to successfully complete this command exchange. <i>Note:</i> When a Supported-Features AVP is used to identify features that have been defined by 3GPP, the Vendor-Id AVP contains the vendor ID of 3GPP (10415). The detailed syntax is as follows: <pre>Supported-Features ::= < AVP header: 628 10415 > {Vendor-Id} {Feature-List-ID} {Feature-List}</pre>

TDF-Application-Identifier

AVP Name	TDF-Application-Identifier
AVP Code	1088
AVP Data Type	OctetString
Description	References the application detection filter to which the PCC rule for Application Detection and Control (ADC) in the PCEF applies. This AVP also identifies the application when reporting application Start/Stop notifications to the PCRF. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

TDF-Application-Instance-Identifier

AVP Name	TDF-Application-Instance-Identifier
AVP Code	2802
AVP Data Type	OctetString
Description	This value is dynamically assigned by the PCEF to correlate the application Start/Stop events for use with the Application Detection and Control (ADC) functionality. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Time-First-Usage

AVP Name	Time-First-Usage
AVP Code	2043
AVP Data Type	Time
Description	Contains the time in UTC format for the first IP packet to be transmitted and mapped to the current service data container.

Time-Last-Usage

AVP Name	Time-Last-Usage
AVP Code	2044
AVP Data Type	Time
Description	Contains the time in UTC format for the last IP packet to be transmitted and mapped to the current service data container.

Time-Quota-Mechanism

AVP Name	Time-Quota-Mechanism
AVP Code	1270
AVP Data Type	Grouped
Description	The OCS may include this AVP in the Multiple-Services-Credit-Control AVP, when granting time quota. The detailed syntax is as follows: <pre>Time-Quota-Mechanism ::= < AVP Header: 1270> {Time-Quota-Type} {Base-Time-Interval}</pre>

Time-Quota-Threshold

AVP Name	Time-Quota-Threshold
AVP Code	868
AVP Data Type	Unsigned32
Description	Contains a threshold value in seconds.

Time-Quota-Type

AVP Name	Time-Quota-Type
AVP Code	1271
AVP Data Type	Enumerated
Description	Indicates the time quota consumption mechanism used for the associated rating group. The following values are defined: ■ DISCRETE_TIME_PERIOD (0) ■ CONTINUOUS_TIME_PERIOD (1)

Time-Usage

AVP Name	Time-Usage
AVP Code	2045
AVP Data Type	Time
Description	Contains the effective used time within the service data container reporting interval.

Traffic-Data-Volumes

AVP Name	Traffic-Data-Volumes
AVP Code	2046
AVP Data Type	Grouped
Description	When a change in charging conditions occurs for this IP-CAN bearer, this container reports the volume count (separated for uplink and downlink). The detailed syntax is as follows: <pre>Traffic-Data-Volumes ::= < AVP Header: 2046> [QoS-Information] [Accounting-Input-Octets] [Accounting-Input-Packets] [Accounting-Output-Octets] [Accounting-Output-Packets] [Change-condition] [Change-Time] [3GPP-User-Location-Info] [3GPP-RAT-Type] [CP-CIoT-EPS-Optimisation-Indicator] [Serving-PLMN-Rate-Control]</pre>

Trigger

AVP Name	Trigger
AVP Code	1264
AVP Data Type	Grouped
Description	Contains the trigger types. The detailed syntax is as follows: <pre>Trigger ::= < AVP Header: 1264 > *[Trigger-Type]</pre>

Trigger-Type

AVP Name	Trigger-Type
AVP Code	870
AVP Data Type	Enumerated
Description	When included in the CCA on the Gy interface, the Trigger-Type AVP indicates the events that cause the credit control client to re-authorize the associated quota. When included in the CCA on the Gx interface, the Trigger-Type VSA is used to notify the PGW when a ULI, Time Zone, or PLMN change occurs and the PGW is required to send the respective event triggers over the Gy interface to the OCS. For more information, see Trigger-Type (VSA) (page A-129).

TWAN-User-Location-Info

AVP Name	TWAN-User-Location-Info
AVP Code	2714
AVP Data Type	Grouped
Description	Contains the UE location in a Trusted WLAN Access Network (TWAN) as TWAN Identifier defined in TS 29.274 [226] (TS 32.299 v15.3). Sent from the TWAG/TWAP to the OCS via the Gy interface. The detailed syntax is as follows: <pre>TWAN-User-Location-Info ::= < AVP Header: 2714 > {SSID} [BSSID]</pre> The Twan-User-Location-Info AVP is configurable in Gy CCR data record templates and appears as psinfotwanuserlocationinfo. The RADIUS Attribute Mapping template must be configured so that the TWAG extracts the SSID/BSSID received in the Called-Station-ID AVP (within the RADIUS Access-Request) and includes it in the Twan-User-Location-Info AVP on the Gy interface. For example: 1 Configure the attributes in the incoming RADIUS messages. <pre>an3000-mcm-slot7cpu1(config)# service-construct radius-mapping template <template-name> attribute calledstationid to-format (.*):(.*) to-list [apbssid apssid]</pre> 2 Associate the RADIUS Attribute Mapping Template with the Session Learner interface. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface gw-session-learner [non-3gpp-access-diameter-client radius-proxy twag- radius-client] <interface-name> radius-server-trigger radius-mapping access-request <template-name></pre> <i>Note: For more information on configuring the RADIUS Attribute Mapping template, see the Affirmed Networks TWAG/TWAP Administration Guide.</i>

UNI-PDU-CP-Only-Flag

AVP Name	UNI-PDU-CP-Only-Flag
AVP Code	3932
AVP Data Type	Enumerated
Description	Indicates whether this PDN connection is applied with the “Control Plane Only flag”. For example, using only the S11-U from the SGW when Control Plane Clot EPS Optimization is enabled. If this AVP is not present, then both the user plane and control plane can be used in UNI for the PDU transfer. For example, using both S1-U and S11-U respectively from the SGW when Control Plane Clot EPS Optimization is enabled. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The following values are defined: ■ UNI-PDU-both-UP-CP (0) ■ UNI-PDU-CP-Only (1)

Unit-Quota-Threshold

AVP Name	Unit-Quota-Threshold
AVP Code	1226
AVP Data Type	Unsigned32
Description	Contains a threshold value in service specific units.

Uplink-Rate-Limit

AVP Name	Uplink-Rate-Limit
AVP Code	4311
AVP Data Type	Unsigned32
Description	Contains the maximum number of NAS Data PDUs per deci hour for this UE for uplink. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i>

Usage-Monitoring-Information

AVP Name	Usage-Monitoring-Information
AVP Code	1067
AVP Data Type	Grouped
Description	Contains the usage monitoring control information. The detailed syntax is as follows: <pre>Usage-Monitoring-Information ::= < AVP Header: 1067 > [Monitoring-Key] [Granted-Service-Unit] [Used-Service-Unit] [Usage-Monitoring-Level] [Usage-Monitoring-Report] [Usage-Monitoring-Support] * [AVP]</pre>

Usage-Monitoring-Level

AVP Name	Usage-Monitoring-Level
AVP Code	1068
AVP Data Type	Enumerated
Description	Used by the PCRF to indicate whether the usage monitoring instance applies to the IP-CAN session or to one or more PCC rules. The following values are defined: ■ SESSION_LEVEL (0) – This value, if provided within an RAR or CCA command, indicates that the usage monitoring instance applies to the entire IP-CAN session. ■ PCC_RULE_LEVEL (1) – This value, if provided within an RAR or CCA command, indicates that the usage monitoring instance applies to one or more PCC rules. <i>Note: The absence of the Usage-Monitoring-Level AVP indicates the value PCC_RULE_LEVEL (1).</i>

Usage-Monitoring-Report

AVP Name	Usage-Monitoring-Report
AVP Code	1069
AVP Data Type	Enumerated
Description	Used by the PCRF to request that the PCEF report the accumulated usage for the usage monitoring key included in the Monitoring-Key AVP (included within the Usage-Monitoring-Information AVP). The following value is defined: USAGE_MONITORING_REPORT_REQUIRED (0) – This value, if provided within an RAR or CCA command, indicates that the PCEF should report accumulated usage regardless of whether a usage threshold is reached for a certain usage monitoring key.

Usage-Monitoring-Support

AVP Name	Usage-Monitoring-Support
AVP Code	1070
AVP Data Type	Enumerated
Description	Used by the PCRF to indicate whether usage monitoring is disabled for a certain Monitoring Key. The following value is defined: USAGE_MONITORING_DISABLED (0) – This value indicates that usage monitoring is disabled for a monitoring key.

User-CSG-Information

AVP Name	User-CSG-Information
AVP Code	2319
AVP Data Type	Grouped
Description	Contains the Closed Subscriber Group (CSG) information associated with CSG cell access, including the CSG ID within the PLMN, the access mode and the CSG membership indication for the user when hybrid access applies. The detailed syntax is as follows: <pre>User-CSG-Information :: = < AVP Header: 2319> {CSG-Id} [CSG-Access-Mode] [CSG-Membership-Indication]</pre>

User-Location-Info-Time

AVP Name	User-Location-Info-Time
AVP Code	2812
AVP Data Type	Time
Description	Contains the NTP time at which the UE was last known to be in the location, which is reported during bearer deactivation or IP-CAN session termination. The User-Location-Info-Time AVP is sent from the PCEF to the PCRF.

UWAN-User-Location-Info

AVP Name	UWAN-User-Location-Info
AVP Code	3918
AVP Data Type	Grouped
Description	Contains the UE location in an Untrusted Wireless Access Network (UWAN). The UWAN User Location Information includes the UE local IP address and optionally UDP source port number (if NAT is detected). The detailed syntax is as follows: <pre> UWAN-User-Location-Info ::= < AVP Header: 3918> { UE-Local-IP-Address } [UDP-Source-Port] </pre>

Volume-Quota-Threshold

AVP Name	Volume-Quota-Threshold
AVP Code	869
AVP Data Type	Unsigned32
Description	Contains a threshold value in octets.

NASREQ/RADIUS AVP Definitions

This section describes the following NASREQ/RADIUS AVPs in the CCR and CCA messages.

- [Called-Station-Id \(page A-102\)](#)
- [Framed-Interface-Id \(page A-102\)](#)
- [Framed-IP-Address \(page A-102\)](#)
- [Framed-IPv6-Prefix \(page A-103\)](#)

Called-Station-Id

AVP Name	Called-Station-Id
AVP Code	30
AVP Data Type	UTF8String
Description	The gateway support node of the GPRS. The identifier for the target network: APN.

Framed-Interface-Id

AVP Name	Framed-Interface-Id
AVP Code	96
AVP Data Type	Unsigned64
Description	The user IPv6 interface identifier.

Framed-IP-Address

AVP Name	Framed-IP-Address
AVP Code	8
AVP Data Type	OctetString
Description	The IP address allocated for this user.

Framed-IPv6-Prefix

AVP Name	Framed-IPv6-Prefix
AVP Code	97
AVP Data Type	OctetString
Description	The IPv6 address prefix allocated for this user.

Vodafone AVP Definitions

This section describes the following Vodafone AVPs in the CCR and CCA messages. The AVPs include Vendor-ID of 12645 (Vodafone).

- [vf-Base-Time-Interval \(page A-105\)](#)
- [vf-Context-Type \(page A-105\)](#)
- [vf-Envelopes \(page A-105\)](#)
- [vf-Envelope-End-Time \(page A-105\)](#)
- [vf-Envelope-Reporting \(page A-106\)](#)
- [vf-Envelope-Start-Time \(page A-106\)](#)
- [vf-Quota-Consumption-Time \(page A-106\)](#)
- [vf-Quota-Holding-Time \(page A-107\)](#)
- [vf-Radio-Access-Technology \(page A-107\)](#)
- [vf-Reporting-Reason \(page A-108\)](#)
- [vf-Rulebase-Id \(page A-108\)](#)
- [vf-Time-Of-First-Usage \(page A-109\)](#)
- [vf-Time-Of-Last-Usage \(page A-109\)](#)
- [vf-Time-Quota-Mechanism \(page A-109\)](#)
- [vf-Time-Quota-Threshold \(page A-109\)](#)
- [vf-Time-Quota-Type \(page A-110\)](#)
- [vf-Trigger \(page A-110\)](#)
- [vf-Trigger-Type \(page A-110\)](#)
- [vf-Unit-Quota-Threshold \(page A-111\)](#)
- [vf-User-Location-Information \(page A-111\)](#)
- [vf-Volume-Quota-Threshold \(page A-111\)](#)

vf-Base-Time-Interval

AVP Name	Base-Time-Interval
AVP Code	276
AVP Data Type	Unsigned32
Description	Contains the length of the base time interval, for controlling the consumption of time quota in seconds.

vf-Context-Type

AVP Name	Context-Type
AVP Code	256
AVP Data Type	Enumerated
Description	This AVP indicates whether the PDP context corresponding to the CC session is a primary or secondary. The following values are defined: ■ PRIMARY (0) ■ SECONDARY (1)

vf-Envelopes

AVP Name	Envelopes
AVP Code	269
AVP Data Type	Grouped
Description	The Envelope AVP reports the start and end time of one time envelope using the Envelope-Start-Time and Envelope-End-Time AVPs. Further details of its usage are described in 3GPP 32.299.

vf-Envelope-End-Time

AVP Name	Envelope-End-Time
AVP Code	271
AVP Data Type	Time
Description	The Envelope-End-Time AVP is set to the time of the end of the time envelope.

vf-Envelope-Reporting

AVP Name	Envelope-Reporting
AVP Code	272
AVP Data Type	Enumerated
Description	This AVP is of type Enumerated and is used in the initial CCA to indicate whether the client reports the start and end of each time envelope, in those cases in which quota is consumed in envelopes. The following values are defined: ■ DO_NOT_REPORT_ENVELOPES (0) ■ REPORT_ENVELOPES (1) ■ REPORT_ENVELOPES_WITH_VOLUME (2)

vf-Envelope-Start-Time

AVP Name	Envelope-Start-Time
AVP Code	270
AVP Data Type	Time
Description	The Envelope-Start-Time AVP is set to the time of the packet of user data which caused the time envelope to start.

vf-Quota-Consumption-Time

AVP Name	Quota-Consumption-Tim
AVP Code	257
AVP Data Type	Unsigned32
Description	This AVP contains an idle traffic threshold time in seconds. This AVP may be included within the Multiple-Services-Credit-Control AVP which also contains a Granted-Service-Units AVP containing a CC-Time AVP (for example, when the granted quota is a time quota).

vf-Quota-Holding-Time

AVP Name	Quota-Holding-Time
AVP Code	258
AVP Data Type	Unsigned32
Description	Contains the length of time (in seconds) for which the system can hold quota during a period of inactivity. A quota expires when no traffic associated with the quota is observed for the value indicated by this AVP. The system stops the timer when it sends a CCR message and resets the timer each time it receives a CCA with a Quota-Holding-Time value.

vf-Radio-Access-Technology

AVP Name	Radio-Access-Technology
AVP Code	260
AVP Data Type	Enumerated
Description	This AVP specifies the radio access technology being used by the user. The following values are defined: ■ UTRAN (1) ■ GERAN (2) ■ WLAN (3)

vf-Reporting-Reason

AVP Name	Reporting-Reason
AVP Code	261
AVP Data Type	Enumerated
Description	The Reporting-Reason specifies the reason for usage reporting for one or more types of quota for a rating group. It can occur directly in the Multiple-Services-Credit-Control AVP, or in the Used-Service-Units AVP. The following values apply to all quota types and are used in the Multiple-Services-Credit-Control AVP: ■ QHT ■ FINAL ■ VALIDITY_TIME ■ RATING_CONDITION_CHANGE The following values apply to one particular quota type and are used in the Used-Service-Units AVP: ■ THRESHOLD ■ QUOTA_EXHAUSTED ■ OTHER_QUOTA_TYPE The following values are defined for the Reporting-Reason AVP: ■ THRESHOLD 0 ■ QHT 1 ■ FINAL 2 ■ QUOTA_EXHAUSTED 3 ■ VALIDITY_TIME 4 ■ OTHER_QUOTA_TYPE 5 ■ RATING_CONDITION_CHANGE 6 ■ FORCED_REAUTHORISATION 7

vf-Rulebase-Id

AVP Name	Rulebase-Id
AVP Code	262
AVP Data Type	UTF8String
Description	Contains the identifier of the currently selected rule base. <i>Note: All charging instances configured under <code>zone <name> gateway apn <name> charging</code> are included in the CCR-I to the OCS. If no charging instances are configured and the system receives charging instances from external interfaces, those charging instances are included in the CCR-I. If no charging instances are available, the data profile charging instance is used.</i>

vf-Time-Of-First-Usage

AVP Name	Time-Of-First-Usage
AVP Code	263
AVP Data Type	Time
Description	This AVP is a time-stamp identifying the date and time of the first packet of usage of the quota being reported for the rating group.

vf-Time-Of-Last-Usage

AVP Name	Time-Of-Last-Usage
AVP Code	264
AVP Data Type	Time
Description	This AVP is a time-stamp identifying the date and time of the last packet of usage of the quota being reported for the rating group.

vf-Time-Quota-Mechanism

AVP Name	Time-Quota-Mechanism
AVP Code	274
AVP Data Type	Grouped
Description	This AVP may be used by the OCS for granting time quota. Time-Quota-Mechanism AVP takes precedence over Quota-Consumption-Time.

vf-Time-Quota-Threshold

AVP Name	Time-Quota-Threshold
AVP Code	259
AVP Data Type	Unsigned32
Description	The AVP contains a value indicating the quota threshold for time quota in seconds.

vf-Time-Quota-Type

AVP Name	Time-Quota-Type
AVP Code	275
AVP Data Type	Enumerated
Description	The Time-Quota-Type is used to indicate which time quota consumption mechanism is used for the associated rating group.

vf-Trigger

AVP Name	Trigger
AVP Code	265
AVP Data Type	Grouped
Description	This AVP is used to arm the triggers in the client and to identify the triggers which caused re-authorization in usage. On receipt of the AVP, the OCS arms all known triggers present in the AVP and resets all known triggers missing from the AVP. The presence of the Trigger AVP without any Trigger-Type AVP in a CCA allows the OCS to disable all the triggers. The presence of the Trigger AVP in the CCR identifies the event(s) triggering the CCR.

vf-Trigger-Type

AVP Name	Trigger-Type
AVP Code	266
AVP Data Type	Enumerated
Description	When included in the CCA, the Trigger-Type AVP indicates the events that cause the credit control client to re-authorize the associated quota. The system supports the following triggers between the system and the OCS. ■ CHANGE_IN_QOS (2) – Specifies whether a QoS change triggers an update. ■ CHANGE_IN_RAT (4) – Specifies whether a RAT Type change triggers an update. ■ CHANGE_IN_SERVING_NODE (61) – Specifies whether a Serving Node change triggers an update.

vf-Unit-Quota-Threshold

AVP Name	Unit-Quota-Threshold
AVP Code	273
AVP Data Type	Unsigned32
Description	This AVP contains a threshold value in service specific units. This AVP may be included within the Multiple-Services-Credit-Control AVP which also contains a Granted-Service-Units AVP containing CC-Service-Specific-Units AVP (that is, when the granted quota is service specific).

vf-User-Location-Information

AVP Name	User-Location-Information
AVP Code	267
AVP Data Type	OctetString
Description	Indicates details of where the UE is currently located (for example, SAI or CGI).

vf-Volume-Quota-Threshold

AVP Name	Volume-Quota-Threshold
AVP Code	268
AVP Data Type	Unsigned32
Description	The AVP contains a value indicating the quota threshold for volume quota in bytes.

Vendor-Specific AVP Definitions

This section describes the following supported Vendor-Specific AVPs.

- ACW-Entitlement (page A-113)
- ACW-Sponsor-URL (page A-113)
- AF-Msisdn (page A-113)
- Assume-Positive-Treatment (page A-114)
- Charging-Gateway-Function-Host (page A-114)
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ACW-Entitlement

AVP Name	acwentitlement
Vendor-ID	9
AVP Code	131501
AVP Data Type	OctetString
Description	Used to send tethering entitlement information from the PCRF to the PGW for the subscriber session. This VSA can be sent in CCA-I/CCA-U and RAR messages on the Gx interface. The following values for this VSA are defined: GTETH/MHS1/TETH/GVC/GVC1/MHSM/TOGGL/TGLSB/TGLSI. GTETH/MHS1/TETH – If one or more of these values are present in the acwentitlement VSA, the MCC will enable tethering (entitlement) for the session, irrespective of the configured qos-flows. However, the session-level AMBR will still be applied. If the VSA contains any value other than these three values or if the PGW does not receive the acwentitlement VSA in the CCR-I, the MCC will disable tethering (entitlement) for the session and the configured qos-flow will take effect. <i>Note: You can control tethering for an individual APN, based on the presence of configurable strings in the ACW- Entitlement VSA received from the PCRF. For more information, see the Affirmed Networks Gateway Administration Guide.</i>

ACW-Sponsor-URL

AVP Name	ACW-Sponsor-URL
Vendor-ID	9
AVP Code	131506
AVP Data Type	UTF8String
Description	Contains the URL of the manifest server.

AF-Msisdn

AVP Name	AF-Msisdn
Vendor-ID	37963
AVP Code	26
AVP Data Type	String
Description	Contains an MSISDN value used to set the MSISDN for the specific UE.

Assume-Positive-Treatment

AVP Name	Assume-Positive-Treatment
Vendor-ID	12951
AVP Code	7001
AVP Data Type	OctetString
Description	Used on the Gx interface in the CCA-Initial message to indicate a subscriber credit control group type (RTR, UC or Prepaid). The MCC PGW uses the credit control group received in the Assume-Positive-Treatment AVP from the PCRF as a key to select a Subscriber Analyzer. Use the following CLI command to configure the Subscriber Analyzer key for the credit control group. <pre>an3000-mcm-slot7cpul(config) # workflow subscriber-analyzer <name> key <name> credit-control-group <name></pre>

Charging-Gateway-Function-Host

AVP Name	Charging-Gateway-Function-Host
Vendor-ID	12951
AVP Code	6068
AVP Data Type	OctetString
Description	This AVP is sent from the PGW to the OFCS and contains the Fully Qualified Domain Name (FQDN) of the Charging Data Function (CDF). When the PGW receives the Charging-Gateway-Function-Host AVP via the S6b interface, the PGW includes this AVP in the Accounting-Request message on the Gz/Rf interface. <i>Supported on the Gz/Rf Interface only.</i>

Charging-Group-ID

AVP Name	Charging-Group-ID
Vendor-ID	12951
AVP Code	6069
AVP Data Type	UTF8String
Description	This AVP is sent from the PGW to the OFCS and identifies the subscriber charging group. When the PGW receives the Charging-Group-ID via the S6b interface, the PGW includes this AVP in the Accounting-Request message on the Gz/Rf interface. <i>Supported on the Gz/Rf Interface only.</i>

Cisco-CC-Failure-Type

AVP Name	Cisco-CC-Failure-Type
Vendor-ID	9
AVP Code	132077
AVP Data Type	Unsigned32
Description	Indicates the OCS failure reasons to the PCRF. <i>Supported on the Gx Interface.</i> The following values are defined: ■ CC_CONECTION_FAILURE (0) ■ CC_RESPONSE_TIMEOUT (1) ■ CC_UNKNOWN_FAILURE (5) ■ CC_RESULT_CODE – Result code value received over the Gy interface

Cisco-Event

AVP Name	Cisco-Event
Vendor-ID	9
AVP Code	131195
AVP Data Type	Grouped
Description	Indicates the OCS failure result code. <i>Supported on the Gx Interface.</i> The detailed syntax is as follows: <pre>Cisco-Event ::= < AVP Header: 131195> [Cisco-Event-Trigger-Type] [Cisco-CC-Failure-Type]</pre>

Cisco-Event-Trigger

AVP Name	Cisco-Event-Trigger
Vendor-ID	9
AVP Code	131193
AVP Data Type	Grouped
Description	Indicates the events that cause the OCS failure. <i>Supported on the Gx Interface.</i> The detailed syntax is as follows: <code>Cisco-Event-Trigger ::= < AVP Header: 131193></code> <code>[Cisco-Event-Trigger-Type]</code>

Cisco-Event-Trigger-Type

AVP Name	Cisco-Event-Trigger-Type
Vendor-ID	9
AVP Code	131192
AVP Data Type	Enumerated
Description	Indicates the events that cause the OCS failure. <i>Supported on the Gx Interface.</i> The following values are defined: ■ NO_CISCO_EVENT_TRIGGERS (0) ■ CREDIT_CONTROL_FAILURE (5)

Cumulative-Acct-Input-Octets

AVP Name	Cumulative-Acct-Input-Octets
Vendor-ID	12951
AVP Code	132044
AVP Data Type	Unsigned64
Description	Contains the total number of input octets used during the entire session. Added to the service data containers generated by the PGW for rating groups. <i>Supported on the Gz/Rf Interface.</i>

Cumulative-Acct-Output-Octets

AVP Name	Cumulative-Acct-Output-Octets
Vendor-ID	12951
AVP Code	132045
AVP Data Type	Unsigned64
Description	Contains the total number of output octets used during the entire session. Added to the service data containers generated by the PGW for rating groups. <i>Supported on the Gz/Rf Interface.</i>

Gx-TMO-Append-IMEI

AVP Name	Gx-TMO-Append-IMEI
Vendor-ID	29168
AVP Code	117
AVP Data Type	Enumerated
Description	Defines whether the IMEI of the mobile station is appended to the redirect URL. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ Disable (0) – (default) Do not append the IMEI. ■ Enable (1) – Append the IMEI.

Gx-TMO-Append-IMSI

AVP Name	Gx-TMO-Append-IMSI
Vendor-ID	29168
AVP Code	116
AVP Data Type	Enumerated
Description	Defines whether the IMSI of the mobile station is appended to the redirect URL. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ Disable (0) – (default) Do not append the IMSI. ■ Enable (1) – Append the IMSI.

Gx-TMO-Append-MSIP

AVP Name	Gx-TMO-Append-MSIP
Vendor-ID	29168
AVP Code	118
AVP Data Type	Enumerated
Description	Defines whether the IP address assigned to the mobile station is appended to the redirect URL. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ Disable (0) – (default) Do not append the MS IP address. ■ Enable (1) – Append the MS IP address. If both IPv4 and IPv6 MS IP addresses are allocated, the MCC appends the active IP address to the redirect URL. For IPv6 IP addresses, the MCC uses the preferred format.

Gx-TMO-Append-MSISDN

AVP Name	Gx-TMO-Append-MSISDN
Vendor-ID	29168
AVP Code	115
AVP Data Type	Enumerated
Description	Defines whether the MSISDN of the mobile station is appended to the redirect URL. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ Disable (0) – (default) Do not append the MSISDN. ■ Enable (1) – Append the MSISDN.

Gx-TMO-Append-Original-URL

AVP Name	Gx-TMO-Append-Original-URL
Vendor-ID	29168
AVP Code	114
AVP Data Type	Enumerated
Description	Defines whether the original URL is appended to the redirect URL. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ Disable (0) – (disable) Do not append the original URL. ■ Enable (1) – Append the original URL.

Gx-TMO-Deactivate-By-Redirect

AVP Name	Gx-TMO-Deactivate-By-Redirect
Vendor-ID	29168
AVP Code	119
AVP Data Type	Enumerated
Description	Defines whether the system performs a one-time redirection. After the one-time redirection, the rule is deactivated. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ Disable (0) – The rule is not deactivated once enforced. ■ Enable (1) – The rule is deactivated once enforced.

Gx-TMO-Redirect-Address-Type

AVP Name	Gx-TMO-Redirect-Address-Type
Vendor-ID	29168
AVP Code	112
AVP Data Type	Enumerated
Description	Defines the address type of the address given in the TMO-Redirect-Server-Address VSA. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ IPv4 Address (0) ■ IPv6 Address (1) ■ URL (2)

Gx-TMO-Redirect-Server

AVP Name	Gx-TMO-Redirect-Server
Vendor-ID	29168
AVP Code	111
AVP Data Type	Grouped
Description	Contains the address information of the redirect server (for example, HTTP redirect server) to which the end user is redirected. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The detailed syntax is as follows: <pre>Gx-TMO-Redirect-Server ::= < AVP Header: 111> [Gx-TMO-Redirect-Address-Type] [Gx-TMO-Redirect-Server-Address] [Gx-TMO-Append-Original-URL] [Gx-TMO-Append-MSISDN] [Gx-TMO-Append-IMSI] [Gx-TMO-Append-IMEI] [Gx-TMO-Append-MSIP] [Gx-TMO-Deactivate-By-Redirect]</pre> Note: If both the Redirect-Information AVP and Gx-TMO Redirect-Server VSA are received, the MCC will use the Gx-TMO Redirect-Server VSA for redirection and will ignore the Redirect-Information AVP provided the Gx-TMO-Redirect-Server VSA is enabled in the CCA data record template.

Gx-TMO-Redirect-Server-Address

AVP Name	Gx-TMO-Redirect-Server-Address
Vendor-ID	29168
AVP Code	113
AVP Data Type	UTF8String
Description	Defines the address of the redirect server (for example, HTTP redirect server) with which the end user is to be connected. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Max-Wait-Time

AVP Name	Max-Wait-Time
Vendor-ID	12951
AVP Code	7103
AVP Data Type	Unsigned32
Description	Indicates the maximum amount of time (in milliseconds) that the originator is willing to wait for a response since the Origination-TimeStamp. This AVP is only provided with the initial authorization request. <i>Supported on the Gx and Gy Interfaces only.</i>

MVNO-Reseller-Id

AVP Name	MVNO-Reseller-Id (affmnvostringid)
Vendor-ID	9
AVP Code	131507
AVP Data Type	UTF8String
Description	This AVP is sent from the PCRF to the PGW/GGSN (via the Gx interface) or from the PGW/GGSN to the OCS (via the Gy interface) and contains the Mobile Virtual Network Operator (MVNO) name or identifier. <i>Supported on the Gx and Gy Interfaces only.</i>

MVNO-Sub-Class-Id

AVP Name	MVNO-Sub-Class-Id (<i>affmvnosubclassid</i>)
Vendor-ID	9
AVP Code	131508
AVP Data Type	UTF8String
Description	This AVP is sent from the PCRF to the PGW/GGSN (via the Gx interface) or from the PGW/GGSN to the OCS (via the Gy interface) and contains the Mobile Virtual Network Operator (MVNO) sub-name or sub-identifier. <i>Supported on the Gx and Gy Interfaces only.</i>

Origination-Time-Stamp

AVP Name	Origination-Time-Stamp
Vendor-ID	12951
AVP Code	7102
AVP Data Type	Unsigned64
Description	Indicates the time that the message was sent, representing the number of seconds since January 1, 1900 00:00 UTC. This AVP is only provided with the initial authorization request. <i>Supported on the Gx and Gy Interfaces only.</i>

Override-Charging-Action-Exclude-Rule

AVP Name	Override-Charging-Action-Exclude-Rule
Vendor-ID	9
AVP Code	132021
AVP Data Type	Octet String
Description	Indicates the name of the service rule to which override control will not be applied. Service rules that are specified in this VSA will still be associated with the rating group based on the workflow configuration. This VSA may be included more than once if more than one rule needs to be excluded.

Override-Charging-Action-Parameters

AVP Name	Override-Charging-Action-Parameters
Vendor-ID	9
AVP Code	132019
AVP Data Type	Grouped
Description	This AVP is sent from the PCRF to the PCEF (via the Gx interface) and is used to override charging parameters for predefined and static rules on the PCEF. The detailed syntax is as follows: <pre> Override-Charging-Action-Parameters ::= < AVP Header: 132019> *[Override-Charging-Action-Exclude-Rule] [Override-Charging-Parameters] </pre>

Override-Charging-Parameters

AVP Name	Override-Charging-Parameters
Vendor-ID	9
AVP Code	132022
AVP Data Type	Grouped
Description	Contains the override charging parameters to be applied to predefined and static rules on the PCEF. The detailed syntax is as follows: <pre> Override-Charging-Parameters ::= < AVP Header: 132022> [Override-Rating-Group] [Override-Online] [Override-Offline] </pre>

Override-Control

AVP Name	Override-Control
Vendor-ID	9
AVP Code	132017
AVP Data Type	Grouped
Description	This AVP is sent from the PCRF to the PCEF (via the Gx interface) and is used to override charging parameters for predefined and static rules on the PCEF. The detailed syntax is as follows: <pre>Override-Control ::= < AVP Header: 132017> [Override-Charging-Action-Parameters] *[Override-Charging-Action-Exclude-Rule] [Override-Charging-Parameters] [Override-Rating-Group] [Override-Online] [Override-Offline]</pre>

Override-Offline

AVP Name	Override-Offline
Vendor-ID	9
AVP Code	132027
AVP Data Type	Enumerated
Description	This VSA overrides the offline value configured for a static/predefined rule. The following values are defined: ■ DISABLE_ONLINE (0) – Indicates that the offline charging interface for the associated service rule is disabled. ■ ENABLE_ONLINE (1) – Indicates that the offline charging interface for the associated service rule is enabled.

Override-Online

AVP Name	Override-Online
Vendor-ID	9
AVP Code	132026
AVP Data Type	Enumerated
Description	This VSA overrides the online value configured for a static/predefined rule. The following values are defined: ■ DISABLE_ONLINE (0) – Indicates that the online charging interface for the associated service rule is disabled. ■ ENABLE_ONLINE (1) – Indicates that the online charging interface for the associated service rule is enabled.

Override-Rating-Group

AVP Name	Override-Rating-Group
Vendor-ID	9
AVP Code	132024
AVP Data Type	Unsigned32
Description	This VSA is used to override the value of the rating group configured for a static/predefined rule. The rating group identifier sent in this VSA must correspond to a predefined rating group quota-id on the MCC (<code>services -> charging -> rating-group -> quota-id</code>).

QoS-Group-Rule-Definition

AVP Name	QoS-Group-Rule-Definition
Vendor-ID	9
AVP Code	132003
AVP Data Type	Grouped
Description	The PCRF uses this VSA to define the parameters for a pre-configured service rule whose name matches the QoS-Group-Rule-Name. For example, if the received QoS-Group-Rule-Definition VSA has a redirect server address, the MCC will apply redirection to the associated flows. <i>Note: The Redirect-Server AVP sent from the PCRF as part of the QoS-Group-Rule-Definition AVP applies to predefined PCC rules only and applies to both online and offline PCC rules.</i> The detailed syntax is as follows: <pre>QoS-Group-Rule-Definition ::= <AVP header: 132003> {QoS-Group-Rule-Name} [Monitoring-Key] [Flow-Status] [Redirect-Server]</pre>

QoS-Group-Rule-Install

AVP Name	QoS-Group-Rule-Install
Vendor-ID	9
AVP Code	132001
AVP Data Type	Grouped
Description	The PCRF uses this VSA to activate predefined service rules on the MCC gateway. The detailed syntax is as follows: <pre>QoS-Group-Rule-Install ::= <AVP header: 132001 > * [QoS-Group-Rule-Definition]</pre>

QoS-Group-Rule-Name

AVP Name	QoS-Group-Rule-Name
Vendor-ID	9
AVP Code	132004
AVP Data Type	OctetString
Description	The MCC tries to match the service rule name, received in the QoS-Group-Rule-Name VSA, to a service rule name configured locally on the MCC gateway (<i>service-construct</i> → <i>service-rule</i>). ■ If a match is found, the MCC activates the matching service rule and all services associated with the received QoS-Rule-Group-Name. ■ If a match is not found, the MCC sends the Charging-Rule-Report AVP with the QoS-Group-Rule-Name VSA to the PCRF.

QoS-Group-Rule-Remove

AVP Name	QoS-Group-Rule-Remove
Vendor-ID	9
AVP Code	132002
AVP Data Type	Grouped
Description	The PCRF uses this VSA to deactivate predefined service rules on the MCC gateway. The detailed syntax is as follows: <pre>QoS-Group-Rule-Remove ::= <AVP header: 132002 > * [QoS-Group-Rule-Name]</pre>

TMO-Transparent-Data

AVP Name	TMO-Transparent-Data
Vendor-ID	29168
AVP Code	1
AVP Data Type	Octet String
Description	Contains information from the TMO-Transparent-Data VSA received from the RADIUS server. The MCC gateway includes this VSA in the CCR message to the OCS when the RADIUS server returns a transparent data value in the Access-Accept message. If the RADIUS Access-Accept message does not return a transparent data value, you can configure the MCC gateway to assign a transparent data value and the MCC gateway will include the configured transparent data value in this VSA to the OCS. For more information on how to configure the MCC gateway to assign a transparent data value, see the <i>Affirmed Networks Gateway Administration Guide</i>.

TMO-Virtual-Gi-ID

AVP Name	TMO-Virtual-Gi-ID
Vendor-ID	29168
AVP Code	15
AVP Data Type	32 Bit Integer
Description	Contains the value of the virtual-ID. See “Processing the TMO-Virtual-Gi-ID VSA” in the <i>Affirmed Networks RADIUS Interface Specification</i> for more information.

Trigger (VSA)

AVP Name	Trigger
Vendor-ID	12951
AVP Code	1264
AVP Data Type	Grouped
Description	Contains the trigger types. The detailed syntax is as follows: <pre>Trigger ::= < AVP Header: 1264 > 1*4 [Trigger-Type]</pre>

Trigger-Type (VSA)

AVP Name	Trigger-Type
Vendor-ID	12951
AVP Code	870
AVP Data Type	Enumerated
Description	When included in the CCA on the Gx interface, the Trigger-Type AVP is used to notify the PGW when a ULI, Time Zone, or PLMN change occurs and the PGW is required to send the respective event triggers over the Gy interface to the OCS. On the Gx interface, the Trigger-Type AVP contains the following values: ■ CHANGE_IN_UE_TIMEZONE (5) – Indicates a change in the UE time zone. ■ CHANGEINLOCATION_MCC (30) – Indicates a change in the MCC of the serving network, which represents the first half of the PLMN. ■ CHANGEINLOCATION_MNC (31) – Indicates a change in the MNC of the serving network, which represents the second half of the PLMN. ■ CHANGE_IN_LOCATION (3) – Indicates a change in the user location. If this Trigger-Type is received, the PGW will send the OCS an updated ULI for all changes, including the PLMN; therefore, this value should not be used in conjunction with the MCC and MNC location change Trigger-Types. Upon receiving the Trigger AVP → Trigger-Type AVP from the PCRF and if the subscriber does not have a Gy session, the PGW sends a CCR-Event message to the OCS via the Gy interface that includes the following AVPs: CC-Request-Type: EVENT_REQUEST (4) Requested-Action: Location Update (4) Service-Information PS-Information 3GPP-MS-Timezone 3GPP-SGSN-MCC-MNC 3GPP-User-Location-Info Note: If the subscriber already has a Gy session, the PGW sends a CCR-U message if a quota request is needed for any of the rating groups.

